Team Sammard: Project AIRAVATA

Team 188: Project Technical Report for the 2025 IREC



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The following document outlines the design, analysis, testing, and production specifications of Team Sammard's rocket, *AIRAVATA*, for IREC 2025. Designed to reach an apogee of 10,000 ft AGL, *AIRAVATA* is our entry into the 10k COTS category of the International Rocket Engineering Competition 2025. COTS solid motors are widely recognized in amateur rocketry for their optimal blend of reliability and performance. This serves as the primary motivation for selecting a solid propulsion system for *AIRAVATA*. The payload is a high-energy platform integrating three advanced subsystems that push the boundaries of rocket flight. A key feature is the Visual-Aided Navigation System—a GPS-independent solution enabling precise guidance across extraterrestrial and uncharted terrains. This system ensures high-accuracy navigation without reliance on satellite signals. The payload also houses two Scintillators, an industry graded and an additively manufactured scintillator, in order to do a comparative analysis on their performance in detecting the muon flux throughout the rocket's trajectory. The data will be then used to carry out simulation studies for muon tomography. This innovative approach to muon tomography provides valuable insights into structural properties during critical supersonic and sonic flight phases, marking a significant leap in aerospace navigation and structural analysis. The mission objective is to reliably collect data from the payload and onboard flight computer throughout the flight, reach the target apogee, and successfully recover the launch vehicle and its components via a soft landing.

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I. Nomenclature

I.A Chemicals

ABS	Acrylonitrile Butadiene Styrene
Al	Aluminum
APCP	Ammonium Perchlorate Composite Propellant
BP	Black Powder
PLA	Polylactic Acid
SS	Stainless Steel

I.B Acronyms

AGL	Above Ground Level
AoA	Angle of Attack
AoS	Angle of Sideslip
$CONOPS$	Concept of Operations
CAD	Computer Aided Design
CFD	Computational Fluid Dynamics
CFRP	Carbon Fiber Reinforced Polymer
CNC	Computer Numerical Control
COTS	Commercial off the shelf
Cp	Centre of Pressure
Cg	Center of gravity
CSV	Comma Separated Values
DoF	Degrees of Freedom
$DTEG$	Design, Test, & Evaluation Guide
dt	Time Step
ESRA	Experimental Sounding Rocket Association
FOS	Factor of Safety
FPS	Frames per Second
FSI	Fluid Structure Interaction
$GFRP$	Glass Fiber Reinforced Polymer
GPS	Global Positioning System
GPU	Graphics Processing Unit
GSM	Grams per Square Meter
GUI	Graphical User Interface
IMU	Inertial Measurement Unit
IREC	International Rocket Engineering Competition
LED	Light Emitting Diode
$Li-ion$	Lithium-ion Battery
$LiPo$	Lithium-ion Polymer Battery
MP	Mega Pixel
PCB	Printed Circuit Board
PDB	Power Distribution Board
SD	Secure Digital
SDL	Space Dynamics Laboratory
SRAD	Student Researched and Developed

I.C Variables

A	Cross sectional area (m^2)
AR	Aspect Ratio
c	Chord
C_d	Coefficient of drag
D_0	Outer diameter (mm)
D_i	Inner diameter (mm)
f, g	Generic functions
F	Force (N)
F_{load}	Force from load (N)
F_{max}	Maximum force (N)
Fx	X component of the resultant pressure force acting on the vehicle
Fy	Y component of the resultant pressure force acting on the vehicle
Fz	Z component of the resultant pressure force acting on the vehicle
h	Height
i	Time index during navigation
j	Waypoint index
p	Pressure (Pa)
r	Radius (mm)
S	Nominal surface area (m^2)
t	Wall thickness of a cylindrical pressure vessel (mm)
V	Velocity (m/s)
V_f	fin flutter velocity
V_d	desired descent velocity
V_I	velocity at main deployment
W	Weight (kg)

II. Introduction

Team Sammard is a vibrant and passionate student aerospace team from the Vellore Institute of Technology, Vellore, India; which aims to enrich its members in the domain of aerospace technology by competing in various competitions like IREC, US CANSAT and multiple other Indian Space-Based Competitions. Team Sammard is recognized as one of the most prestigious technical teams in VIT Vellore. Team Sammard also takes pride in delivering strong hands-on experience along with technical knowledge to its members and has prominent national visibility by frequenting various industrial expositions and conferences. Furthermore, Team Sammard hosts a plethora of enlightening webinars, courses, and sessions, not just for its members, but also to enthusiastic students of the university to empower the community. The team not only aims to develop state-of-the-art technology but also yearns to inspire young minds to pursue aerospace related careers.

A. Academic Program

Team Sammard, a distinguished sounding rocket team based at Vellore Institute of Technology (VIT), India, benefits from a range of university amenities, including access to funding requests, financial system management, and facilities. While the university provides essential funding for the team through their quarterly funding requests, the team also secures its budget via external sponsorship and donations to the team. Composed primarily of engineering students, Team Sammard utilizes Project Labs' state-of-the-art manufacturing and testing facilities to explore and refine rocket concepts. Embracing a multidisciplinary approach, the team actively recruits members from diverse academic backgrounds, mirroring the collaborative environment found in industry settings. This diversity fosters the development of complex systems and encourages innovative problem-solving. The team's overarching goal is to bridge academic knowledge with real-world applications, offering members invaluable experience relevant to future careers in the industry. Furthermore, Team Sammard is committed to nurturing the talents of younger students, providing them with early exposure to technical and communication skills essential for success in their academic journeys.

This year the team consists of 46 members comprising of Mechanical, Electronic, Electrical, Computer Science, and Instrumentation Engineering graduates. Even though Vellore Institute of Technology does not have an aerospace program, our members through a process of rigorous self-learning, bridge the gap between learning and application. This offers our members a healthy knowledge in a multitude of domains.

Since the its inception, Team Sammard has always aimed to focus on challenging traditional manufacturing techniques while operating in an extremely tight budget constraint. This year in the IREC 2025, the team aims to bring a competitive rocket and win the International Rocket Engineering Competition hosted by the Experimental Sounding Rocket Agency (ESRA) and will also submit unique and challenging payload for the Space Dynamics Laboratory Payload Challenge.

B. Stakeholders

Team Sammard, being primarily new in the field of rocketry, relies primarily on funding from Vellore Institute of Technology, Vellore, to support its endeavors. The commitment and passion of its members often extend beyond graduation, with many continuing to support the team long after their time at the university. In addition to university funding, the team benefits from financial contributions from individuals and dedicated alumni who are invested in the team's success. Moreover, team members themselves contribute financially to sustain the operations of the team, demonstrating their dedication and commitment to its mission. The invaluable support of the university and monetary providers has been instrumental in the team's development journey. Their contributions serve as the foundation for current projects and pave the way for future endeavors. Furthermore, alumni play a crucial role by offering technical expertise, solutions, and guidance, serving as mentors to current team members as they navigate challenges and pursue innovation in rocket design and manufacturing.

VIT®
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

Image 1. University Logo

C. Team Management Strategies

Team Sammard utilizes a structured executive leadership to facilitate the continuous growth of the team. The Captain and the Head of Operations act as anchors to the university by acting as the crucial interface between the team and Vellore Institute of Technology's Student Welfare Office. They handle important operations like obtaining funds, managing finances, and obtaining test permissions from the university. The competition/project leads look into the development of the rocket for the International Rocket Engineering Competition (IREC 2025) and other competitions like US CANSAT. Each department has a department head to oversee all tasks in each department happen smoothly within the required deadlines. They act as the backbone of the team by guiding the members and juniors in their technical learning.

Team Sammard thrives on principles like accountability, punctuality, and technical competence. Open communication is paramount and is facilitated through digital, oral, and written channels. Virtual communication thrives on WhatsApp Messenger, where each department and sub-department manages dedicated groups for discussions and inquiries. At the onset of each project season, senior management and department heads collaborate to outline a comprehensive work timeline and define both long-term and short-term goals. Daily meetings at 9:30 PM in the university's Project Labs foster collaboration across departments, with weekly reviews to ensure alignment with objectives. To foster skill development and knowledge transfer, regular training sessions, technical presentations, and individual/group tasks are conducted by sub-department leads. After all competitions and the project season reflection is key at the end of each season, with senior management evaluating achievements and areas for improvement. Insights from these reviews inform adjustments to the team structure and project requirements, with leadership roles transitioning to younger members to sustain continuity and fresh perspectives.



Image 2.Team Night Lab Discussions

D. Team Structure

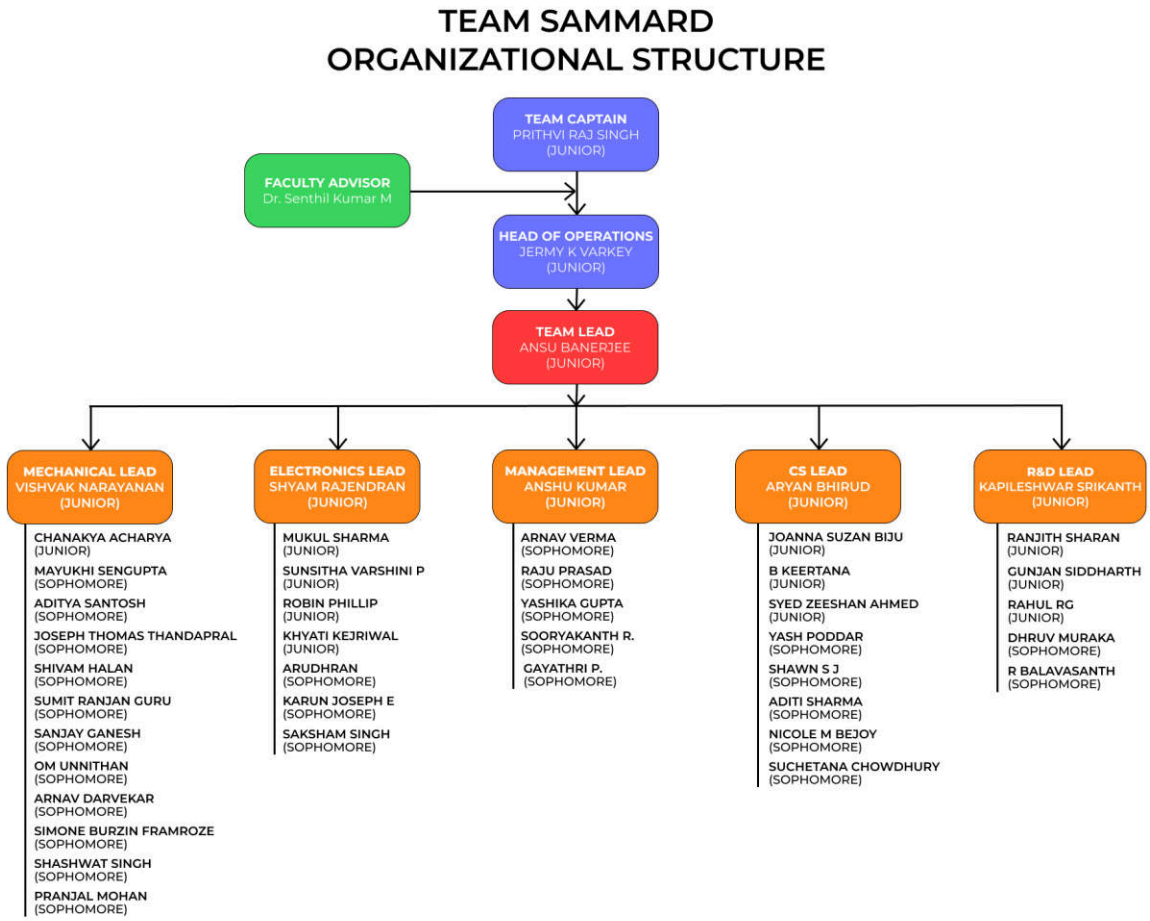


Image 3. Team Structure

III. System Architecture Overview

Airavata is the fourth launch vehicle developed by Team Sammard and serves as the official entry for the 10k COTS category at the Experimental Sounding Rocket Association's (ESRA) International Rocket Engineering Competition (IREC). Powered by the Aerotech M1845NT-PS motor, *Airavata* is designed to achieve an apogee of 10,000 feet. Its design integrates numerous improvements based on lessons learned from previous missions and features a non-deployable scientific payload, a state-of-the-art avionics bay, and an innovative recovery system.

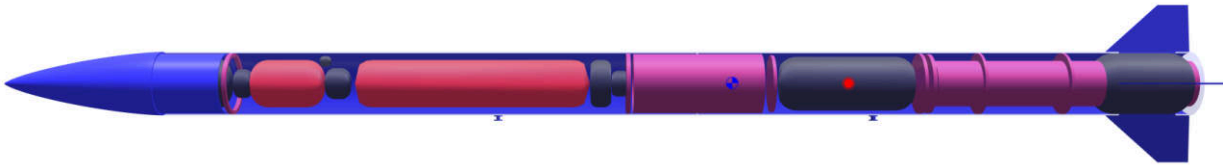


Image 4. Rocket 3D Figure Image

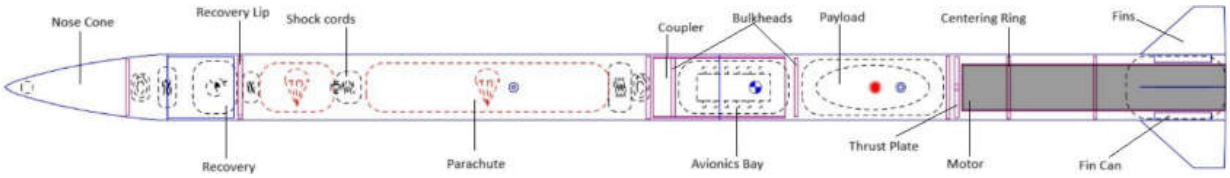


Image 5. Labelled Rocket Image

The basic overview of the rocket is provided in the table below:

Rocket Parameters	Measurements
Airframe Length with Nosecone	282cm
Airframe Length without Nosecone	245cm
Airframe Diameter	14.8cm
Number of Fins	4
Fin Length	Root Chord 19.5cm, Tip Chord 70cm
Vehicle Weight	25612g
Propellant Type	COTS Solid
Propellant Letter Characterization	M
Propellant Weight	6682g
Number of Stages	1

Table 1. Rocket Parameters

A. COTS Propulsion

Airavata features a M Class motor which sits securely inside a body tube of length 1250 mm, which also accommodates the payload positioned directly above the motor. This arrangement not only ensures an efficient use of internal space. The compact vertical stacking of the motor and payload within the body tube allows for simplified structural integration and ease of assembly.

Motor Details

The Aerotech M1845NT-PS was chosen for *Airavata* primarily due to its Blue Thunder propellant formulation, which delivers a high-thrust profile ideal for achieving the performance requirements of the 10K COTS competition category. In addition to its impressive thrust-to-weight characteristics, the motor was selected for its broad availability, cost-effectiveness, and reliability when compared to other motors within the same total impulse class, making it a practical and efficient choice for the mission profile.
Table <> provides a detailed overview of the motor's specifications which includes key parameters such as total impulse, burn time, and average thrust and Figure illustrates the motor's thrust curve, offering insight into its performance characteristics over time and highlighting the thrust profile critical to trajectory planning and simulation accuracy.

PROPERTY	VALUE
Casing Length	23.50 inches (597mm)
Hardware Required	RMS-98/7680
Total Impulse	8307 N-sec
Average Thrust	1875 N
Peak Thrust	3081 N-sec
Burn Time	4.4 seconds
Delay Time	Plugged
Propellant Weight	3772 grams
Motor Weight	6682 grams

Table 2. Motor Details

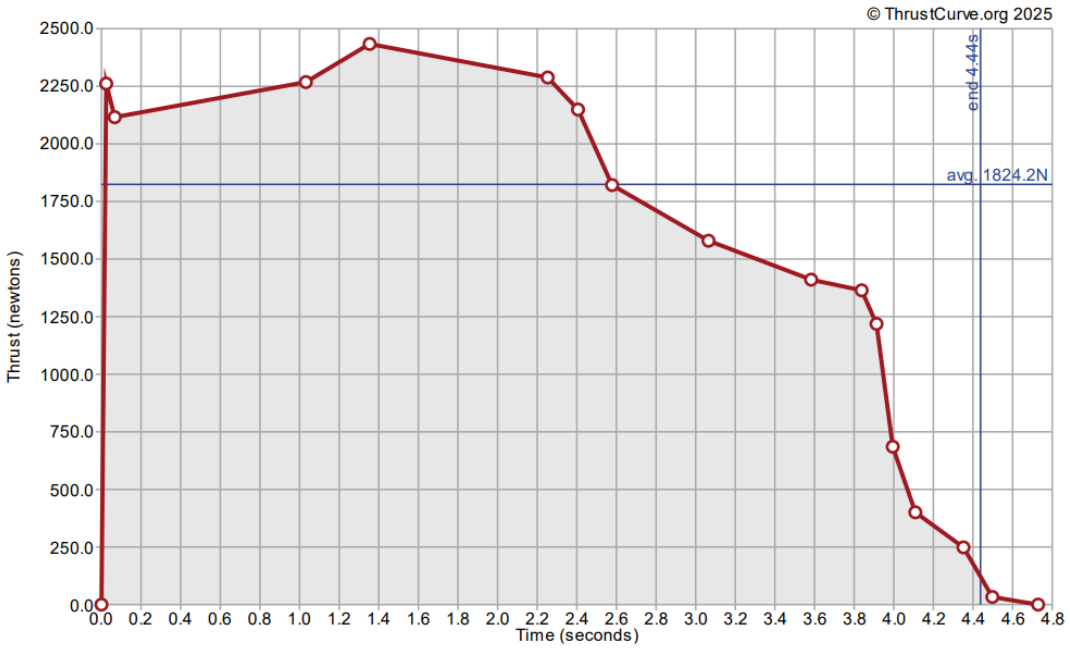


Image 6. Thrust Curve

Motor Selection Criteria

A comprehensive comparative analysis was conducted to evaluate the performance of various motors with similar specifications. Based on this assessment, the M1845 NT-PS motor was selected as the optimal choice for our current configuration. This decision was primarily driven by its superior performance metrics, including higher total impulse, as well as greater maximum and average thrust when compared to its competitors.

In addition to its favorable technical characteristics, the M1845 NT-PS also benefits from our team's prior experience with the motor, having successfully employed it in last year's launch. Its proven reliability and consistent performance under similar conditions further reinforced our confidence in selecting this motor for the current mission profile.

Motor	M1845NT-PS	M1545 Green3	M1790 Skidmark
Manufacturer	Aerotech	Cessaroni	Cessaroni
Propellent Type	Blue Thunder	Green3	Skidmark
Diameter	75mm	75mm	75mm
Total Impulse	8307 N-sec	8186 N-sec	8088 N-sec
Maximum Thrust	3081 N	1840 N	2092 N
Average Thrust	1875 N	1547 N	1786 N
Burn Time	4.4s	5.29s	4.53s
Cost	641.99 USD	572.72 USD	828.32 USD

Table 3. Motor Comparison

B. Flight Mechanics

Flight simulations were conducted using the OpenRocket software, which employs an Extended Barrowman method along with a six degrees of freedom (6-DoF) Runge-Kutta method to simulate the launch vehicle's flight dynamics. To ensure the accuracy of the simulation results, predicted physical conditions at the launch site were carefully input into the software. These included the altitude of Saragosa, the average wind speed of 7m/s, and the international standard atmospheric temperature.

The launch angle was set at 6 degrees, and the launch rail height was configured at 17 feet. OpenRocket was initially used to develop the base design of the rocket and to generate an early apogee estimate for the vehicle's flight. Following this, the entire design underwent an iterative feedback loop, incorporating results from Computational Fluid Dynamics (CFD) and Finite Element Analysis (FEA) performed on the complete rocket and its individual components. This approach allowed for continuous improvements to the design, ensuring optimized performance and structural integrity.

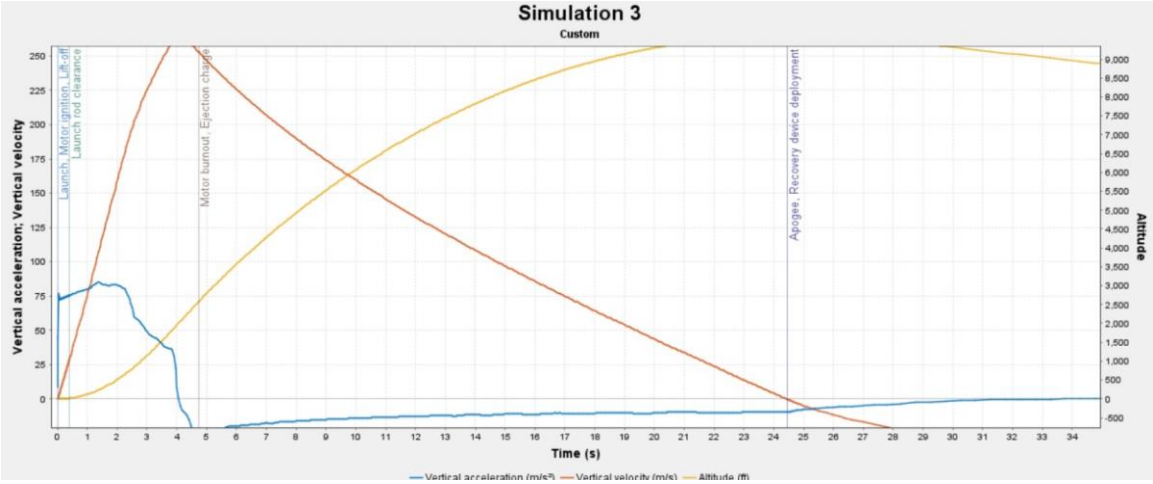


Image 7: Altitude, Vertical Velocity, Vertical Acceleration vs Time

During design, it is carefully ensured that launch rod clearance velocity of the rocket remains below 30 m/s as stated in the launch guidelines. The rocket is set to achieve a peak acceleration of 9.75m/s^2, peak velocity of 271 m/s and apogee of 9640ft.

Stability Margin:

Stability of a rocket can be classified into three types:

1. Under stability - a lower than optimum stability margin can cause understability to the rocket and thereby make it tumble after clearing the launch pad.
2. Optimum Stability - an optimally stabilized rocket will have a stable flight without deviating from its planned flight path.
3. Overstability - a higher than optimum stability margin can cause overstability to the rocket and can thereby lead to weathercocking. Weathercocking is a phenomenon where the rocket runs into the wind which ultimately reduces stability and altitude.

The value of stability margin is given by:

$$Stability\ Margin = \frac{C_p - C_g}{D}$$

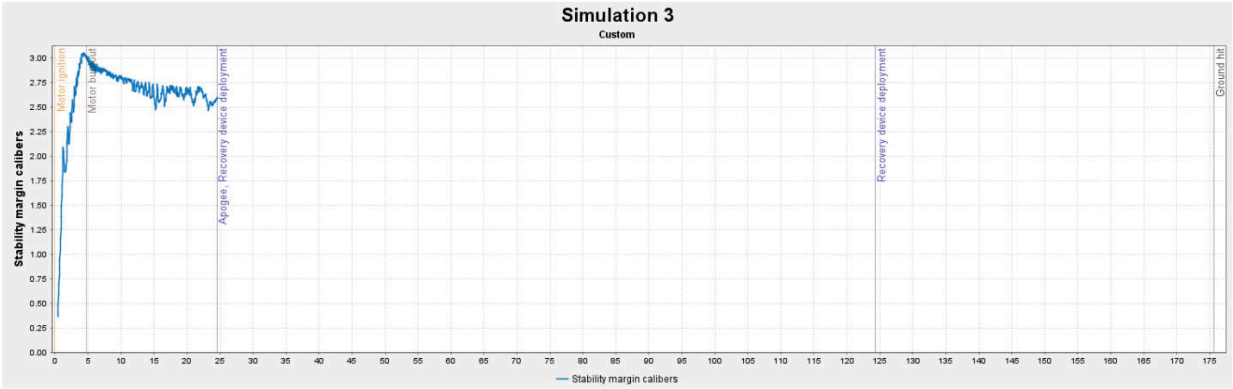


Image 8: Stability Margin Calibers vs Time

The rocket is designed in such a way that the rocket neither undergoes understability nor overstability at any point of time during the entire flight. The maximum stability achieved by the rocket stands at 3.2 calipers

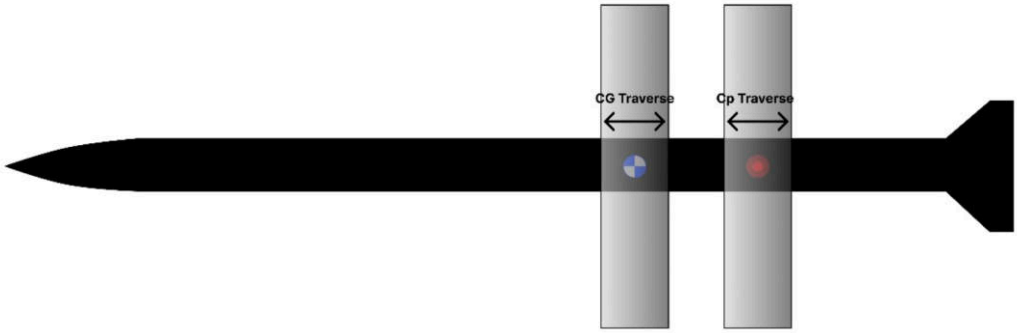


Image 9. CP CG Location

The figure shows the traversal of the center of gravity and center of pressure in the launch vehicle throughout the rocket's flight. The Centre of Pressure being adequately below the Centre of Gravity ensures that the rocket remains stable throughout the flight.

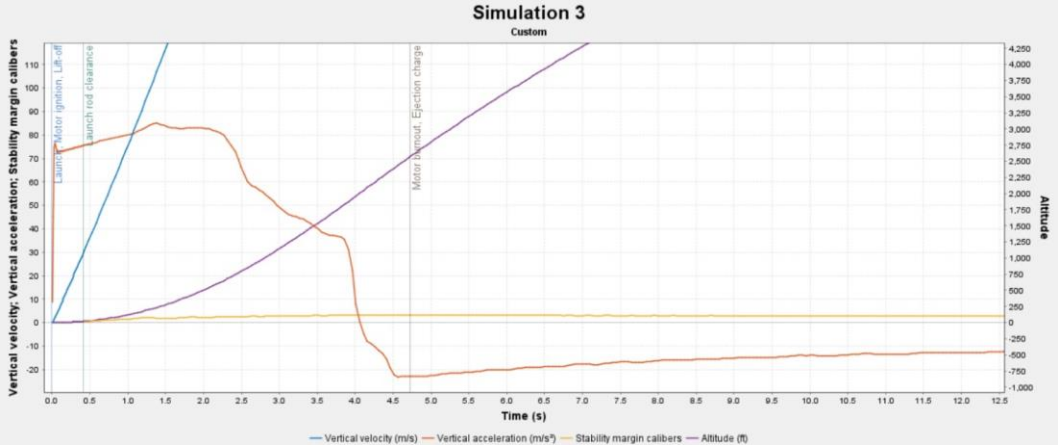


Image 10. Stability Calipers Graph during flight

CFD Simulations of the Rocket:

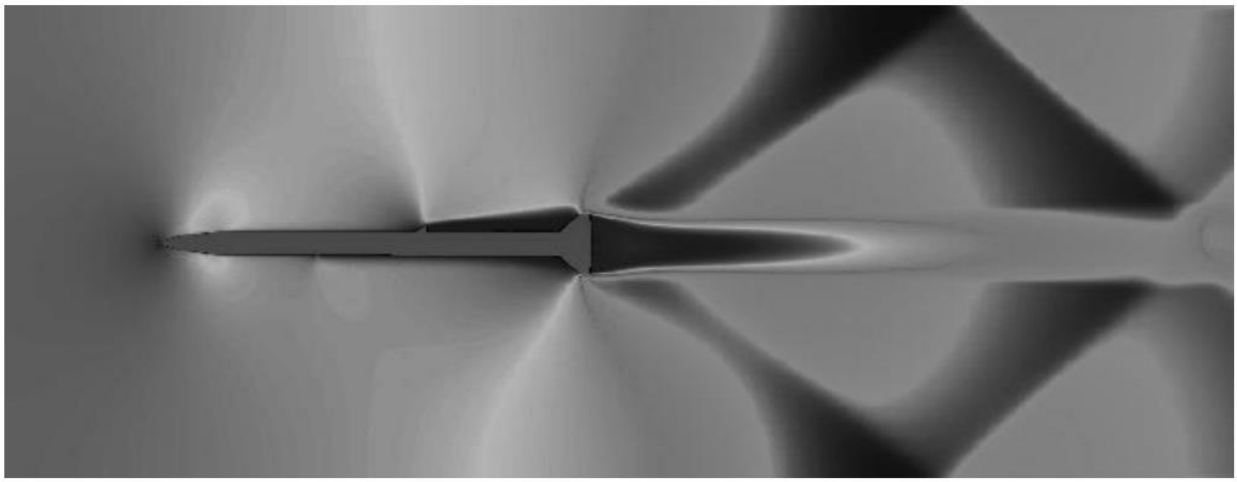


Image 11. Shock Wave Formation

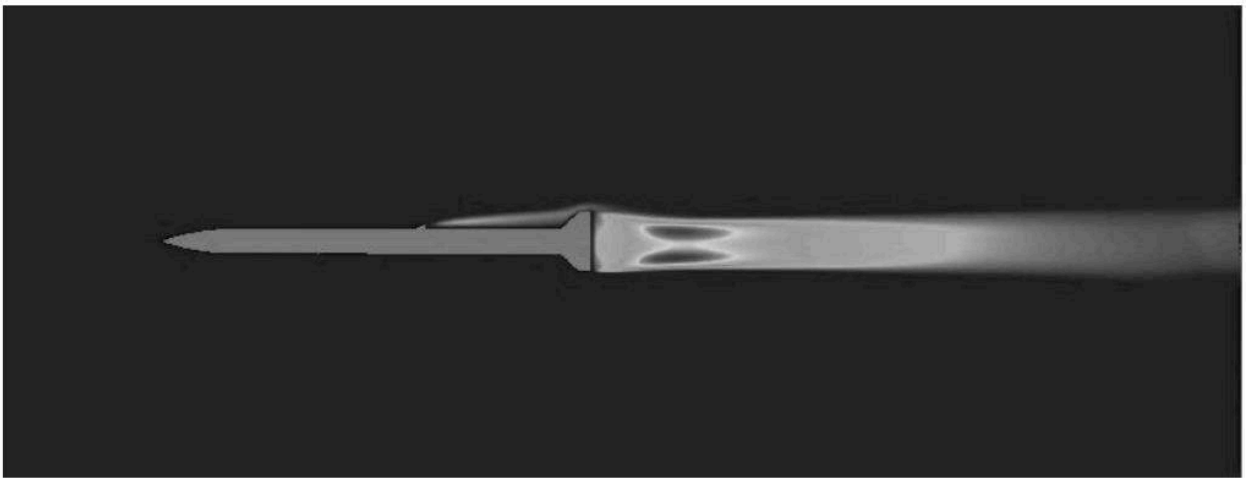


Image 12. Turbulence Wake Region Formation



Image 13. 3D Flight Visualization

C. Aerostructures Subsystem

In *Airavata*, the Aerodynamics and Structures team implemented an integrated design framework aimed at optimizing the vehicle's performance to reach a target apogee of 10,000 feet. Concurrently, the methodology ensured structural robustness under dynamic flight loads. The design strategy was grounded in maintaining aerodynamic stability and ensuring structural integrity throughout all flight phases to uphold the flight safety requirements.

The following software were used for the design and simulation of the rocket:

- OpenRocket - Open-source software used to design the 2D layout of the rocket and to run basic simulations to get an approximate apogee and ground hit velocity.
- SolidWorks 2024 - 3D CAD modelling software used to model the entire rocket.
- ANSYS 2024 student -The Static Structural module is employed to perform stress analysis on various rocket components using the finite element method. Aerodynamic forces during flight are computed using the FLUENT module, which simulates external flow around the rocket via the finite volume method. The Explicit Dynamics module is utilized to evaluate impact loading effects on the fins. Modal Analysis is conducted to determine the payload's natural frequencies and vibrational behavior. The Transient Thermal module supports thermal response simulations, while the Coupled Field Transient module is applied for fluid-structure interaction (FSI) analysis of the fins
- FinSim v5.7 - To calculate fin flutter velocity and deflection.

The following table is a list of all screws, pins and bolts utilized for the assembly of *Airavata*

Dimension	Type	Reference standard	Material /grade
M10	EYE BOLT	DIN 580	SS-316
M4X150	THREADED RODS		SS-304
M4X10	HEX NUTS	DIN 934	GRADE 8
M4X16	HEX NUTS	DIN 934	GRADE 8
M4X20	HEX NUTS	DIN 934	GRADE 8
M3X16	HEX NUTS	DIN934	GRADE 8
M3X12	GRUB SCREWS		GRADE 8
M4X15	SHEAR PINS		NYLON
M9X50	UNC BOLT		

Table 4. Screw, Pin, Bolt List

1. Airframes

Airavata features a robust yet lightweight structure designed to endure both flight-induced loads and the demands of a black powder-based ejection recovery and a solid propellant motor. A key component of this structure is its two cylindrical body tubes, which form the outer shell and structural spine of the vehicle. The rocket's body tubes are primarily tasked with withstanding axial compressive forces generated during motor thrust, as well as tensile loads during parachute deployment and landing impacts. Additionally, these components must maintain structural integrity under aerodynamic loading throughout the ascent and descent phases.

Configuration:

The rocket consists of two primary body tubes made from Glass Fiber Reinforced Polymer (GFRP) for its strength, affordability, and ease of manufacture. The 1.2-meter upper tube houses the recovery subsystems, including the nose cone and parachute deployment mechanisms, and features a reinforced, aerodynamically designed camera mount. The 1.1-meter lower tube holds the rocket motor and primary payload bay, maintaining consistent mechanical and thermal properties. Both tubes are joined by an in-house fabricated GFRP coupler, produced via wet layup and vacuum bagging, which also houses the avionics bay and ensures strong axial load transfer during high-thrust phases.
All three cylindrical sections have an outer diameter of 148 mm, an inner diameter of 142 mm, and a uniform 3 mm wall thickness.

Manufacturing:

Glass Fiber Reinforced Polymer (GFRP) was selected for its high strength-to-weight ratio, corrosion resistance, and RF transparency, essential for onboard signal transmission. The tubes were manufactured using a wet layup process followed by vacuum bagging to enhance fiber alignment and minimize voids. Post-processing ensured dimensional accuracy and aerodynamic stability, with a uniform wall thickness maintained for predictable mechanical behavior.
Alternative materials such as Carbon Fiber Reinforced Polymer (CFRP) and aluminum were evaluated. CFRP was ruled out due to its higher cost, radio transparency, and galvanic corrosion risks, while aluminum was avoided due to lower fatigue resistance and added mass. GFRP offered a balanced solution, providing sufficient structural integrity, ease of fabrication, and compatibility with electronic payloads due to radio transparency.

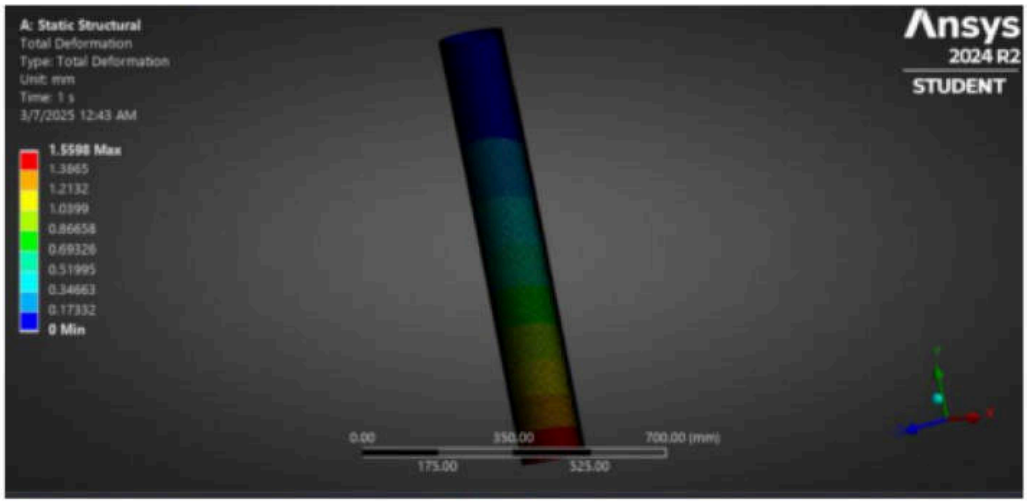


Image 14. Deformation under loading of 2.5 kN

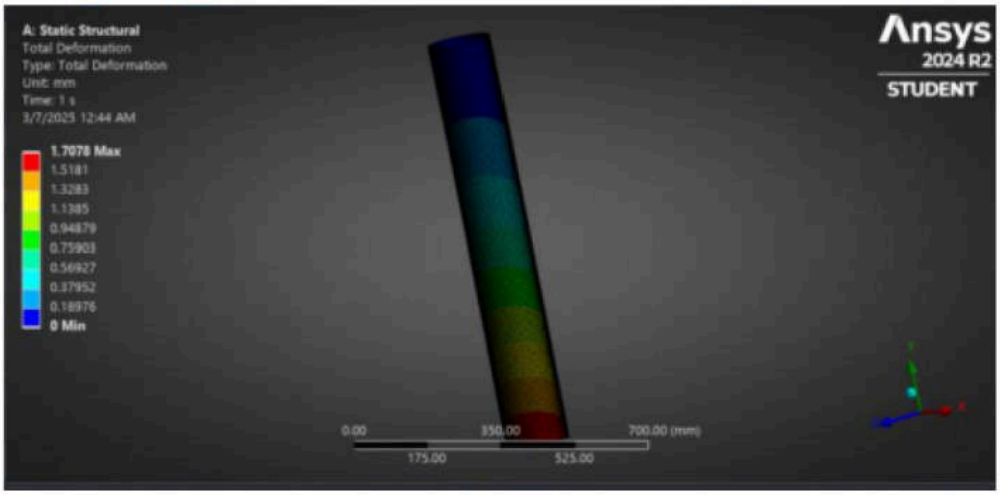


Image 15.Deformation under loading of 5 kN

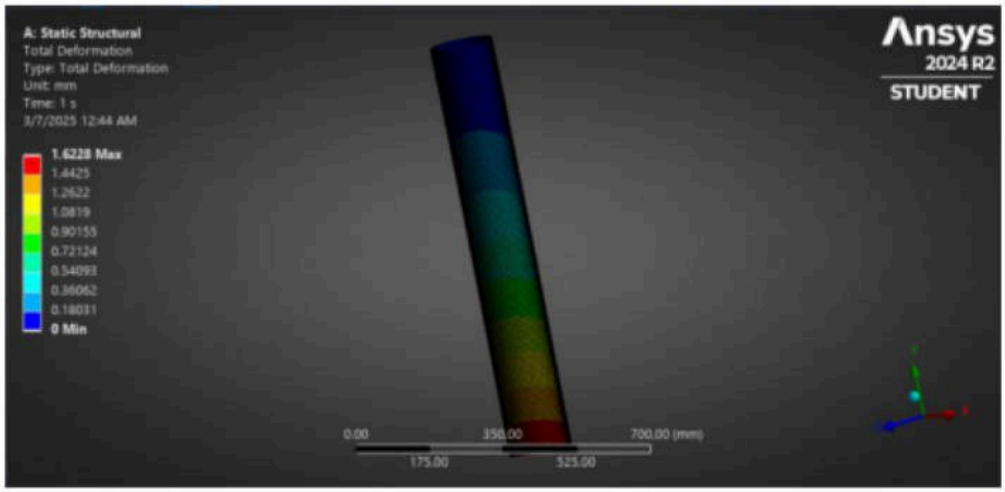


Image 16. Deformation under loading of 7.5 kN

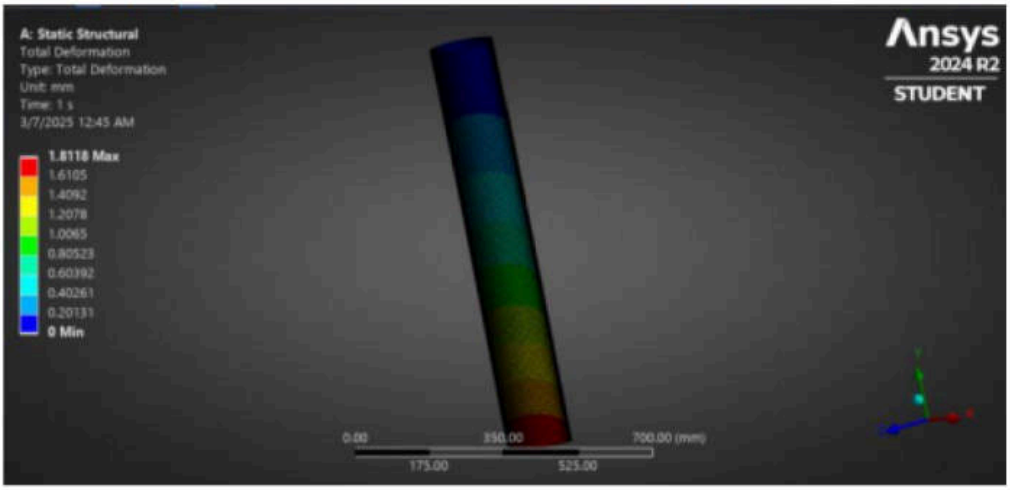


Image 17. Deformation under loading of 10 kN

The ANSYS analysis evaluates their structural integrity under axial loads up to 10 kn. The analysis uses a reduced yield strength for GFRP of 24.8 MPa (80% of the nominal 31 MPa) to ensure a conservative safety margin.

For applied forces of 2.5 kN, 5 kN, 7.5 kN, and 10 kN, the calculated factors of safety (FOS) for GFRP are 4.7, 3.37, 2.62, and 2.14, respectively.

This demonstrates that even at the maximum expected load, the body tube maintains an FOS above 2, indicating adequate structural safety for flight and recovery events. These results confirm that the GFRP tubes are structurally suitable for the intended application, with the analysis supporting their use in the rocket's design.



Image 18. Manufactured Body Tubes

Aerodynamics:

An ANSYS Fluent simulation was conducted to evaluate the aerodynamic behavior of the rocket's upper and lower body tubes with the camera mount and rail buttons attached, replicating the flight-ready configuration. A density-based solver was employed to model compressible high-speed flow, using steady-state conditions and the SST k-omega turbulence model for accurate wall-bounded and free-stream flow representation. The energy equation was enabled, and air properties were modeled with the ideal gas law and Sutherland's viscosity model. Simulation conditions were based on the point of maximum aerodynamic pressure (MAXQ), with atmospheric values obtained from OpenRocket simulations. Boundary conditions included a pressure far-field inlet, a pressure outlet, and a zero operating pressure to match atmospheric conditions. A second-order upwind scheme was used, and the Courant number was reduced to 1.5 for improved stability and convergence.

Simulation results, including contour plots for pressure, turbulence, density, and velocity, indicated localized pressure and turbulence near the rail buttons and camera mount, but overall stable flow. The mass flow rate approached zero,

confirming convergence. The computed drag coefficient (Cd ≈ 0.2) was consistent with earlier component-level analyses, demonstrating minimal aerodynamic impact from external features.

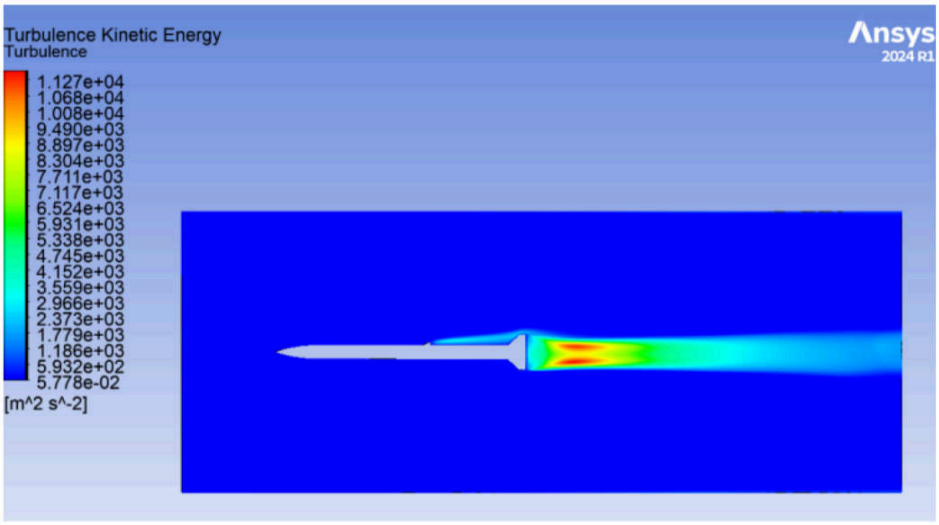


Image 18. Turbulence Simulation

2. Nose Cone

The nose cone is a vital aerodynamic component designed to minimize drag and enhance the flight performance of *Airavata*, especially during its ascent through the transonic regime. At these speeds (Mach 0.8–1.2), the rocket begins to encounter wave drag caused by the formation of shockwaves around the nose cone. These shockwaves result in a significant pressure increase at the front and a pressure drop at the rear of the nose cone, creating aerodynamic resistance. To reduce this effect, the nose cone was carefully streamlined to minimize wave drag and delay shockwave formation. Additionally, smooth transitions between the nose cone and body tubes were incorporated to reduce pressure drag and skin friction drag, both of which are influenced by surface area and flow separation.

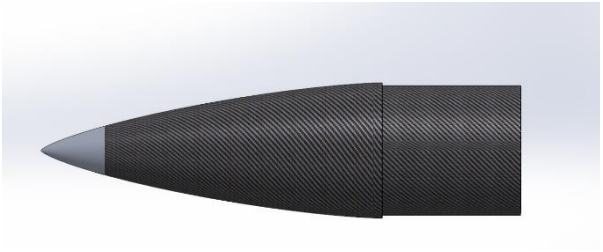


Image 19: Nose Cone

Integration:

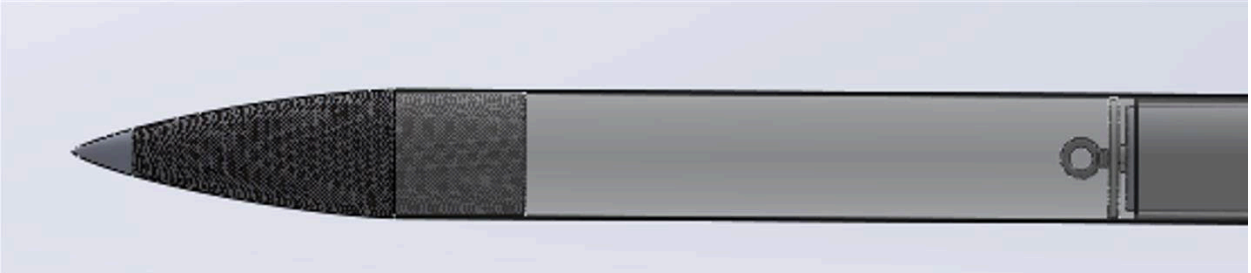


Image 20: Nose Cone Integrated with assembly of Airavata

During assembly, the nose cone's shoulder, a cylindrical extension at its base, is inserted into the upper body tube to achieve a flush and aerodynamically continuous interface. Precision alignment ensures minimal discontinuity at the junction, preserving aerodynamic efficiency. Pre-drilled holes in the body tube and shoulder are aligned, and Nylon shear pins, are inserted to secure the nose cone mechanically. A friction fit, augmented by a controlled interface between the shoulder and the inner surface of the body tube, enhances stability, mitigating wobble or misalignment during handling and launch preparations.

Throughout the rocket's ascent, the nose cone remains securely attached via the combined action of shear pins and the friction fit. These mechanisms withstand aerodynamic drag, turbulence, and vibrational loads induced by high-speed flight. The shoulder's precise fit within the body tube maintains concentric alignment, ensuring structural integrity and preventing aerodynamic instability that could compromise the rocket's trajectory or performance.

At apogee, a piston ejection system, activated by a pyrotechnic charge, i.e, black powder, the internal pressure generated by the pyrotechnic charge causes the piston to eject the nose cone outward. This pressure exerts an outward force on the nose cone, exceeding the shear strength of the pins and overcoming the frictional resistance of the shoulder's interface. The shear pins fracture, and the nose cone detaches cleanly from the body tube. This controlled separation enables the sequential deployment of the drogue and main parachutes for recovery.

Nosecone Profile:

LD-Haack Series with a shape parameter of 0.0 and a fineness ratio of 2.5:1 to ensure drag reduction and smooth functioning of the recovery ejection mechanism, is used for the nose cone profile of Airavata. It is proven that this parameter develops the least drag and achieves maximum apogee.

First, different nose cone profiles were analyzed to determine the best possible design that produces the least drag and maximum altitude. Airavata's flight regime lies between Transonic regime having a Mach number of 0.856 at maxQ. Three different nose cone profiles: LD-Haack, Parabolic, Conical, and Ellipsoidal - were evaluated using OpenRocket software. By changing the nose cone profile, we observed how these variations affected the rocket's apogee and drag. These findings from OpenRocket served as a foundation for a more in-depth study where Computational Fluid Dynamics (CFD) simulations were conducted.

Nose Cone Profile	Altitude (ft)	Max Velocity (m/s)	AoA (degrees)	Drag Coefficient	Ease Of Fabrication
LD Haack (2.5:1)	9640	288	Small Blunt	0.202	Normal
Conical	9257	283	Sharp Point	0.310	Easiest
Parabolic	9443	282	Rounded	0.213	Normal
Ellipsoidal	9103	279	Rounded	0.312	Normal

Table 6. Nose Cone Profile Comparison

A density-based setup was used for the boundary value problem of an axisymmetric model of nose cones. The gauge temperature and pressure at maxQ were calculated with the isentropic flow equation given by:

$$\frac{P}{Pt} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\frac{-\gamma}{\gamma - 1}}$$

$$\frac{T}{Tt} = \left(\frac{P}{Pt}\right)^{\gamma/(\gamma-1)}$$

Where, Pt is the total pressure, P is the static pressure, Tt is the total temperature, T is the static temperature, γ is the ratio of specific heats and M is the Mach number at maxQ. Assuming γ = 1.4, all the necessary calculations were done for the nose cone boundary conditions.

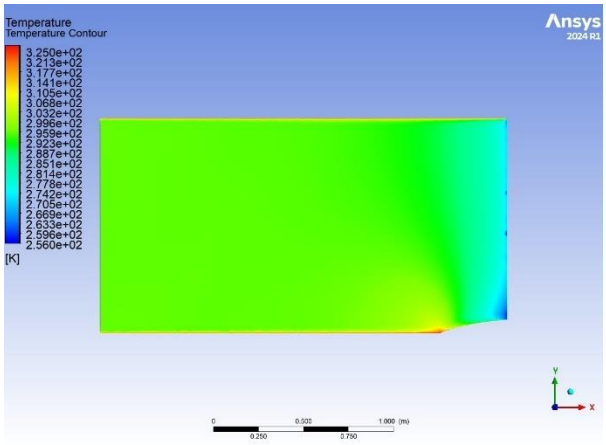


Image 21: Temperature Contours

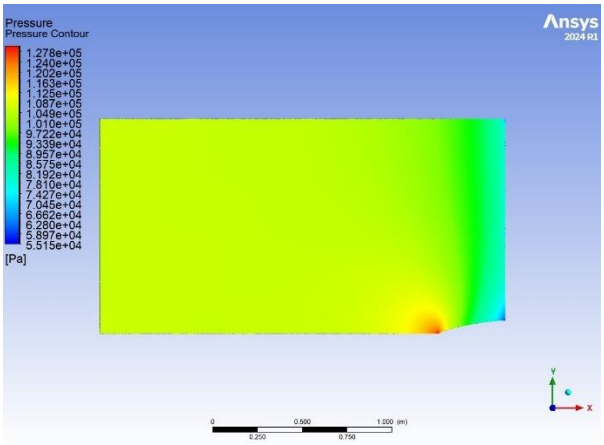


Image 22: Pressure Contours

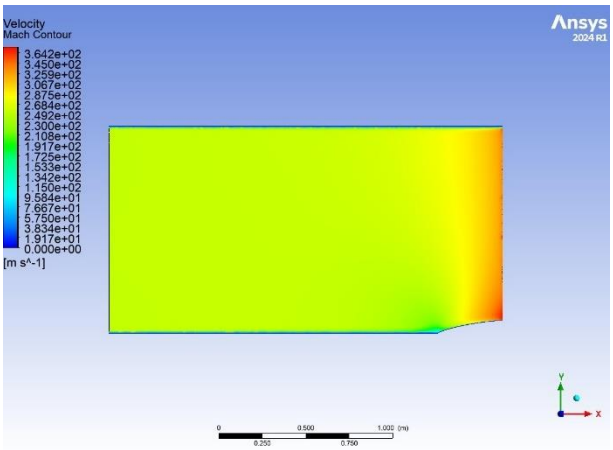


Image 23: Velocity Mach Contours

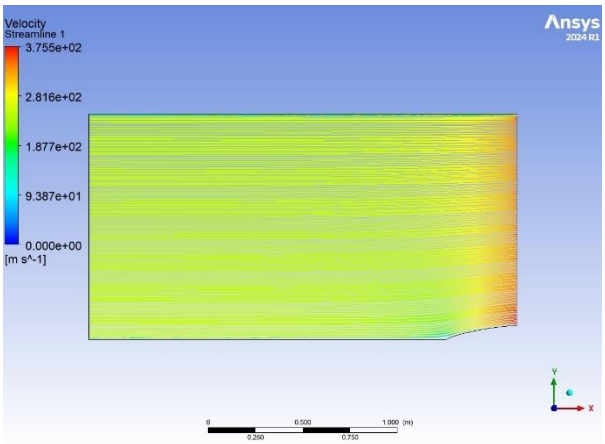


Image 24: Velocity Streamline Contours

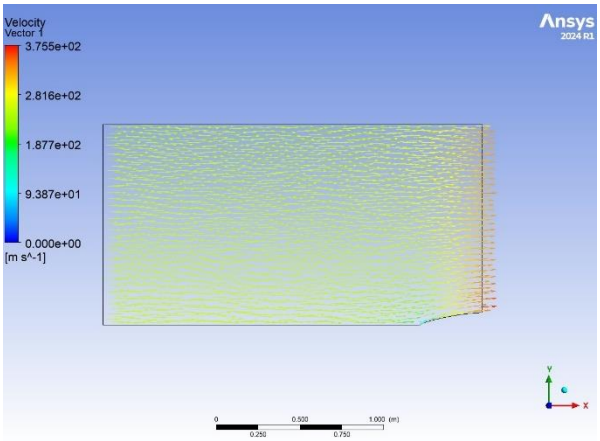


Image 25: Velocity Vector


Image 26: Manufactured Nose Cone

Material Selection:

Material	Tensile Strength (MPa)	Compressive Strength (MPa)	Density g/cm^3	Stiffness (GPa)	Thermal resistance	Impact Resistance
Carbon fibre	3,000-7,000	1,000-3,000	1.6-2.0	200-600	~450 Degrees	Moderate
Fibreglass	1,000-3,000	500-1,500	2.5-2.6	70-90	~220 Degrees	Good
Aluminium	200-600	200-500	2.7	70	~600 Degrees	Poor
Kevlar	2,500-4,000	200-500	1.4	100-150	~200 Degrees	Excellent
Epoxy Composites	Varies (1,000-7,000)	Varies (500-3,000)	1.4-2.6	70-600	~150-200 Degrees	Varies

Table 7. Nose Cone Material Comparison

The nose cone of a vehicle or projectile requires materials with specific physical properties to withstand aerodynamic forces, heat, and structural demands. This comparison evaluates carbon fiber, fiberglass, aluminum, Kevlar, and epoxy-based composites based on their physical properties, including strength, compressibility, material composition, and other relevant properties. Ultimately, for the following reasons Carbon Fiber was chosen for Nose Cone manufacturing.

1. Superior Strength-to-Weight Ratio:
Carbon fiber exhibits a high tensile strength ranging from 3,000 to 7,000 MPa, coupled with a low density of 1.6 to 2.0 g/cm³. This exceptional strength-to-weight ratio renders it an optimal material for nose cone applications where minimizing structural weight is paramount, thereby enhancing overall rocket performance and payload capacity.

2. High Stiffness
With a modulus of elasticity spanning 200 to 600 GPa, carbon fiber provides the necessary rigidity to preserve the aerodynamic profile of the nose cone during exposure to high-speed airflow and significant structural loads. This stiffness ensures stability and integrity throughout the flight envelope.

3. Adequate Compressive Strength
Although carbon fiber's compressive strength (1,000 to 3,000 MPa) is lower than its tensile strength, it remains sufficient to withstand the compressive forces encountered in most nose cone applications, ensuring reliable performance under operational conditions.

4. Thermal and Environmental Resistance
Nose cones are subjected to elevated temperatures due to aerodynamic heating during ascent. Carbon fiber demonstrates robust thermal resistance, withstanding temperatures up to approximately 200°C when paired with epoxy resin matrices, or higher with advanced matrix systems. Additionally, its low thermal expansion coefficient ensures dimensional stability, mitigating deformation under thermal loads and maintaining structural integrity in challenging environmental conditions.

3. Fins

The fins are essential aerodynamic components designed to ensure the passive stability of the rocket throughout its flight. By generating aerodynamic forces that counteract deviations from the intended flight path, they help maintain alignment with the planned trajectory, particularly during coasting and descent phases. Their placement and orientation are optimized to provide consistent stabilizing forces without requiring active control mechanisms.

While the fins are designed to operate primarily within the subsonic regime, the rocket may briefly pass through transonic conditions during ascent. In such cases, the vehicle may experience minor effects of transonic buffeting, a phenomenon caused by unstable shock wave formation and interaction on the rocket's surface. Although these effects are limited due to the subsonic design of the fins, the structure must still be capable of withstanding potential vibrational forces and pressure fluctuations.

To meet these demands, the fins are constructed to be both lightweight and structurally robust. The geometry and material selection are carefully chosen to minimize aerodynamic drag while maintaining rigidity and resistance to stress. This ensures that the fins contribute effectively to the rocket's aerodynamic performance and structural reliability throughout all phases of flight.

The fin design process followed the following steps:

1. Stability Requirement: The primary requirement of positioning the center of pressure (Cp) below the center of gravity (Cg) throughout the flight profile was kept in mind while designing the fins.

2. Geometry Selection: Computational Fluid Dynamics (CFD) analysis of various available fin shapes was performed to determine the fin geometry with the least drag and optimum lift in the flight regime. Hence, trapezoidal fins were selected following the analysis results.

3. Number of Fins: A 4-fin design provides symmetrical stability keeping the static stability margin within the required limits of 1 to 2 calibers.

4. Thickness: The limits of vacuum bagging process of composite materials and fin flutter were considered while choosing the optimum thickness for the fin. Fin flutter velocity was calculated for various magnitudes of thickness to match the criteria of it being 1.5 times greater than the peak velocity of the rocket and the magnitude which passed the criteria was selected.

5. Aeroelastic Analysis: A comprehensive Two-Way Fluid Structure Interaction (FSI) was performed for deep analysis of in-situ performance of the fins.

The dimensions of the fins are given below

Type	Trapezoidal
Material	Carbon Fiber
Number of Fins	4
Root Chord	7.67717'' (195mm)
Tip Chord	2.75591'' (70mm)
Semi-Span	4.29134'' (109mm)
Sweep Angle	44.5
Sweep Length	4.80315" (122mm)
Thickness	0.15748" (4mm)
Fin Cross Section	Rounded

Table 8. Fin Dimensions

Fin Material Selection:

A few materials were considered as potential options to manufacture the fin. The properties of the materials are listed below in the table.

Material	Density [g/cm^3]	Manufacturability	Yield Strength [MPA]	Cost
Carbon Fiber	2	Hard	600	High
Fiberglass	2.55	Easy	250	Low

Table 9. Fin Materials

Considering the material properties of carbon fiber (high strength to weight ratio) and its supreme performance in the transonic-supersonic regime it was selected over fiberglass by accepting a few tradeoffs.

CFD Analysis:

A detailed CFD analysis was performed for various fin geometries mainly to determine the drag coefficient, distance of the turbulence region from the rocket body and various other parameters to ensure a highly efficient aerodynamics performance of the rocket.

Additionally, attention was paid to the stability margin not exceeding 2 calibers during liftoff and it was maintained between 0.9 and 2.7 throughout the flight profile in order to avoid under stability or extensive weathercocking.

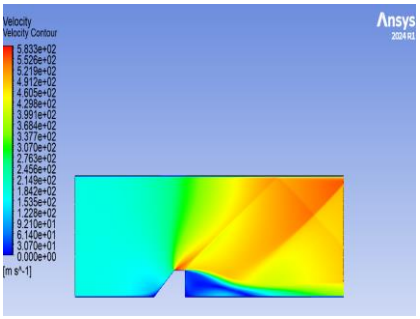


Image 28: Clipped Delta Fin

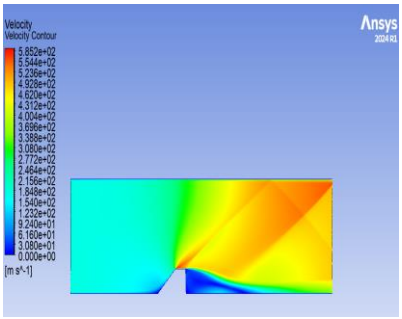


Image 29: Trapezoidal Fin

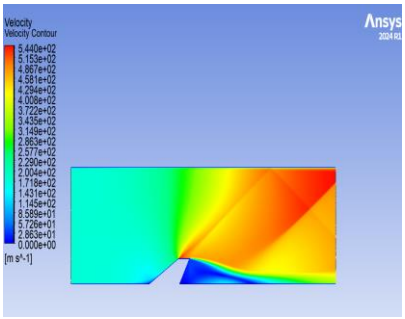


Image 30: Swept Back Fin

As a result of CFD analysis, the drag coefficients of trapezoidal, clipped delta and swept back were found out to be 0.074, 0.075, 0.701.

Analysis of the selected fin design

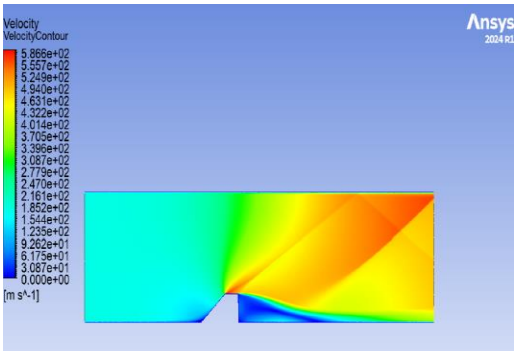


Image 31. Velocity Contour

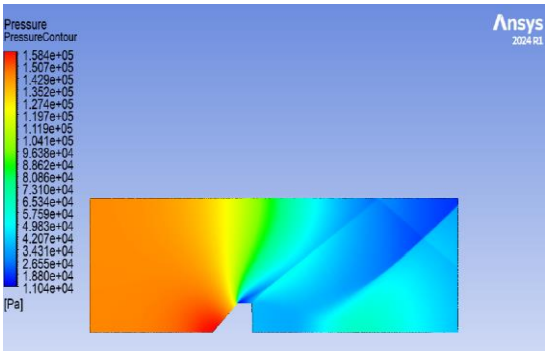


Image 32. Pressure Contour

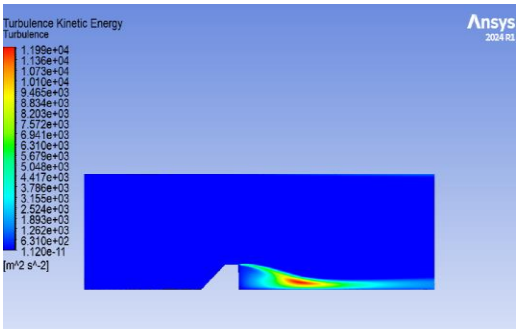


Image 33. Turbulence Kintetic Energy

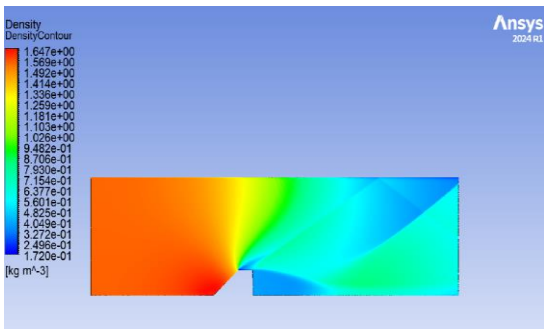


Image 34. Density Contour

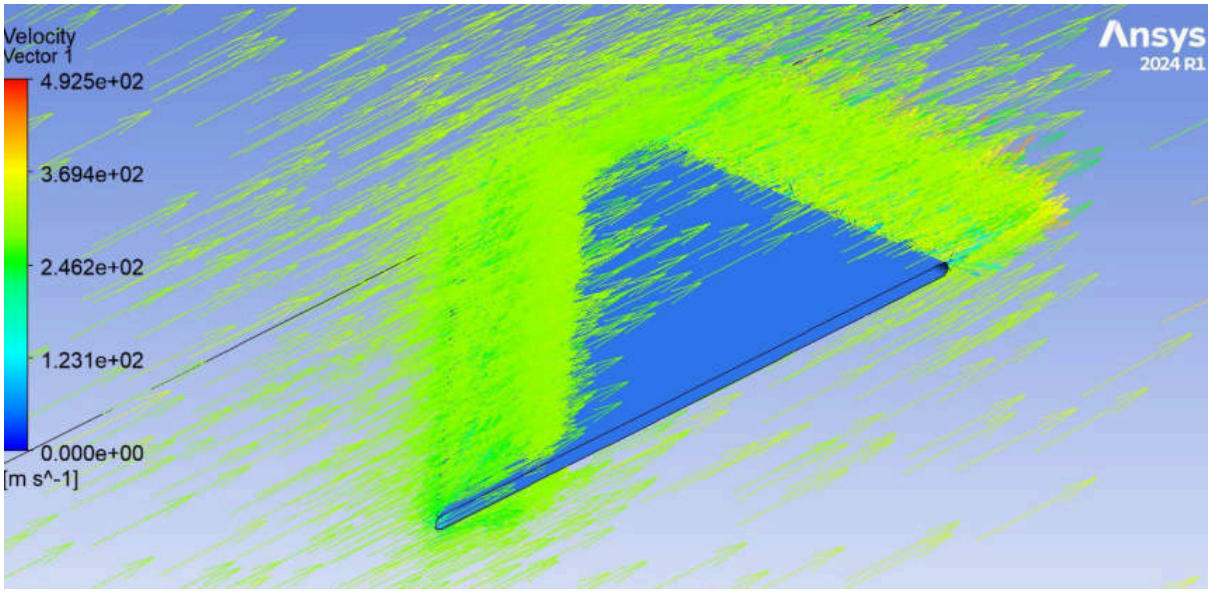


Image 35. Flow Boundary Visualizations over fin

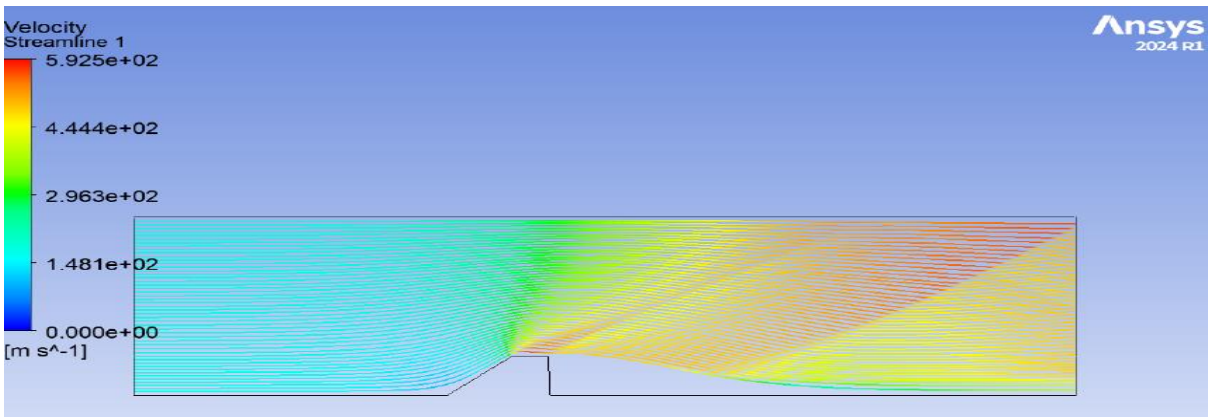


Image 36. Velocity Streamline

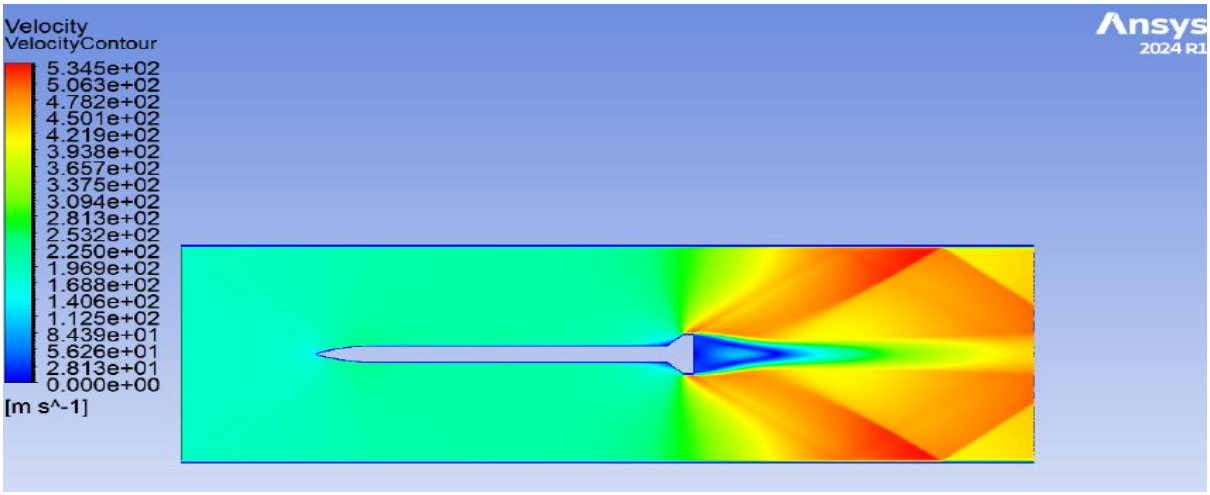


Image 37. Velocity Contour

Fin Flutter Analysis:

Fin Flutter is a dynamic aeroelastic instability where aerodynamic forces cause a fin to oscillate, potentially leading to structural damage or failure. For assured safety, the fin flutter velocity should be 1.5 times the peak velocity of the launch vehicle as mentioned in the competition guidelines.

$$V_f = a\sqrt{\frac{G}{\frac{1.337AR^3P(\lambda+1)}{2(AR+2)(\frac{t}{c})^3}}}$$

Using the equation, the fin flutter velocity was found out to be 545.37 m/s (1.59 Mach) thereby satisfying the safety criteria.

After determining the material and the dimensions of the fin, a comprehensive Two-Way Fluid Structure Interaction [FSI] is required to further analyze the fin's structural response throughout the flight. A Two-Way FSI analysis couples the results of structural and fluid analysis of the fin to get the dynamic response of the fin's structure to aerodynamic pressure forces caused by the airflow during the flight.

The workflow for the two-way FSI analysis is given below:

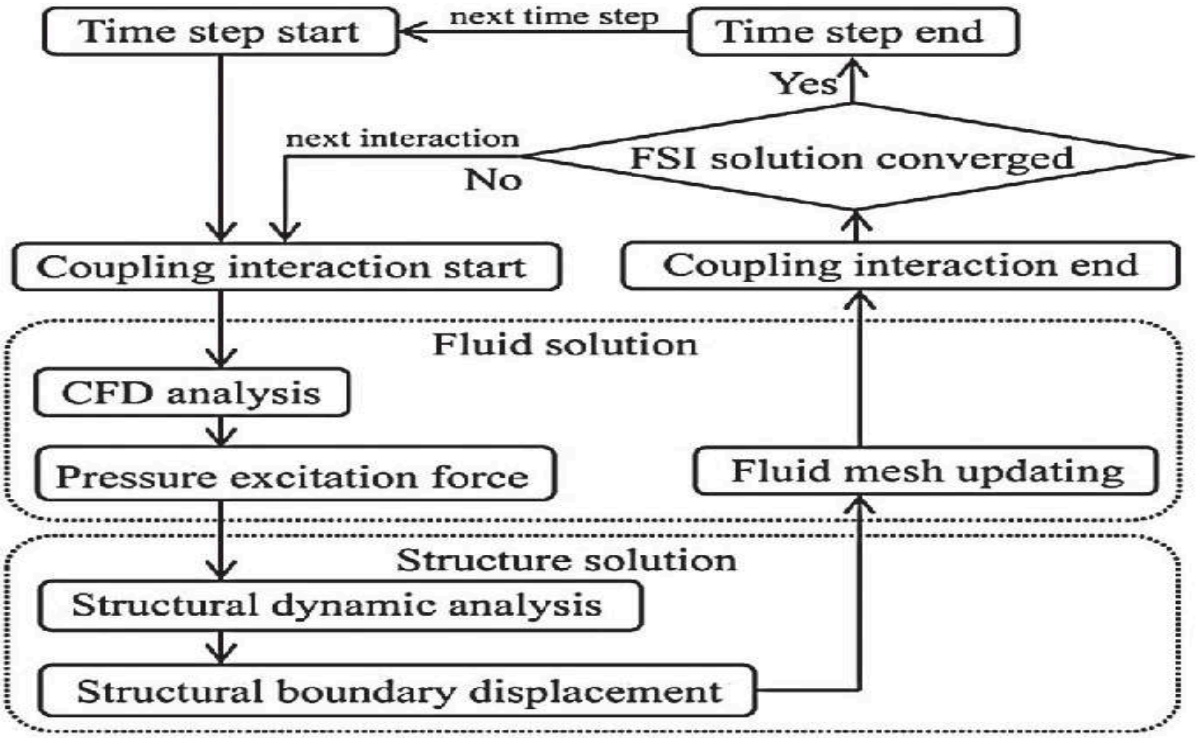


Image 38. 2 way FSI analysis workflow

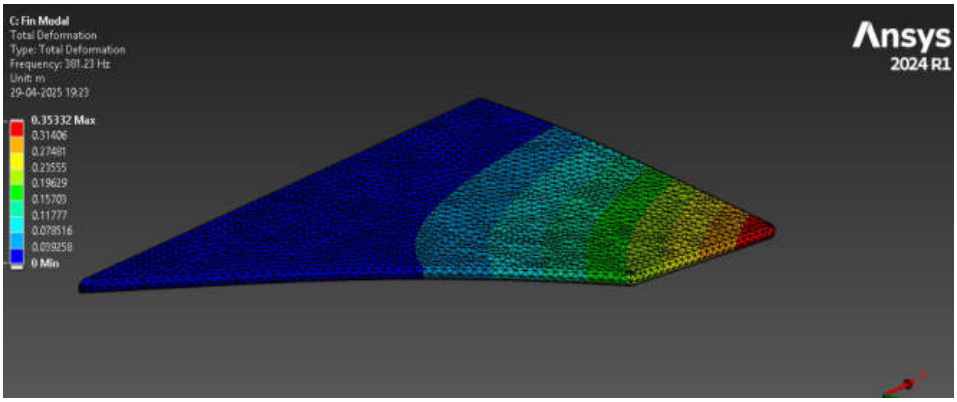


Image 39: Modal Response of the fin to the least natural frequency

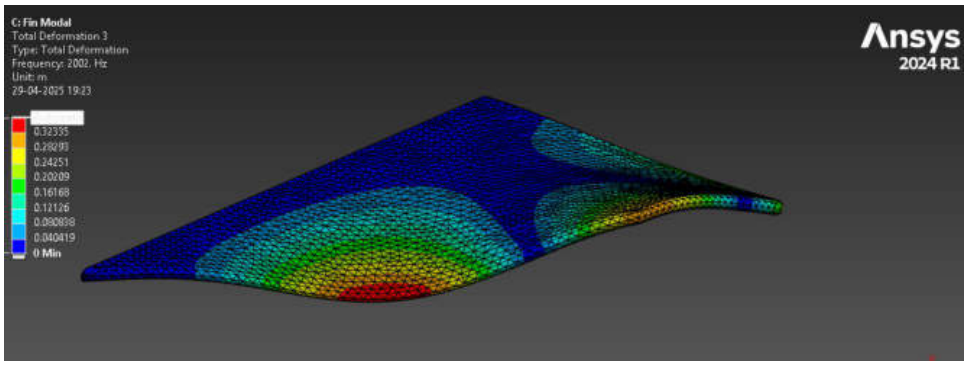


Image 40: Modal Response of the fin to the third natural frequency

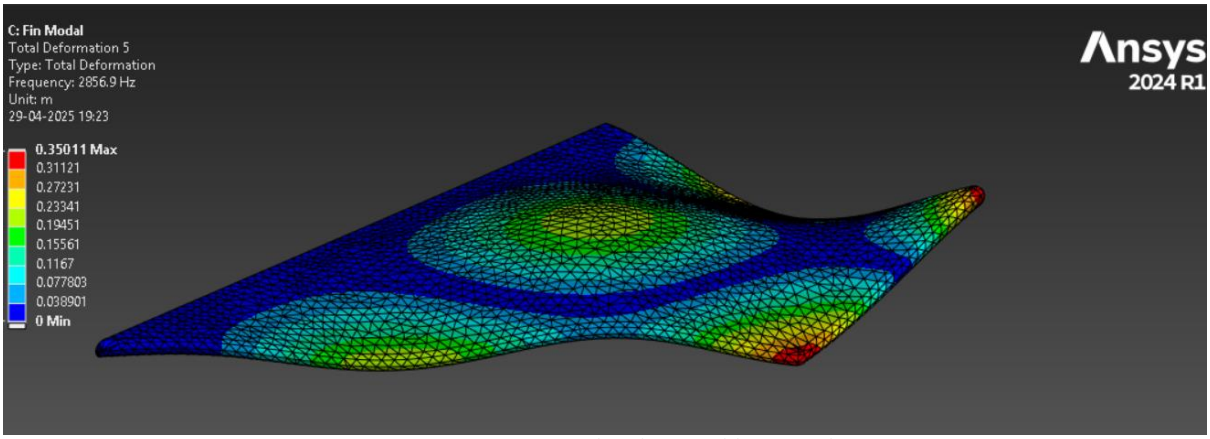


Image 41: Modal Response of the fin to the fifth natural frequency

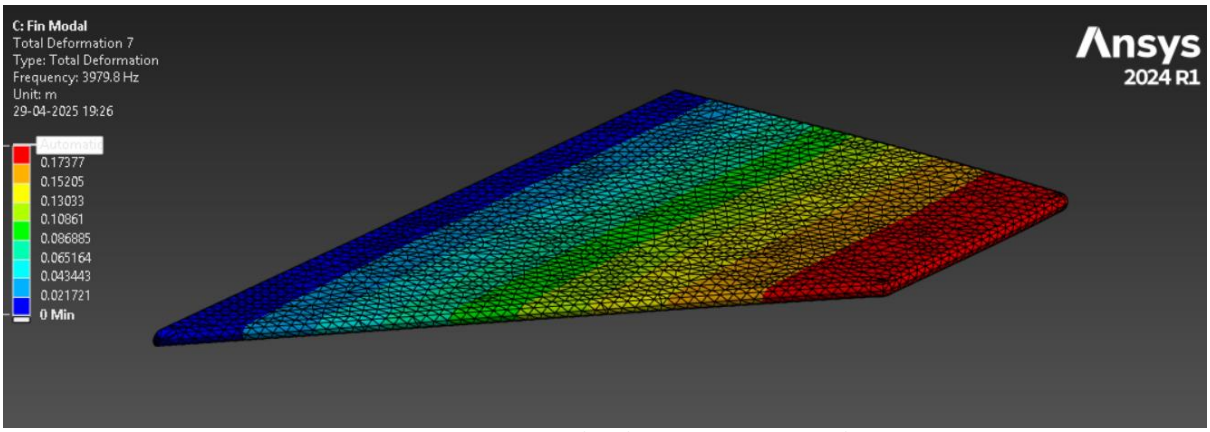


Image 42: Modal Response of the fin to the seventh natural frequency

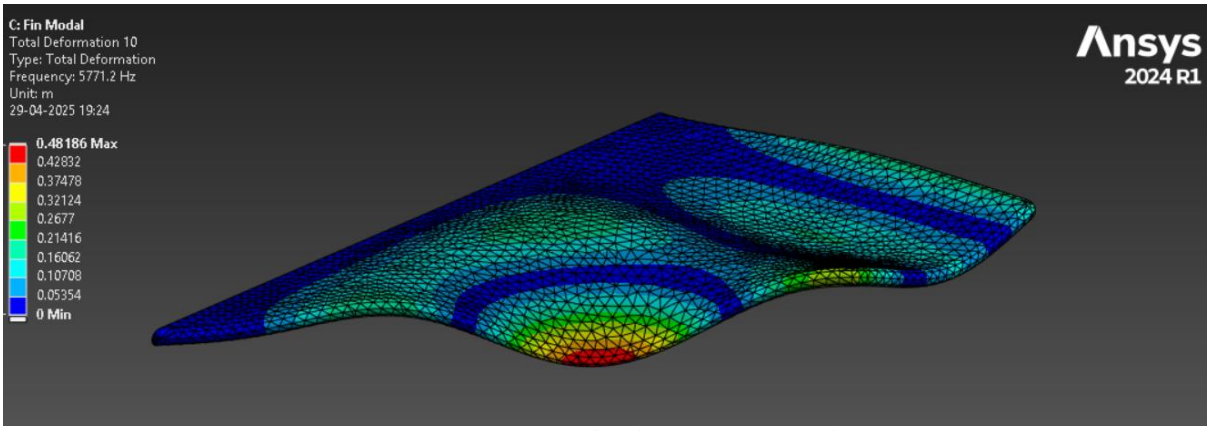


Image 43: Modal Response of the fin to the tenth natural frequency

Using the Two Way FSI and Modal solution, the rocket's fin flutter and vibrational response was thoroughly analyzed to determine how aerodynamic pressure affects the fin's structure. Both static and dynamics analysis were performed to reveal the natural frequencies of the fin's structure under applied pressure and the critical points of equivalent stress.

4. Fin Can

The fin can is a critical structural component located at the aft (rear) end of the rocket's body tube. It serves as the mounting interface for the rocket's fins, which are essential for providing aerodynamic stability during ascent. The fin can ensure that the fins remain securely attached and properly aligned throughout all phases of flight, despite the intense mechanical stresses encountered during launch and atmospheric flight. Its design directly influences the rocket's trajectory stability, control response, and overall flight performance

Design Objectives and Considerations:

The design of the fin can was influenced by the following key requirements:

1. Structural Integrity – The fin can must withstand high axial, lateral, and torsional loads without yielding or deforming, especially during rapid acceleration phases.
2. Aerodynamic Alignment – Accurate and repeatable fin alignment is critical to minimizing induced drag and preventing undesired yaw, pitch, or roll.
3. Weight Optimization – Excess weight at the aft end of the rocket adversely affects the center of gravity, making mass efficiency a top priority.
4. Ease of Assembly and Modularity – The fin can must allow for easy installation and removal of fins, for rapid assembly during launch operations.
5. Thermal and Corrosion Resistance – Materials must withstand moderate heat and resist corrosion due to outdoor exposure and plume heating during motor burn.

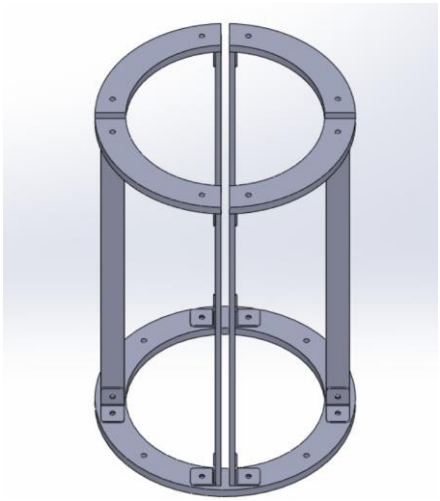


Image 44: Fin Can Design



Image 45: Fin Can Assembled

Material Selection:

After evaluating multiple materials, aluminum alloy 1060 was selected as the primary material for all major fin can components. Aluminum 1060 is a commercially pure alloy with excellent corrosion resistance, a high strength-to-weight ratio, and good workability. The corrosion-resistant properties of 1060 aluminum ensure long-term durability. Its low density contributes to the lightweight design of the fin can, which helps maintain an optimal center of gravity for the rocket.

Structural Design and Fabrication:

The fin can consist of several key components, including:

- Mounting baseplate
- Vertical support tabs
- L-brackets for attachment
- Attachment hardware (M4, M3 screws and locking nuts)

The structural layout was developed through an iterative process that included CAD modeling, finite element analysis (FEA), and physical prototyping. Areas subjected to concentrated stress—such as where the fins join the mounting plate—were reinforced with thicker aluminum sections. In contrast, regions that primarily served as enclosures or minor support were fabricated using thinner sheets to reduce weight. All parts were fabricated using laser cutting, which provided exceptional precision and edge finish quality. Laser cutting was particularly advantageous for achieving tight tolerances on mounting holes, slot features, and alignment guides. These tolerances ensured proper mating with fins and body tube components, simplifying assembly and reducing the likelihood of misalignment.

Attachment Mechanism:

Fins are affixed to the fin can using L-brackets and M3 screws, a method chosen for its robustness and modularity. This configuration allows the fins to be bolted securely to the fin can, while also enabling their removal or replacement if necessary. The use of M3 metric fasteners was a deliberate choice, balancing strength and ease of sourcing. To prevent loosening under vibration, fasteners are secured using lock nuts.

Integration with the Rocket Body:

The fin can is designed to interface seamlessly with the rocket's aft body tube. It slides into the airframe, and is secured using internal brackets. Once installed, the fin can maintain precise angular positioning of the fins relative to the rocket's longitudinal axis, which is essential for symmetric aerodynamic behavior. The design supports four-fin configurations, spaced evenly at 90 degrees apart. This arrangement offers aerodynamic stability in all axes, and the symmetrical spacing ensures balanced control moments. Keyed slots are used to guide the fin can into its correct orientation during assembly.

5. Bulkheads

Bulkheads are critical structural components used to separate internal sections within the rocket airframe. In Project *Airavata*, bulkheads play a vital role in housing and supporting avionics, payloads, and recovery systems while maintaining the structural integrity of the rocket during flight.

Material Selection and Manufacturing:

The bulkheads are manufactured from fiberglass using the wet layup process. Fiberglass was chosen due to its excellent mechanical strength, resistance to deformation, and lightweight properties—key considerations for aerospace applications. The wet layup technique enables precise control over layer thickness and fiber orientation, ensuring uniform strength across the bulkhead surface.
Each bulkhead in Airavata is fabricated to a thickness of 8 mm, providing a balance between structural robustness and mass efficiency.

Placement and Functionality:

A total of five bulkheads are used in the rocket. Their specific placements and functions are as follows:

- Bulkhead 1: Located within the nose cone, this bulkhead includes an eyebolt to which one end of the shock cord is attached.
- Bulkhead 2: Positioned at the end of the upper body tube, it also features an eyebolt for securing the opposite end of the shock cord.
- Bulkhead 3: Placed at the end of the first body tube, it serves as one of the anchoring points for the avionics bay.
- Bulkhead 4: Located at the top of the second body tube, this bulkhead supports the other end of the avionics bay.
- Bulkhead 5: Situated just above the thrust plate, it is responsible for securing the payload section.

Structural Integration:

Each bulkhead is integrated into the airframe using four M4 screws, which ensure a firm and reliable connection with the body tube. This screw-based attachment method allows for easy assembly and disassembly during pre-flight checks and maintenance procedures. To validate the design, a static structural analysis was performed on all bulkheads and their corresponding mounting screws. The simulation results confirmed that the bulkheads can withstand the expected loads during all phases of flight, with all stress and deformation values well within the acceptable safety limits.

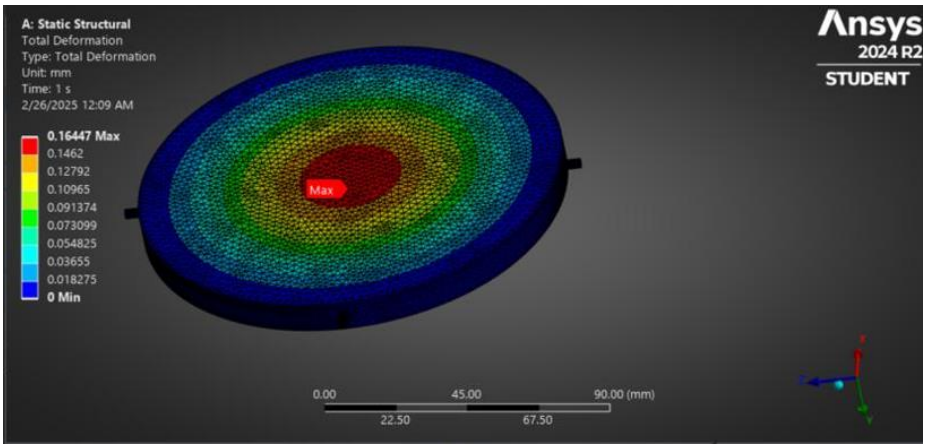


Image 46: Bulkhead Deformation

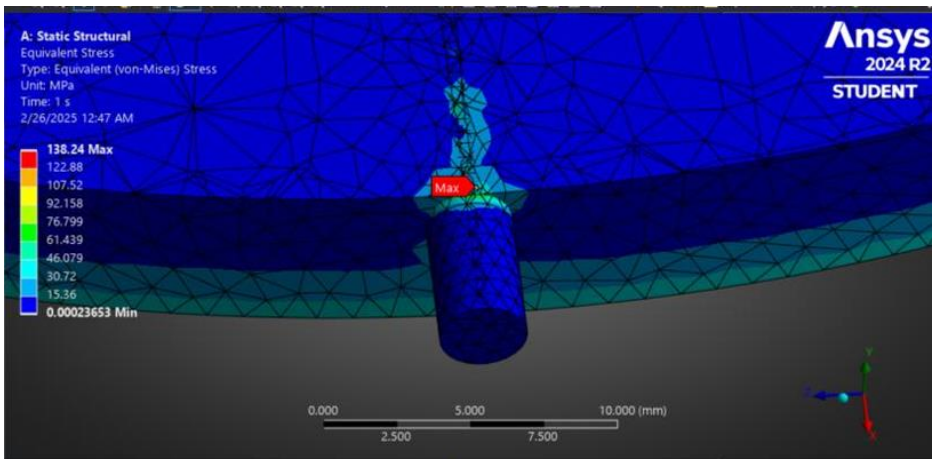


Image 47: Stress on screw

6. Coupler

The glass fiber composite coupler used in Airavata are manufactured in-house to ensure strict control over dimensional accuracy and material properties. The coupler is designed with an internal diameter of 136 mm and a uniform wall thickness of 3 mm. Initial fabrication attempts using PVC mandrels were discontinued due to inconsistencies in achieving the required geometry. To overcome this limitation, the team adopted 3D-printed ABS mandrels, which offer high precision and repeatability. These mandrels are wrapped in plastic and coated with wax to produce a smooth internal surface on the finished coupler.



Image 48: Coupler

The hand layup process is employed for coupler fabrication. Low-viscosity epoxy resins are selected for their ability to thoroughly impregnate the fiberglass reinforcement and for their extended working time, which is advantageous when producing large or complex parts. During fabrication, standard safety protocols are strictly followed, including the use of gloves and respiratory masks to protect against resin exposure. A peel-ply layer is applied during layup to absorb excess resin and prevent surface contamination, simplifying post-curing processing. Once cured, the couplers are removed from the mandrels and undergo sanding to achieve a smooth and uniform outer finish, ready for integration into the rocket structure.

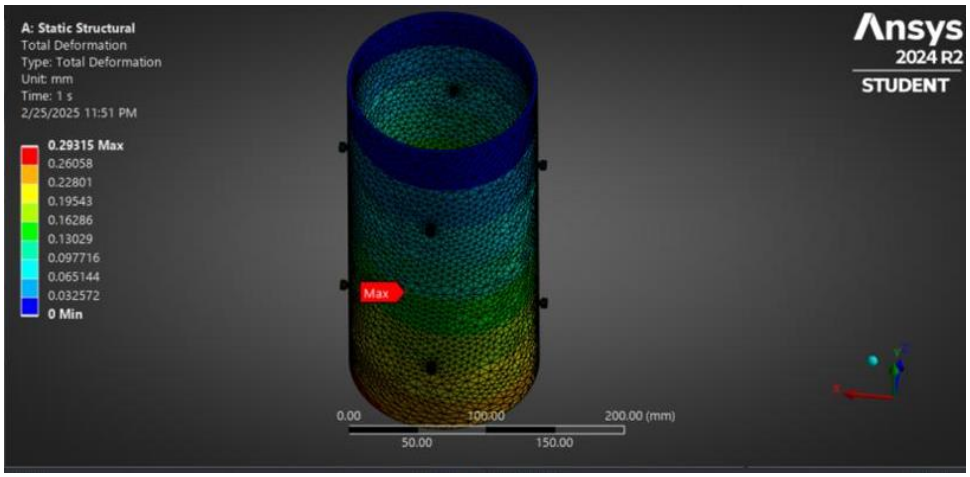


Image 48: Deformation on Coupler

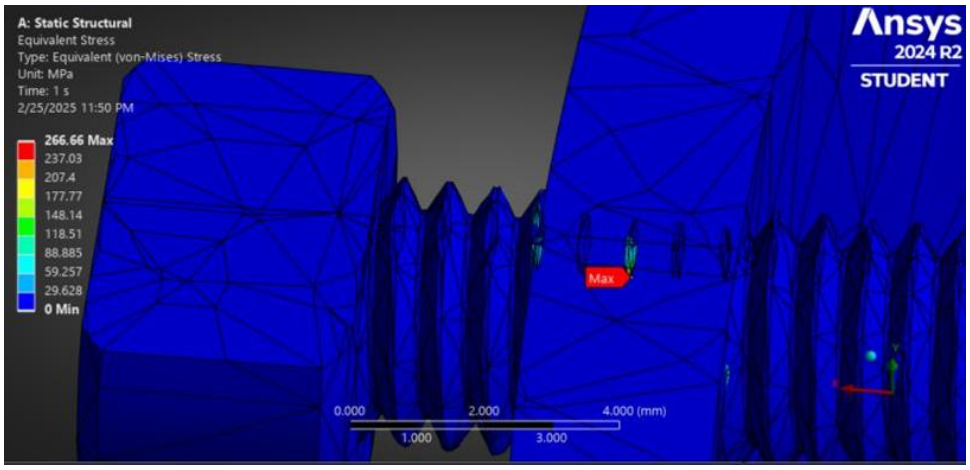


Image 49: Deformation on Screws

7. Centering Rings

Centering rings are used to keep the motor casing aligned within the body tube, ensuring everything is straight and balanced. This is essential as even a slight misalignment of the motor can result in thrust vector misalignment, which will cause a significant deviation from the intended flight trajectory. It is designed according to the dimensions of the motor casing, ensuring the motor does not wobble during flight which could reduce aerodynamic performance and stability.

Position and Manufacturing:

Airavata uses two centering rings of 7mm thickness which are 3-D printed using ABS due to its excellent strength-to-weight ratio and heat resistance, which is important as they will have to withstand the high temperatures produced by the motor. They are integrated into the body tube using epoxy. Structural analysis has been performed to ensure their structural integrity during motor burn phase, as the rings will experience significant shearing force from the

motor. It was found that the centering rings will be able to withstand the forces which are expected to act on them with an appropriate factor of safety.



Image 50: Manufactured Centering Rings

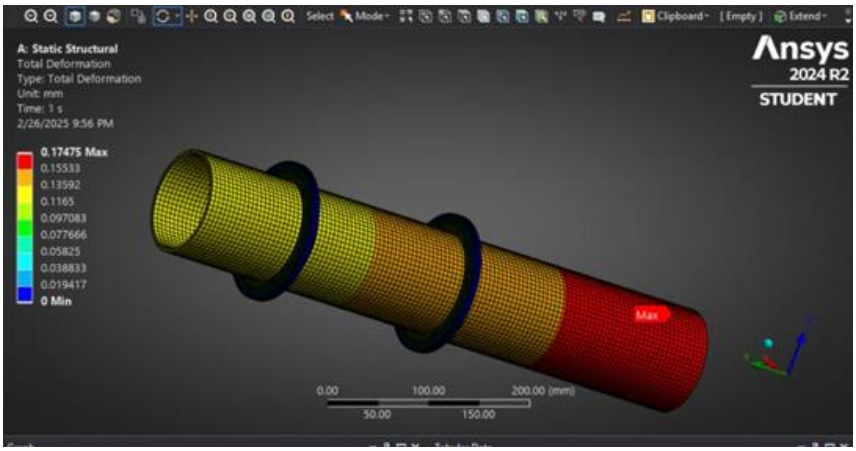


Image 50: Ansys Deformation Simulation on Integrated Centering Rings

8. Thrust Plate

The thrust plate is a structural element responsible for the transfer of motor's thrust to the rocket body. It serves as a physical blockage, which ensures that the motor does not move upward into the rocket at the time of ignition and flight.



Image 51: Thrust Plate



Image 50: Thrust Plate with Motor Casing

Position and Manufacturing:

The thrust plate is made from a solid aluminum billet because of the high strength-to-weight ratio of the material and its capacity to resist heavy mechanical and thermal stress. It is placed directly over the motor and attached to the airframe by eight M4 screws, providing a firm and rigid mounting. Furthermore, one M10 UNC bolt secures the motor casing to the thrust plate with a centralized load path for thrust transfer. The assembly configuration promotes structural integrity during high-thrust conditions and contributes to the overall stability and performance of the rocket at launch.

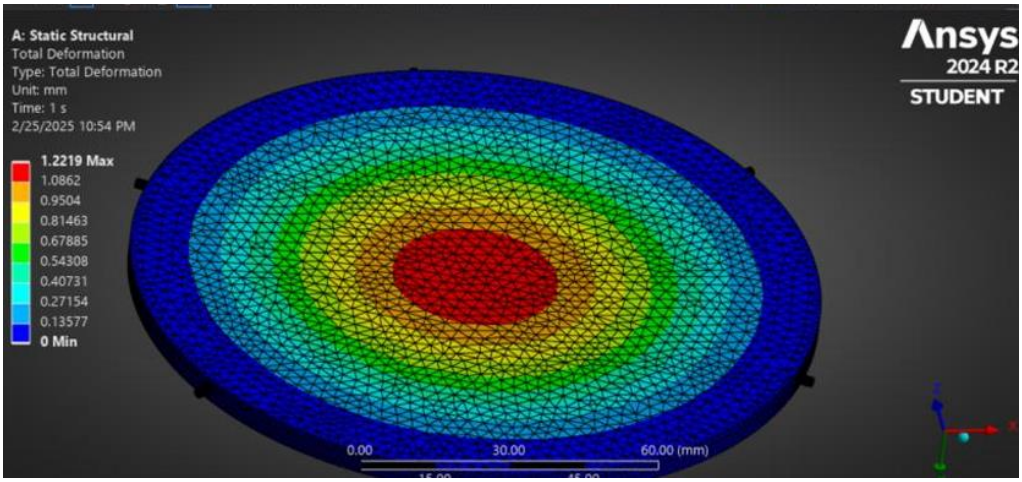


Image 53: Thrust Plate Deformation

9. Boat Tail

Project Airavata features an in-house manufactured boat tail; a critical aerodynamic component designed to minimize base drag and improve overall rocket stability. Fiberglass was selected as the base material for the boat tail due to its excellent mechanical strength, aerodynamic efficiency, and cost-effectiveness. Furthermore, fiberglass offers versatile layup options, making it an ideal choice for custom fabrication.

Material Selection:

Criteria	Fiberglass	Carbon Fiber	PLA (3D Printed)	Aluminium
Mechanical Strength	High	Very High	Moderate	High
Weight	Low	Very Low	Low	Moderate
Cost	Low	High	Low	High
Ease of Manufacturing	Easy layup and shaping	Requires high precision and complex layup	Easy to print but brittle	Requires CNC/machining
Machinability	Moderate	Difficult (needs special tools)	Poor (brittle under sanding/drilling)	Excellent
Surface Finish Quality	Smooth finish achievable with polishing	Very smooth with expertise	Rougher, may need post-processing	Very smooth
Durability under Stress	High	Very High	Moderate to low	High
Adhesion to Epoxy	Excellent	Excellent	Poor	Not applicable
Thermal Resistance	Good	Excellent	Poor	Excellent
Suitability for Vacuum Bagging	Excellent	Excellent	Not suitable	Not suitable
Availability and Handling	Easily available, safe to handle	Limited availability, requires expertise	Easily available	Readily available, heavy equipment needed
Repairability	Easy	Difficult	Difficult	Difficult

Table 10. Boat Tail Material Analysis

Mould Fabrication:

To achieve the desired boat tail geometry, a 3D-printed mould was created using the Creality Ender K3 VE printer. The mould was designed in SolidWorks with precise dimensional control to match the required curvature. Acrylonitrile Butadiene Styrene (ABS) was selected as the printing material over PLA due to its higher mechanical robustness and ease of post-processing. ABS allowed for sanding and drilling, which facilitated fine-tuning of the mould surface. Moreover, it showed good compatibility with commonly used release agents such as wax, PVA, and silicone-based sprays, making it an excellent choice for composite layups.

Prior to layup, chunks of paraffin wax were melted using a double boiling method and uniformly applied over the surface of the 3D-printed mould as a release agent. This ensured easy separation of the fiberglass structure after curing. The use of captain sheets was intentionally avoided due to their poor adhesion to the ABS surface, which had previously led to air bubble formation during test runs.

Layup Process:

The fiberglass layup process involved cutting sheets of fiberglass fabric and applying them continuously over the prepared mould surface. The epoxy resin used for bonding the fibres was mixed in the specified ratio and applied carefully to ensure uniform saturation. The aim was to achieve a final thickness of 3 mm for the fiberglass shell.

After the layup was completed, peel ply and breather cloth were applied on top to facilitate vacuum bagging. The assembly was then sealed and left under vacuum overnight. This process helped eliminate air bubbles trapped within the composite and ensured optimal resin distribution, leading to a structurally sound and aerodynamically clean finish.

Assembly:

The boat tail was structurally designed to accommodate two aluminum rings that are critical to its integration with the rocket body. The top aluminum ring was inserted into the upper cylindrical portion of the boat tail and secured using four M4 grub screws from the sides. This ring provided the interface for attaching the boat tail to the lower plate of the fin can. Similarly, the bottom aluminum ring was installed in the lower cylindrical section and fastened in the same manner. This ring plays a vital role in supporting the rocket motor during flight, ensuring structural stability under thrust loads.

Design Rationale:

The boat tail follows a conical design with an overall length of 80 mm and a divergence angle of 8.06°. The fore diameter of 148 mm matches the outer diameter of the body tube, while the aft diameter tapers down to 131 mm. To facilitate easier integration, cylindrical features were incorporated at both the top and bottom ends of the cone.

The conical geometry was chosen based on the recommendation of team mentor Mihika Pant. Compared to contoured boat tails, a conical profile is significantly easier to manufacture and analyze. The taper angle provides a predictable location for flow separation, which is vital for understanding aerodynamic behavior and managing base drag. The balance between manufacturing simplicity and aerodynamic predictability made the conical boat tail an ideal choice for Project Airavata.

10. Rail Buttons

The rocket incorporates two COTS 1515 Rail Buttons from Apogee Components. The acquired package consists of two rail buttons which are made up of Delrin-plastic, 10-32 x ¾” machine screws and weld nuts. The positioning strategy of the rail buttons considered the ease of a well-directed liftoff from the launch lug with the center of gravity of the rocket and the wind speed at the launch site being the decisive parameters. The rail buttons are secured to the rocket body using an M5 bolt.

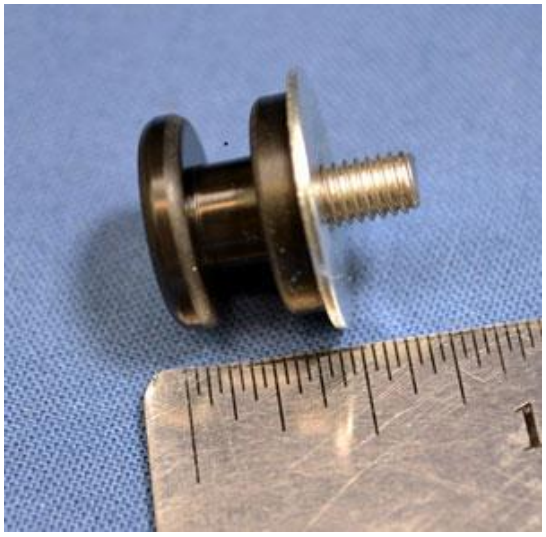


Image 54: COTS 1515 rail button

D. Recovery Subsystem

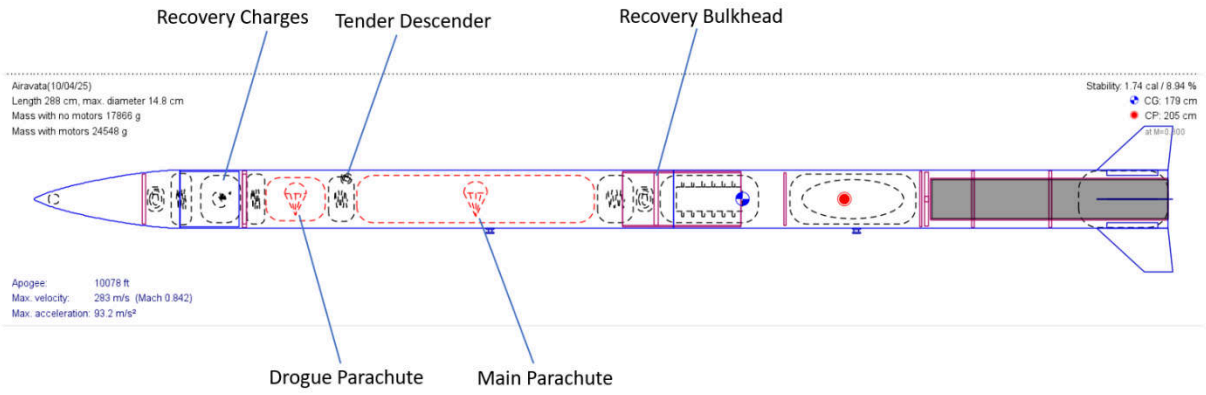


Image 55: Labelled Airavata Diagram

The rocket uses a dual-deployment system to ensure efficient recovery. In the primary setup, the drogue parachute is deployed at apogee, with the main parachute set to deploy at an altitude of 1,490 feet. The secondary system is configured to deploy the drogue parachute a few seconds after apogee and the main parachute a few seconds after passing 1,500 feet. Both deployments in each system are triggered using a black powder ejection mechanism. The primary recovery system utilizes the MissileWorks RRC3 altimeter, while the secondary system employs the FeatherWeight Raven 4 altimeter. The recovery system includes Main and Drogue Parachutes, Recovery charges, a Tender Descender, shock cords, quick links, swivels, eyebolts, a recovery lip for the charges to sit on and a Bulkhead. The drogue and main parachute bays are both located within the upper body tube of the rocket. This internal arrangement supports efficient packing, clean deployment, and minimizes the risk of entanglement during the recovery sequence.

1. Parachute and Shock Chords:

Both the drogue and main parachutes are hemisphere parachutes which have been manufactured in-house using ripstop nylon, a lightweight and tear-resistant material. Each parachute consists of 12 gores, providing stable inflation and descent characteristics. The drogue parachute has a diameter of 100 cm, while the main parachute has a diameter of 260 cm, offering appropriate drag forces and descent velocity at different stages of descent.

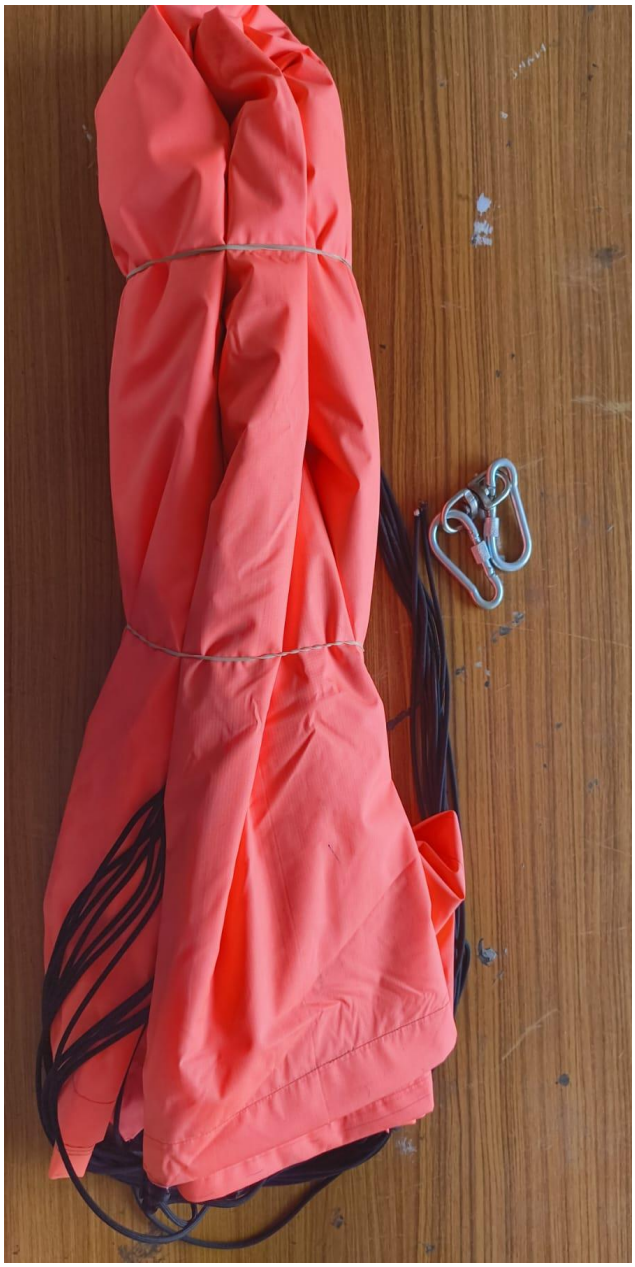


Image 56: Main Parachute

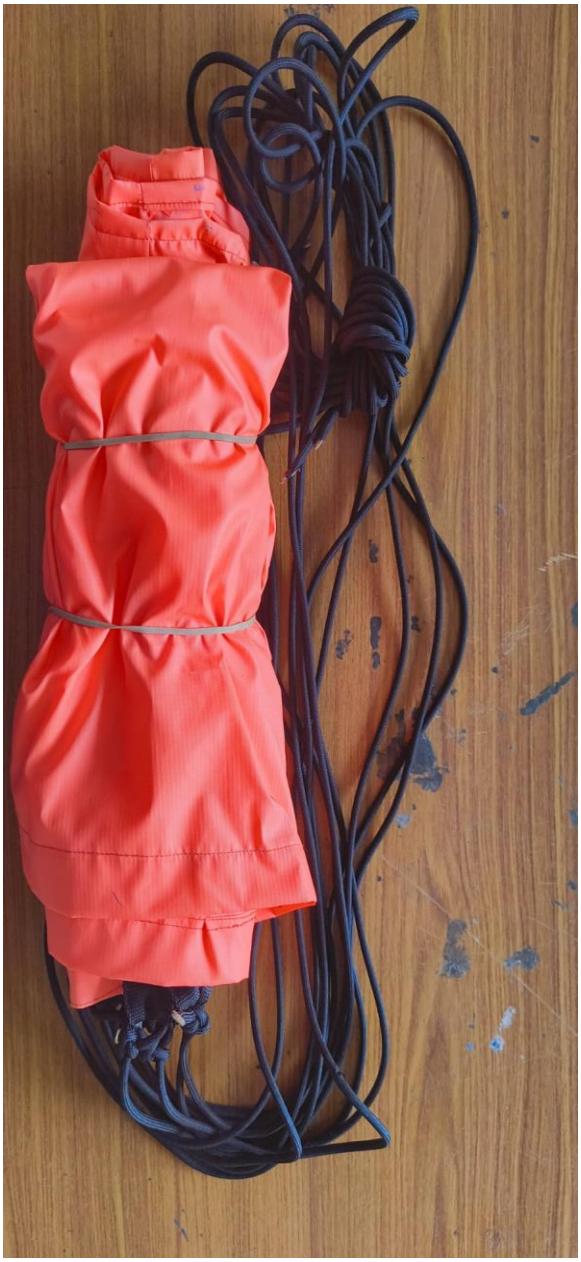


Image 57: Drogue Parachute

2. Calculations:

The parachute diameter is calculated using the formula:

$$A = \frac{2mg}{\rho V^2 Cd}$$

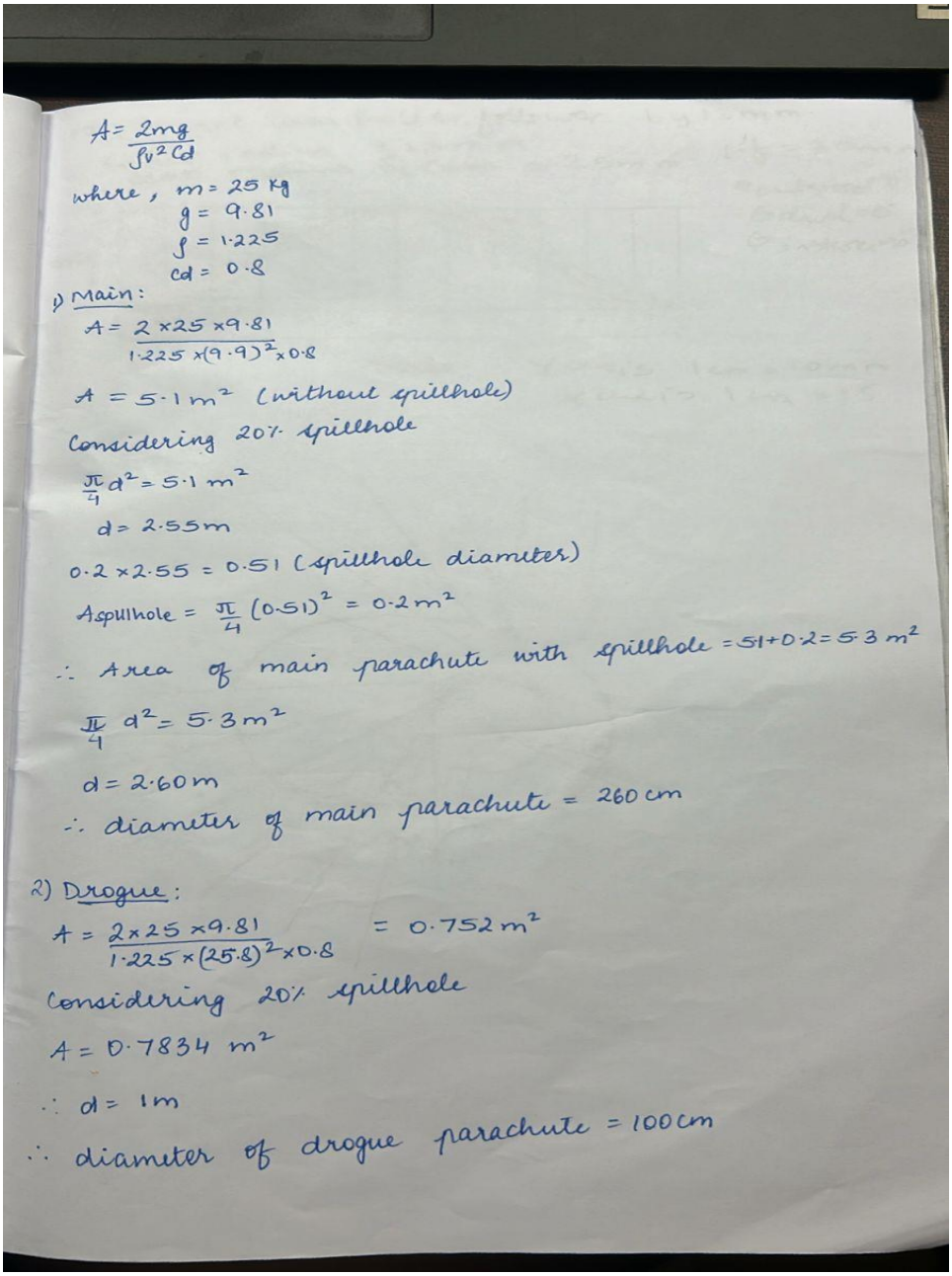


Image 58: Parachute Diameter Calculations

The recovery system of the rocket utilizes nylon flat webbing shock cords with a width of 38 mm, and a 10mm diameter rope selected for their strength and flexibility. These shock cords extend continuously from the bulkhead located within the nose cone to the bulkhead at the base of the upper body tube, ensuring a secure connection and effective load distribution during parachute deployment.
The rope is used for the tender descender connection to the recovery bulkhead as it has high tensile strength of 17.5kN. After drogue deployment a lot of tensile force would be acting on the tender descender connection line.

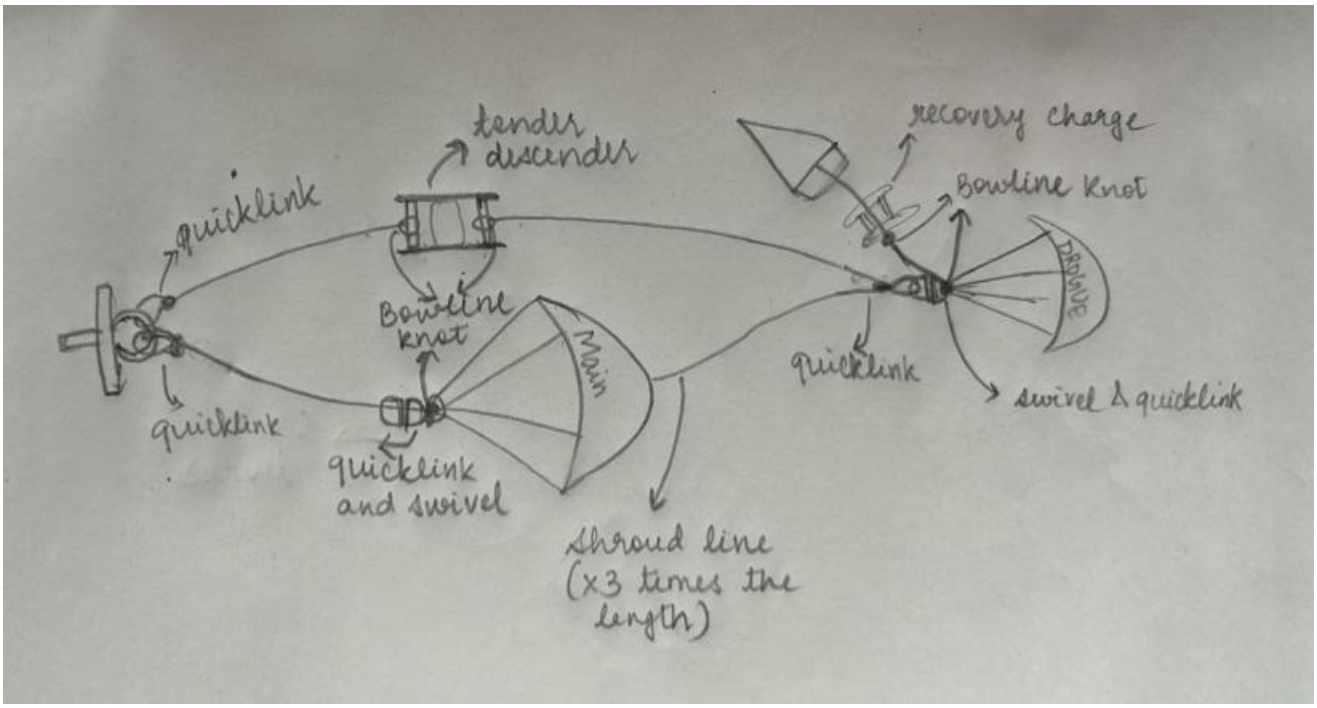


Image 59: Deployment Methodology

3. Recovery Avionics

Apogee Deployment:

The primary recovery system uses the Missileworks RRC3 altimeter to deploy the drogue parachute at apogee. If the RRC3 fails, the Raven altimeter will activate the deployment charges 2 seconds later. If both systems fail, the Sirius Flight Computer will attempt deployment 2 seconds after the Raven4, providing two layers of redundancy to ensure successful drogue chute deployment.

Main Deployment:

At 1,500 feet, the RRC3 triggers the deployment of the main parachute via the Tender Descender. If it fails, the Raven4 will attempt deployment after 2 seconds, followed by the Sirius Flight Computer if necessary. This sequence ensures a highly reliable main parachute deployment.
Power & Wiring:

- RRC3: Powered by a 9V logic battery and uses a separate 3.7V 18650 (2S1P) pyro battery to fire both deployment charges.
- Raven4: Powered by a 9V battery, which also fires its connected charges.
- Sirius: Powered by a 3.7V 18650 (2S1P) battery, which also fires the charges.

Each system is connected to two deployment charges: one in the Recovery Module and one in the Tender Descender

4. Recovery Charges:

The recovery mechanism, shown in the images below, is made of aluminium and contains two black powder charges: a 7g primary charge of FFFF black powder and a 5g redundant charge. The upper section of the charge fits into the shoulder of the nose cone, while the lower section rests against the recovery module's lip. When the black powder ignites, pressure rapidly builds up inside the charge, causing the upper section to separate. This separation pushes against the bulkhead of the nose cone, breaking the shear pins and resulting in nose cone ejection.





Image 60 & 61: Machined Recovery System

5. Tender Descender:

The Tender Descender is a compact, pyrotechnic-based separation device that plays a critical role in the recovery sequence. In this system, it is attached along the tether line between the drogue parachute and the airframe's eyebolt. At the designated altitude of 1,500 feet, the Tender Descender is triggered to release the main parachute. This activation is achieved using a black powder charge.



Image 62: Tender Descender

Black Powder Mass

Due to restrictions on purchasing black powder in India, standard FFFFg black powder could not be sourced for testing purposes. To address this, the team developed an in-house version of black powder using a fixed ratio of charcoal, sulphur, and potassium nitrate. Successful recovery tests were carried out using this homemade composition.
For calculations involving FFFFg powder, experimental validation of the forces involved was conducted. Standard black powder equations were used to determine the required mass of the charge. The team decided to use 7 grams for the primary and 5 grams for the redundant nose cone separation and drogue deployment and 4 grams for the main deployment. This accounts for potential inefficiencies in black powder combustion at high altitudes and includes a built-in safety margin.

For FFFFg calculations, forces were verified experimentally and black powder equations were used to calculate the mass required. A fuel characterization graph was obtained to compare our in-house manufactured with the standard FFFFg black powder.

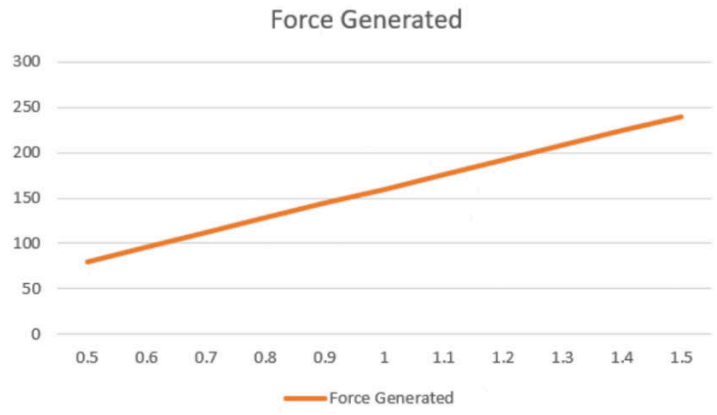


Image 63: Force Generated Graph

E. Payload Subsystem

1. Payload Overview

The proposed payload is designed in a 3U CubeSat form factor with a total mass under 2 kg. It is tailored for integration into the rocket *Airavata* and serves as the official entry for the Space Dynamics Laboratory Payload Challenge. This multi-subsystem payload demonstrates compact and modular experimentation in the fields of autonomous navigation, particle physics, and structural health monitoring.

The payload integrates three key subsystems: a vision-based navigation unit (ViziNav), a muon detection module, and a structural monitoring system using strain gauges. Together, these subsystems enable real-time navigation, cosmic particle analysis, and in-flight structural health assessment. The payload reflects a progressive approach to space instrumentation development, with a focus on cost-efficiency, compactness, and cross-disciplinary functionality.

2. Visual Aided Inertial Navigation System (ViziNav)

The ViziNav subsystem occupies the first part of the payload and is engineered as a vision-based navigation platform. At its core is an NVIDIA Jetson Nano, a powerful edge-computing module selected for its high-performance GPU capabilities and compatibility with deep learning frameworks.
In the absence of GPS and external navigation systems, particularly during suborbital rocket flights or extraterrestrial missions, onboard pose estimation becomes critically important. Conventional methods using only inertial sensors often suffer from drift over time, leading to inaccurate navigation and trajectory records. To address this, our team has developed ViziNav (Visual Aided Inertial Navigation)—a compact, self-contained payload module that integrates a monocular camera and an inertial measurement unit (IMU) to perform real-time 6-DoF pose estimation and provide accurate 3D trajectory reconstruction of a sounding rocket's suborbital flight.
ViziNav is designed to serve two primary objectives. First, it enables the rocket to estimate its position and orientation in real time, compensating for the limitations of inertial-only systems. Second, it provides a post-flight visualization of the rocket's path, capturing the trajectory in three-dimensional space with high temporal and spatial resolution. In contexts such as extraterrestrial landings, where GPS signals are either unavailable or unreliable, such a self-contained system has significant implications. It supports autonomous descent, navigation, and mapping on planetary surfaces like the Moon or Mars, where navigation must rely solely on onboard sensory data.

Specifications:

Our payload processes two main input streams. The first is a live video feed from the Arducam IMX219 Wide Angle Camera Module that operates at 30 frames per second. This provides rich visual data from the rocket's environment during ascent, coast, and descent phases. The second input is high-frequency (~100 Hz) inertial data from the LSM9DS1 9-DOF IMU, which includes linear acceleration and angular velocity.
The output of the system is primarily a 6-DoF pose graph consisting of the rocket's position in 3D space (x, y, z) and its orientation (roll, pitch, yaw). Over time, these pose estimates are used to construct a continuous 3D trajectory, which can be visualized using tools such as RViz. Additional internal outputs include real-time IMU bias correction and visual feature tracks, which support both system stability and offline analysis.
The core modules involved in this system include a visual-inertial frontend, a nonlinear optimization-based backend, and optional loop closure and pose graph optimization components. All these modules are derived from the VINS-Mono architecture, which we have adapted for high-speed suborbital flight conditions.

Algorithm:

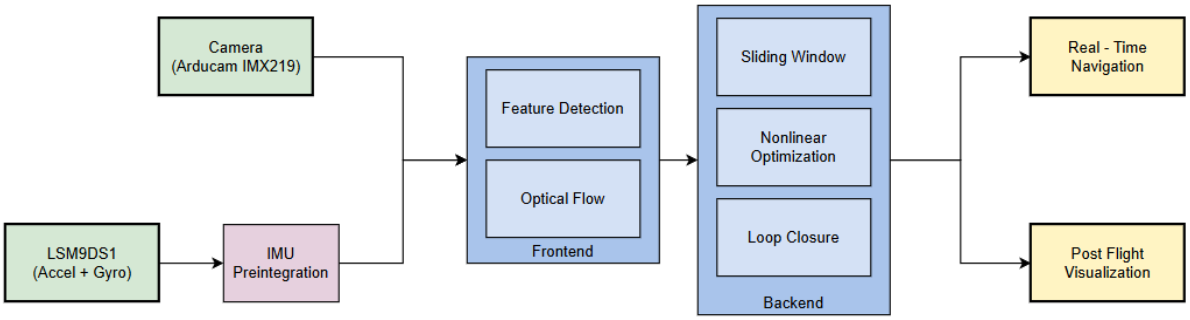


Image 64: ViziNav Algorithm

The pose estimation process in ViziNav begins with the visual-inertial frontend, which processes image and IMU streams concurrently. In each camera frame, corner features are detected using the FAST (Features from Accelerated Segment Test) algorithm and tracked across subsequent frames using KLT (Kanade-Lucas-Tomasi) optical flow. These feature correspondences allow us to infer how the rocket is moving relative to its surroundings. Concurrently, IMU data is processed using midpoint integration, which combines accelerometer and gyroscope readings to predict short-term motion between image frames.

This fusion of visual and inertial data is passed to the backend optimization module, which operates as a sliding window estimator. In this module, the system maintains a recent history of camera poses (keyframes) and performs nonlinear optimization over the following variables: the poses of the keyframes, linear velocities, IMU biases, and the 3D positions of tracked features. The optimization aims to minimize a cost function that combines reprojection errors (from vision) and inertial residuals (from IMU preintegration). This tightly coupled approach allows us to reduce drift and estimate motion more accurately, even during high-speed flight dynamics.

The system includes a loop closure module based on DBoW2 (Bag-of-Words) place recognition and a pose graph optimization layer using G2O. These modules help correct long-term drift by detecting when the system revisits previously seen locations and re-aligning the trajectory accordingly. While loop closure may not be critical in short-duration sounding rocket flights, it provides valuable support in longer missions such as planetary rovers or orbital modules.



Image 65: Arducam IMX219 Wide Angle Camera



Image 66: LSM9DS1 9-DOF IMU



Image 67: Raspberry Pi 5

In our system, the camera and IMU are synchronized using a custom Python script that publishes data as ROS topics. The extrinsic calibration between the camera and IMU is precomputed and loaded into the VINS-Mono

configuration file. The camera and IMU messages are received by the VINS-Mono node, which handles data fusion and pose estimation. The output poses are published and recorded for visualization and analysis.
The Raspberry Pi 5 now serves as the computational backbone, handling image decoding, feature tracking, IMU integration, and bundle adjustment in real time. With its quad-core Cortex-A76 CPU and upgraded GPU (VideoCore VII), it is a powerful yet compact platform well-suited for lightweight, embedded payloads like ours.

Hardware and Software Integration:

ViziNav is deployed on the Raspberry Pi 5 for in-flight operation which provides a reliable embedded environment with hardware acceleration for vision processing. The system is implemented using ROS Noetic, which provides modularity and interoperability between the sensing, processing, and visualization nodes.

The camera is connected via the CSI (Camera Serial Interface) ribbon cable to the Raspberry Pi 5, while the IMU is connected through the I2C interface. A time-synchronization script ensures consistent timestamps across both sensors. The VINS-Mono package, compiled from source, handles all visual-inertial processing. Additional packages used include OpenCV for image handling, Ceres Solver for nonlinear optimization, and GTSAM/G2O for pose graph management. The system is tested in real time and validated through simulations before deployment on physical rocket flights.

3. Muon Detection Subsystem & Comparative Scintillation Study

Occupying the second part of the payload is a muon detection experiment designed to observe high-energy cosmic muons and to evaluate cost-effective methods for particle detection in constrained environments**.**

Scientific Motivation:

Muon tomography is an imaging technique that uses naturally occurring cosmic muons to generate internal structural maps of objects. As muons pass through materials, their trajectories are slightly altered depending on the material's density and composition. By tracking these deflections, it is possible to construct detailed 3D images of the interior of dense structures—like X-ray imaging but on a much larger and deeper scale. This payload aims to leverage muon detection as a foundational step toward developing compact, flight-compatible muon tomography systems.

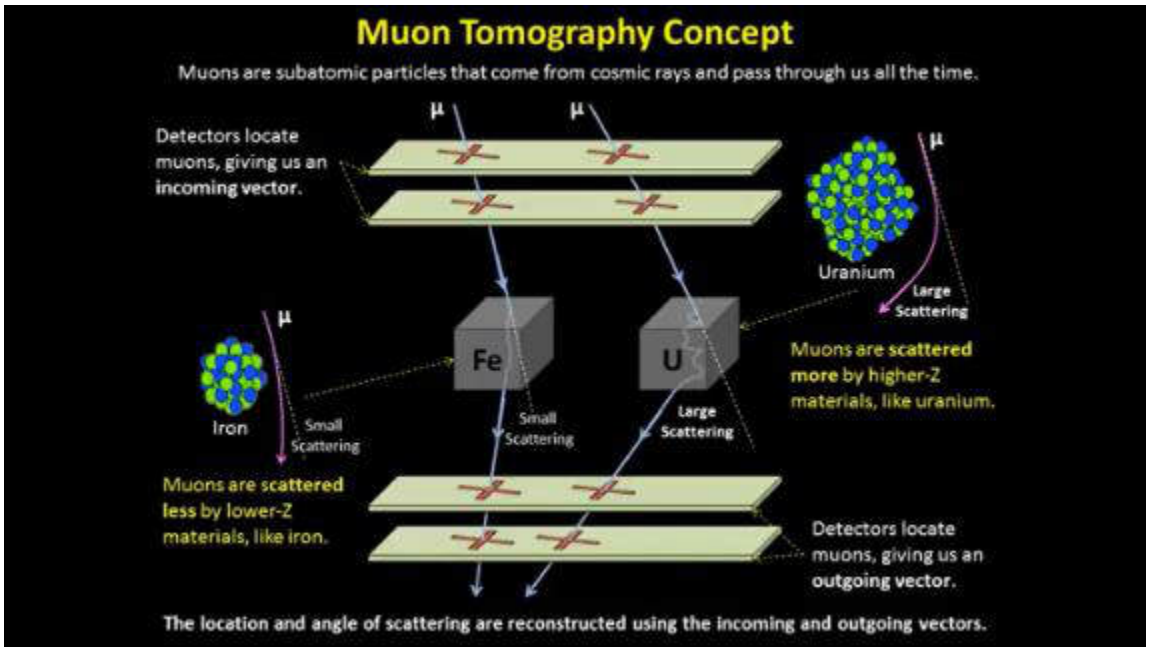


Image 68: Muon Tomography Concept

The original ambitious concept was to create a system capable of imaging two cubes made from the same material — one of which contains a defect (a hole) — using scattering muon tomography. In this approach, atmospheric muons pass through pairs of scintillators and the cube. As the muons scatter, they strike the lower scintillator layers. By analysing the vectors and entry/exit points of the muons at the scintillators, and applying algorithms and equations that account for the atomic numbers of the materials, imaging can be achieved.

However, due to the high cost of scintillators and Silicon Photomultiplier (SiPM) sensors in our country, the decision was made to scale down the system. Instead, the revised system focuses on measuring muon particle counts at an altitude of approximately 10,000 feet. This data is then used in simulation studies using Geant4 — a particle physics simulation toolkit developed by CERN — to determine whether the muon flux at that height is sufficient to produce 2D or 3D images of the test cube. Another key objective is to compare a commercially available BC-408 plastic scintillator with a custom 3D-printed scintillator (developed in collaboration with IIT Roorkee), enabling a cost-effectiveness and performance trade-off analysis. The data from both detectors will be recorded during ascent and descent, allowing evaluation under varying altitudes and atmospheric densities.

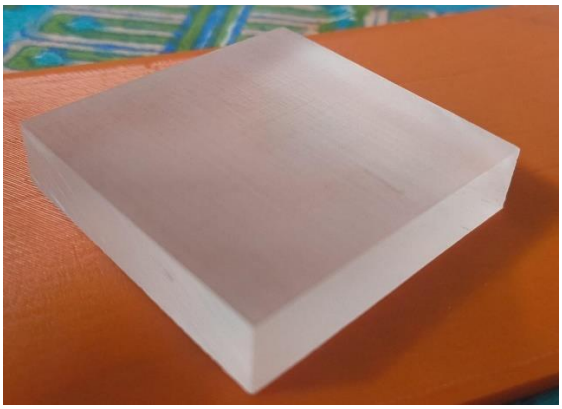


Image 69: Commercial BC408 purchased for the experiment

In the long term, this technology could support the implementation of inbuilt muon tomography systems in space exploration hardware. Such systems would allow non-destructive testing (NDT) of space structures in real time, offering the ability to detect microfractures, material stress points, and internal defects in components exposed to extreme mechanical, thermal, and radiation environments.
This could significantly enhance predictive maintenance, structural monitoring, and material optimization for crewed and uncrewed missions.

Additionally, on planetary missions, muon tomography could be used for subsurface exploration (e.g., scanning lava tubes on the Moon or Mars), supporting the identification of stable habitats or resource-rich zones. In essence, the muon detection system in this payload not only demonstrates in-flight cosmic particle sensing but also lays the groundwork for versatile, scalable space diagnostics technologies.

Detector Architecture:

1. Commercial Scintillator Setup
   - Scintillator: BC-408 plastic scintillator (known for its high light yield and fast decay time)
   - Photo-detector: 3mm × 3mm BROADCOM SiPM (Silicon Photomultiplier)
   - Associated Electronics

2. Experimental 3D Printed Scintillator
   - Developed in collaboration with IIT Roorkee, the 3D-printed scintillator explores the feasibility of using custom-fabricated materials for radiation detection.
   - Coupled with identical SiPM and amplifiers, this enables direct performance comparison between the two scintillation methods.

Detailed Subsystem Description:

Muon Detection Using BC-408 Scintillator

BC-408 Plastic Scintillator is a versatile and widely used plastic scintillator material for detecting high-energy particles, particularly muons. When a high-energy particle, like a muon, passes through the scintillator, it excites the material and emits light (scintillation) in response. This light is then captured by a photodetector (SiPM).

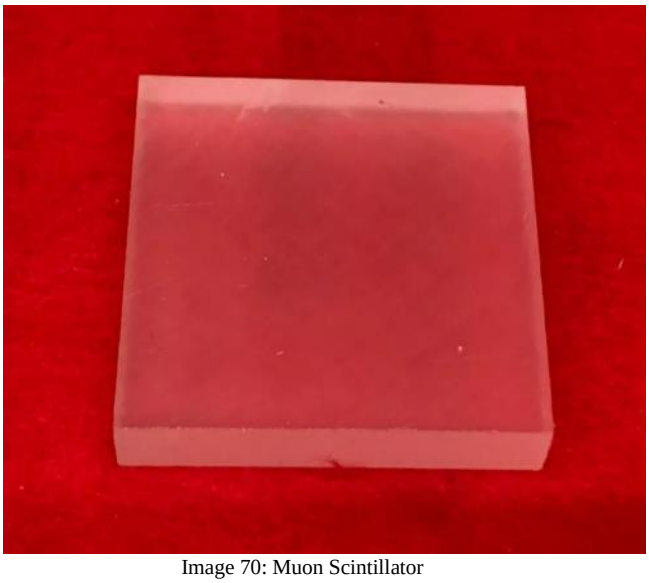


Image 70: Muon Scintillator

The muon travels through the scintillator, depositing energy and causing the material to emit flashes of light. The amount of light produced is proportional to the energy deposited by the particle. By measuring the intensity of this light, the system can detect the passage of a muon through the scintillator.

Scintillation Light Collection: The emitted light is typically detected using a photodetector, usually a Silicon Photomultiplier (SiPM), which is highly sensitive to low light levels and is compact enough to be used in space or high-altitude environments like rocketry.

3D Printed Scintillator (Developed with IIT Roorkee):

By 3D printing the scintillator, the design can be customized for optimal geometric configuration and performance, which may help increase the detection efficiency. For example, different geometries can be explored for better light collection and for creating a more compact design. This payload, however, has the printed scintillator of the same dimensions as the one commercially sourced to have a more accurate comparison of detection capabilities
The collaboration with IIT Roorkee focuses on optimizing the design and manufacturing process, ensuring that the 3D printed scintillator is both efficient and effective in detecting muons while maintaining a lightweight structure suitable for rocket payloads.

SiPM (Silicon Photomultiplier):

The SiPM is crucial for detecting the faint light produced by the BC-408 scintillator. It works by converting the light into an electrical signal, which can then be amplified and processed. SiPMs are favored in particle detection systems because they are compact, have low power consumption, and perform well in the harsh conditions of rocketry or space environments. SiPMs provide good timing resolution, which is important when detecting fast-moving particles. Additionally, their high gain and efficiency make them suitable for low-light detection.

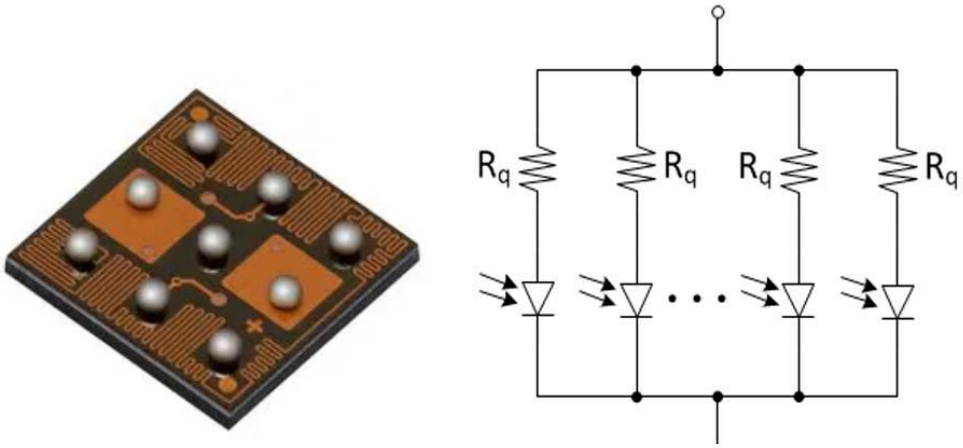


Image 71 & 72: SIPM Circuit

The SiPM used in this part of payload BROADCOM's AFBR-S4N33C013 which contains 9815 microcells and has a peak efficiency at 420 nm. As the scintillator produces the output light at this range, we are integrating this in as our detector module. The peak detection efficiency is 54 percent which is a good parameter for muon detection. The fall time is around 48 ns which is sufficient for muon detection and analysis at lower atmosphere.

Readout Electronics:

The output signal from the SiPM is typically weak, so an amplifier circuit is needed to boost the signal strength. This is crucial for ensuring that the signal is strong enough for further processing by the microcontroller. The amplifier circuit is designed to match the characteristics of the SiPM, ensuring it can handle the specific voltage and current

requirements of the photodetector while also providing a clean, noise-free signal to the onboard microcontroller. The circuit is designed so as to capture and maintain the peak amplitude of a fast analog input to be accurately analyzed.

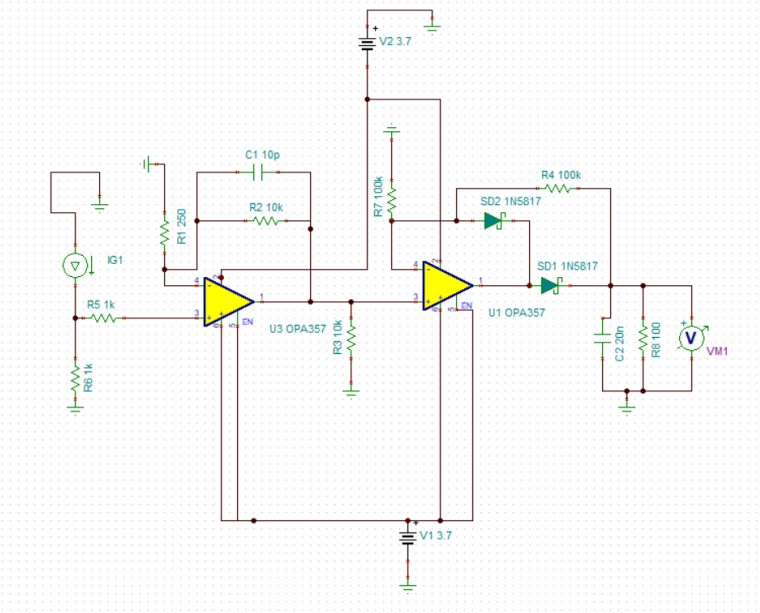


Image 73: Amplifier Circuit

A circuit schematic has been designed for the readout electronics of the SiPM output signal. The readout electronics SPICE model has been evaluated in TINA (Texas Instruments) software.

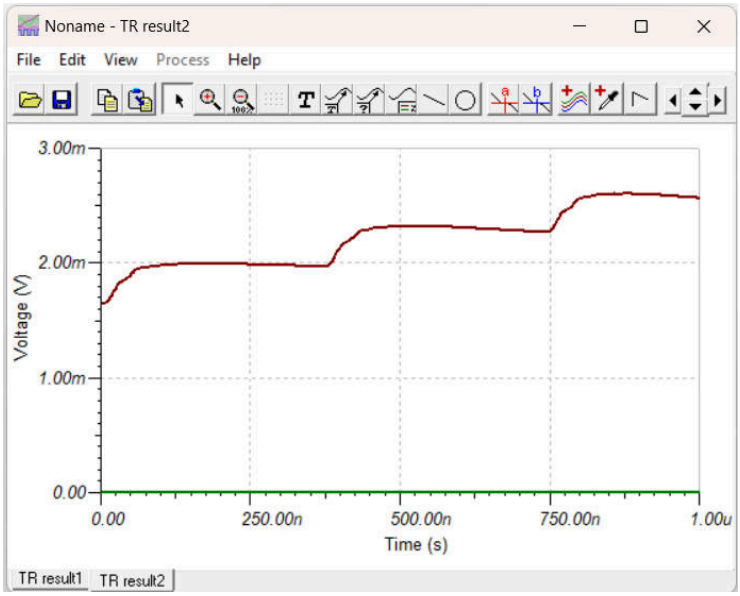


Image 74: Simulation 1

Simulation1: simulated for pulse train with amplitude 2.3uA which is the dark current output of BROADCOM's AFBR-S4N33C013.

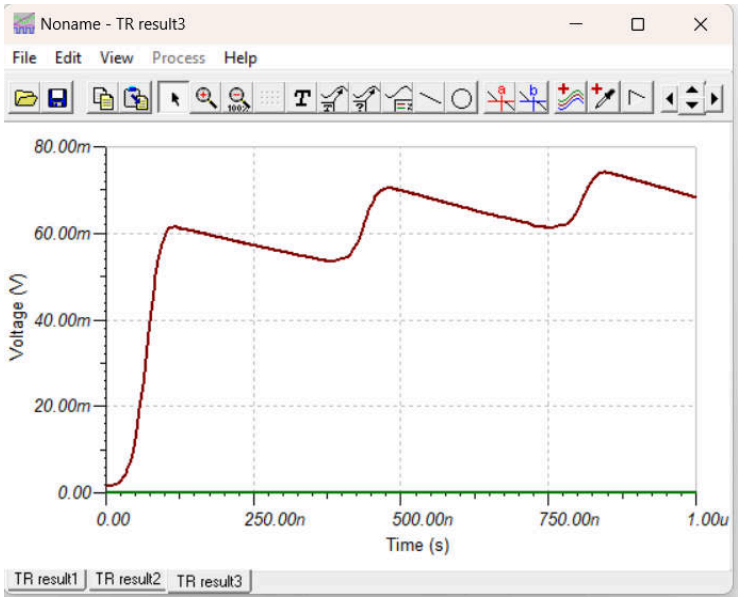


Image 75: Simulation 2

Simulation2: simulated for pulse train with amplitude 10uA output from BROADCOM's AFBR-S4N33C013.

OPA356 op-amp have been used for this purpose because of its high speed, low power, high slew rate (250 V/us), rail-to-rail output, very low quiescent current, fast settling time and its capability to drive large output currents.

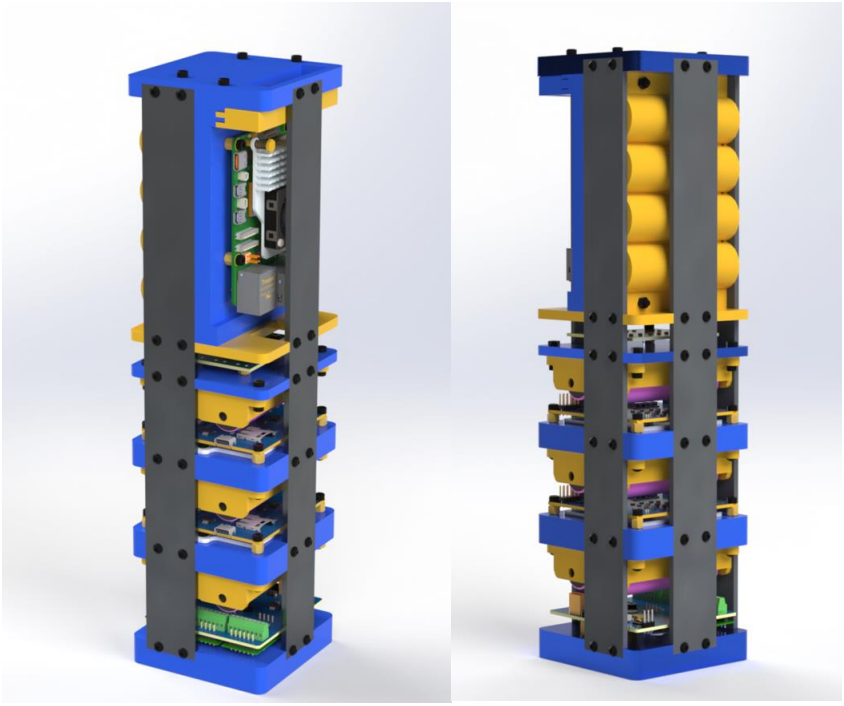


Image 76: Payload Front and Back

Comparative Study and Data Handling:

The payload records muon counts separately for both scintillators during all phases of the flight. The goal is to:

1. Compare detection efficiency, noise susceptibility, and signal fidelity
2. Evaluate cost-to-performance trade-offs
3. Assess suitability for rugged, scalable detector arrays

Structural Health Monitoring Subsystem

The remaining payload volume is utilized for a structural health monitoring (SHM) system aimed at understanding mechanical stresses experienced by the rocket's airframe.

Subsystem Purpose:

Rocket structures are subjected to extreme loads during ascent, maximum dynamic pressure, and recovery impact. Capturing in-situ strain data allows for:

1. Validation of FEA simulations
2. Early detection of over-stressed zones
3. Design iteration for weight-to-strength optimization

Implementation:

A series of strain gauges are strategically bonded to the inner walls of the rocket body tube at high-strain regions. These sensors interface with a custom-designed printed circuit board (PCB). This data will be cross-referenced with simulation data from pre-flight FEA (Finite Element Analysis) models to identify discrepancies and improve future material and structural decisions.

Integration and Future Applications:

Each subsystem is designed to be modular and independent while collectively contributing to a broader set of research objectives. All components are housed in 3D-printed structures reinforced with aluminum, offering enhanced modularity and flexibility in design and construction. This approach also simplifies the assembly process for the subsystems.

F. Avionics Subsystem

The Avionics system features a modular design philosophy that prioritizes component accessibility for rapid field maintenance while maintaining operational integrity. The architecture implements redundant systems, sensor fusion, and autonomous recovery mechanisms to ensure mission success in variable flight conditions. Structured around a hybrid approach, the system utilizes both Commercial Off-The-Shelf (COTS) components and Student Research and Developed (SRAD) components, delivering operational redundancy while allowing for technical innovation within a controlled development framework. This architecture is built upon four core design principles: Modularity enables quick field replacement of all components; Reliability is ensured through multiple redundant systems for critical functions; Autonomous Recovery incorporates self-contained decision systems for parachute deployment; and Real-time Telemetry provides continuous data transmission to ground operations. These principles work in concert to create a robust architecture that maintains operational integrity while allowing for rapid maintenance and adaptation to changing flight conditions.

1. Hardware Integration Design:

A four-layer stack-up design has been implemented to house the SRAD components. From the base to the top layer, the component placement is in the following manner:

1. Layer 1 (Base): Contains the Bessie 2.0
2. Layer 2: Accommodates the redundant Sirius Flight Computer

3. Layer 3: Accommodates the primary Sirius Flight Computer
4. Layer 4 (Top): Integrates the Master Pull Pin switch

A sled design has been developed for mounting all COTS components vertically. This sled design allows for rapid removal and replacement of entire COTS subsystem, independent testing of the COTS package, simplified field diagnostics and consistent orientation of sensors

SRAD Components:

The system architecture incorporates four custom-designed Printed Circuit Boards (PCBs), each with specific operational responsibilities:

1. Sirius (Flight Computer)
2. Bessie 2.0 (GPS Tracker)
3. Shrota 2.0 (Ground Station Module)
4. Master Switch PCB (Pull-pin mechanism)

COTS Components:

The system integrates the following three industry-standard components to enhance reliability:

1, Missileworks RRC3
2. Featherweight Raven4
3. Featherweight GPS Tracker

2. COTS System

Missileworks RRC3

RRC3 is a barometric dual-deploy altimeter developed by Missileworks which will be used as the primary recovery module for parachute deployment. RRC3 was chosen because of its good track record of consistent results and ease of use. The E-matches are connected to both Drogue and Main channels without any separate pyro-batteries. The RRC3 is configured to deploy Drogue at Apogee and Main at 1500 ft AGL altitude by firing the E-matches at the respective altitudes. The altitude is measured using its inbuilt barometer.
A standard 9V alkaline battery is used to power the RRC3. The altimeter has undergone extensive testing and validation, it has proven to be reliable and easy to configure and mount. It is securely mounted on the avionics bay using stand-offs.

Primary Features:

1. Drogue and Main outputs
2. Configurable Deployment operations
3. User-selectable beep frequency
4. Configurable third "Aux" output
5. 15 flight memory @ ~28 minutes per flight
6. Live data streaming and telemetry capability

Specifications:

1. Microcontroller - 16MHz 16-bit MSP430 Series mCU
2. Onboard Flight memory - 8Mbit SST Flash Memory
3. Pressure/Temperature sensor - MSI MS5607 Pressure sensor with 24-bit ΔΣ ADC
4. Operational Ranges - Standard Unit (40K/100K MSL) / Xtreme Unit (100K MSL)
5. Arming Mode – Barometric
6. Minimum altitude for arming - 100 feet (adjustable)
7. Battery 3.5 volts to 10 volts
8. Test Current - 17µa to 50 µa (battery dependent)
9. Firing Current - 3 amps for 1 second (Drogue / Main)
10. Dimensions - 23.5mm x 99.5mm (0.925” x 3.917”)

Featherweight Raven 4

The Raven 4 is used as the secondary recovery module for recovery. It is configured using the Featherweight Interface Program (FIP) via USB-mini connection. The altimeter detects apogee using both barometric and accelerometer data, providing robust detection even in challenging flight profiles. E-matches for Drogue and Main are connected directly into the Apo and Main output channels respectively. Since this is a redundant altimeter, the Drogue is deployed 2 seconds post-apogee and Main is deployed 2 seconds after 1500 ft AGL altitude during descent.
A single 9V battery powers both the altimeter and the deployment charges. The Raven is securely mounted with its long axis parallel to the direction of flight. After flight, the Raven 4 provides audible feedback of peak altitude and stores detailed flight data for download and analysis. The altimeter was tested multiple times on ground through the inbuilt simulation feature via the Featherweight wFIP application.

Primary Features and Specifications:

1. Outputs - 4 fully configurable pyro channels
2. Configuration - USB-mini, Featherweight Interface Program (FIP)
3. Data Logging - Up to 400 Hz (acceleration), 20 Hz (altitude)
4. Sensors - Barometric (MS5607), 3-axis accelerometer (H3LIS331DL)
5. Power - 3.8–16 V DC (9V battery typical), up to 20 V for pyro
6. Dimensions - 1.8" x 0.8" (45.7 mm x 20.3 mm)
7. Max Altitude - 100,000 ft (standard), custom firmware for higher flights
8. Mounting - Parallel to flight axis, auto-orientation
9. Memory - Onboard, accessible via USB

Featherweight GPS Tracker

The Featherweight GPS tracker a very compact COTS module contains a uBlox SAM M8Q GPS tracker. The GPS Tracker is powered using a single 18560 cell (3.7 – 4.2 V). It is also accompanied by a ground station receiver that is powered by a rechargeable 3.7v 400mAh LiPo battery which connects to smartphone through Bluetooth and the location data can be tracked using the smartphone application. Both the tracker and ground station receiver are equipped with a 915 MHz stub antenna. The GPS tracker has been tested on ground extensively using the mobile application and has provided reliable and satisfactory tracking data.

Primary Features:

1. Compact dimensions - 1.67" x 0.8" (excluding antenna)
2. High-rate data logging - Records GPS position, velocity, and signal strength at 10 samples/second during ascent
3. Community relay system - Automatically shares location data with other Featherweight trackers for recovery assistance
4. Bluetooth integration - Downloads flight data to smartphones for post-flight analysis
5. Dual-frequency operation - Supports 915 MHz and 433 MHz bands for global compatibility
6. Specifications
7. Max altitude - 262,000 ft
8. Data logging - 6,144 samples (ascent: 10 Hz, descent: 1 Hz)
9. dl range - 50+ miles line-of-sight
10. Antenna options - Stub antenna (standard) or wire whip for bulkhead mounting
11. Compatibility - Featherweight UI smartphone app
12. Mounting - 4-40 screw

COTS Battery specifications:

Batteries used are of the same specifications as those specified by the manufacturers of the RRC3, Raven 4, and Featherweight GPS. This ensures that the COTS subsystem operates as intended, without any malfunctions, and meets the standby requirements. The RRC3 and Raven 4 requires a nominal input voltage of 9V, while the Featherweight

GPS require an input voltage of 3.7V, as specified by their respective manufacturers. Additionally, a single 18650 Li-ion battery, which functions as a pyro battery, can be used alongside the RRC3 and Raven 4.

The table below provides detailed information about the batteries and their chemistries:

Specifications	Missileworks RRC3	Featherweight Raven4	Featherweight GPS Tracker
Chemistry	Alkaline	Alkaline	Li-Ion
Nominal Voltage	9V	9V	3.7V
Capacity	330 mAh	330 mAh	3500 mAh

Table 11: Battery Chemistries

3. SRAD System

1. Sirius

Sirius is our custom-designed flight computer providing comprehensive avionics capabilities for advanced rocketry applications. The system integrates a high-performance microcontroller module while also incorporating multiple redundant sensor arrays, dual pyrotechnic deployment channels, and resilient data handling, all housed within a very compact 70mm x 70mm form factor board to ensure mission success across a wide range of dynamic flight conditions. Each component has been meticulously selected to deliver optimal performance within its role, while contributing to overall system redundancy, fault tolerance, and autonomous operation.

Microcontroller:

At the heart of the Sirius flight computer lies the ESP32-S3-WROOM-1U, a highly integrated wireless MCU module developed by Espressif Systems. Designed for low-power operation with deep-sleep modes and excellent operation speed, the ESP32-S3-WROOM-1U ensures energy-efficient data acquisition and decision-making even during extended missions. The module's rich I/O set and hardware encryption support add layers of flexibility and security making it the perfect microcontroller for a flight computer.

Barometric Pressure Sensors:

The MS5611, manufactured by TE Connectivity, functions as the primary altitude sensor. With an operational pressure range of 10–1200 mbar, absolute accuracy of ±1.5 mbar, and ultra-fine resolution (up to 0.012 mbar), it enables high-precision altitude determination critical for apogee detection, trajectory profiling, and event sequencing under dynamic flight conditions. The BMP280 from Bosch Sensortec serves as the secondary altitude determination sensor, offering ±0.12 hPa relative accuracy (~±1 m altitude), excellent for redundancy and validating primary sensor data during key flight phases.

Inertial Measurement Units:

The BMI088, developed by Bosch Sensortec, acts as the primary inertial measurement unit. Engineered for superior vibration resistance and stability under high dynamic loads, it offers accelerometer ranges up to ±24 g and gyroscope ranges up to ±2000°/s, with robust filtering to maintain precise motion tracking during powered flight and rapid attitude changes. The LSM6DSL from STMicroelectronics provides additional inertial sensing redundancy, with low-power operation, programmable ranges, and a high-speed FIFO buffer, ensuring uninterrupted data capture even during peak computational loads.

Magnetometers:

The LIS3MDLTR magnetometer, manufactured by STMicroelectronics, is utilized as the primary heading and orientation sensor. Offering selectable measurement ranges up to ±16 gauss and low-noise outputs, it delivers reliable, real-time magnetic field tracking necessary for accurate 3D attitude estimation and navigation. The QMC5883L acts as the secondary magnetic field sensor, supporting orientation redundancy and offering continuous measurement mode for consistent heading determination even under fluctuating magnetic environments.

GNSS:

The u-blox NEO-M8N module provides high-accuracy GPS tracking capabilities, supporting concurrent reception of multiple GNSS constellations including GPS, Galileo, GLONASS, and BeiDou. With acquisition times as low as 1 second (cold start ~26s) and a position accuracy of up to 2.5 meters CEP, it enables reliable vehicle tracking throughout ascent, apogee, and descent phases. The NEO-M8N interfaces with the ESP32-S3-WROOM-1U over UART, allowing seamless integration into the Sirius flight computer architecture. NEO-M8N also offers altitude and precision time data which is used in conjunction with the data received from other sensors.

Precision Time:

The DS3231 temperature-compensated real-time clock module ensures precise timestamping and event synchronization, maintaining ±2 ppm accuracy over the full -40°C to +85°C range. Its backup battery system provides resilience against power interruptions during long pre-launch or recovery operations. Time is also maintained by the NEO-M8N thus providing two very accurate sources for tracking time throughout the mission duration.

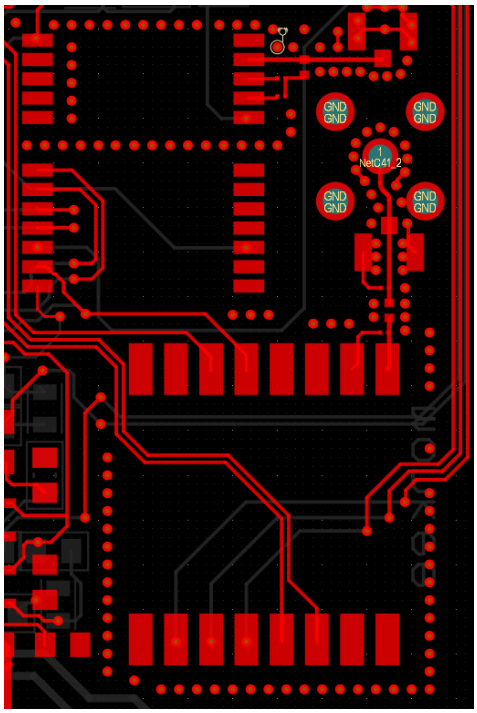


Image 77: Sirius Layout

Communication System:

Operating in the 915 MHz ISM frequency band, the RFM95W LoRa radio module enables long-range, low-power telemetry essential for mission-critical real-time updates and remote system diagnostics. Under optimal atmospheric conditions, communication ranges exceeding two kilometres have been achieved. To ensure superior RF performance,

the PCB layout implements controlled impedance matching, designed specifically for a two-layer FR4 PCB stack-up (dielectric constant $\varepsilon_r \approx 4.6$, 1 oz copper). A 50-ohm microstrip transmission line was achieved, resulting in a calculated trace width of 12.603 mils. Additionally, to further mitigate electromagnetic interference and maintain signal integrity, the RF traces are shielded using closely spaced ground stitch vias on either side, effectively forming a quasi-coaxial environment that suppresses parasitic coupling and external noise.

The RFM95W is interfaced with a 915 MHz omnidirectional dipole antenna, providing uniform azimuthal radiation. This configuration guarantees that, regardless of the rocket's instantaneous orientation or dynamic manoeuvres, telemetry data remains consistently within the ground station's reception envelope.

Data Storage:

Primary non-volatile storage is provided by the W25Q128 NOR flash memory chip from Winbond, offering 128 Mbit (16 MB) of storage capacity via a high-speed SPI interface. This module supports rapid data logging rates of up to 80 MB/s, easily accommodating the high-volume data streams generated by multi-sensor fusion during dynamic flight phases. In addition to flash storage, an SD card interface has been incorporated into the design to provide expandable, removable storage capability. This allows an easy method for retrieving the data, which would be stored into the SD Card, without any additional step. Furthermore, SD Card storage facilitates expandable storage to facilitate for high frequency logging operations. The SD card is interfaced via SPI.

Pyro Channels:

Sirius integrates two independent pyrotechnic channels specifically designed for the controlled activation of E-matches. Each channel features an integrated continuity verification circuit that provides real-time status indication of the E-matches.

For continuity checking, the system establishes a precisely regulated low-current path through each E-match. This sub-threshold current (<10mA) is insufficient to ignite the E-match but provides adequate signal level for monitoring. A comparator is implemented for producing discrete voltage that defines the status of the E-match, which is then read by the MCU and is also indicated through status LEDs.

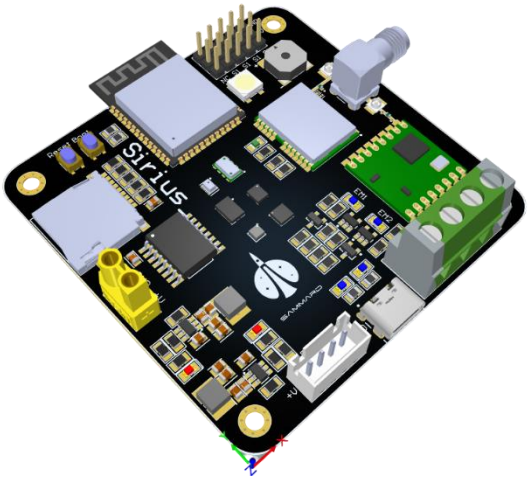


Image 78: 3D CAD Model of Sirius

2. Bessie 2.0

The BESSIE v2.0 tracker represents a fully custom-designed Student Researched and Designed (SRAD) module engineered specifically for consistent and efficient rocket retrieval throughout flight operations and post-landing recovery. This advanced tracking system is built around the high-performance ESP32-S3 microcontroller, selected for its extensive capabilities including enhanced SPI flash memory, octal SPI interface support, PSRAM compatibility, and configurable cache architecture—making it ideally suited for the demanding tasks of real-time sensor fusion, comprehensive data logging, and telemetry management.

At the core of BESSIE V2.0's tracking functionality lies the NEO-M8N GPS module, renowned for exceptional performance characteristics including -161 dBm tracking sensitivity that ensures accurate positional data acquisition even during high-speed flight, extreme altitudes, or in compromised signal environments. The module's concurrent reception capability across multiple Global Navigation Satellite Systems (GPS, Galileo, GLONASS, and BeiDou) significantly enhances positioning accuracy and resilience against signal degradation, while maintaining backward compatibility with NEO-7 and NEO-6 series for future upgrade flexibility.

Beyond GPS capabilities, BESSIE v2.0 incorporates an MPU-9250 9-Degrees-of-Freedom Inertial Measurement Unit paired with an HMC5883L magnetometer, effectively functioning as a digital compass to provide critical orientation and heading data during recovery operations. This integrated approach enables ground crews to precisely determine the rocket's position upon landing, substantially reducing search time and maximizing operational efficiency. Temporal precision is maintained through the DS3231 high-precision Real-Time Clock module, which provides accurate timestamp recording for flight events, telemetry synchronization, and chronological data preservation even during power interruptions.

For communication requirements, BESSIE v2.0 employs a LoRa module operating in the 902-926 MHz frequency band, delivering long-range, power-efficient wireless data transmission. This ensures continuous monitoring of flight parameters and location data throughout all mission phases.

Signal quality is optimized through implemented L-pad impedance matching, effectively reducing noise and enhancing overall transmission efficiency.

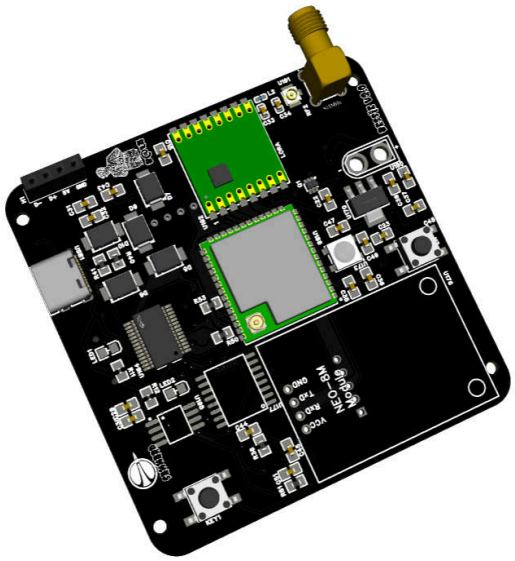


Image 79: 3D CAD Model of Bessie 2.0

3. Shrota 2.0

Shrota, is a Sanskrit word which translates to “a listener”. It is our custom-designed and developed telemetry receiver

and tracker system built for reliable long-range performance. Powered by a single 18650 lithium-ion battery, it refreshes position data at a rate of 5 Hz. The system uses a high-sensitivity 50-channel GPS receiver that operates on the L1 band and supports augmentation systems such as WAAS, EGNOS, and MSAS for improved accuracy. Designed for robustness, Shrota maintains full functionality across a wide operating temperature range of –24°C to 70°C. Shrota is a standalone module that houses an OLED screen, which displays real-time information such as coordinates, satellite lock status, and system health, offering instant visual feedback to the user without the need for an external interface.

To ensure consistent long-distance data transmission on the 915 MHz band, Shrota is paired with an 18 dBi high-gain outdoor Yagi antenna. This antenna operates across a frequency range of 700–2700 MHz, offers a gain of 18 dBi, and has an impedance of 50 ohms. Its directional design allows for data transmission over line-of-sight distances up to 12 kilometres. This makes Shrota 2.0 an ideal solution for remote tracking applications where signal strength, accuracy, and durability are critical.

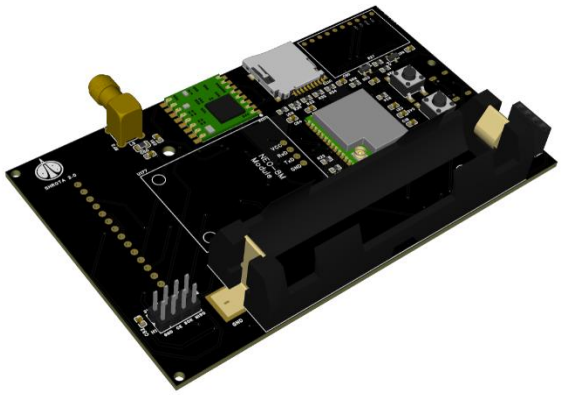


Image 80: 3D CAD Model of Shrota 2.0

4. Master Switch

All power lines from individual modules are routed through a Master Mechanical Switch using XT30 connectors, selected for their compact footprint and secure locking mechanism. This centralized design ensures clean wiring, ease of debugging, and robust pre-launch safety. Once the Master Switch is engaged, power flows from the avionics stack back to the Battery Bay, where all power sources are physically isolated and mounted for easy access. Limit switches are used alongside N-Channel Mosfets for controlling the flow of current for each individual channel.

Master Switch shares power from the redundant Sirius for powering on the MOSFETs and LEDs. It does not require a dedicated power supply unit.

SRAD System Battery Configuration

Specifications	**Primary Sirius**	**Redundant Sirius**	**Bessie 2.0**	**Shrota 2.0**	**Master Switch**
Chemistry	Li-Ion	Li-Ion	Li-Ion	Li-Ion	-
Nominal Voltage (per single battery)	3.7V	3.7V	3.7	3.7V	-

Capacity (per single battery)	3500 mAh	3500 mAh	3500 mAh	3500 mAh	-
Battery Configuration	2S	2S	2S	1S	-

Table 12: SRAD Battery Configuration

Avionics Bay Wiring Strategy

The batteries within the battery bay are connected using ring terminal and are crimped properly at one end. The other end is connected to XT30 that connects to the components and battery bay switch ensuring a tight fit throughout the flight duration.

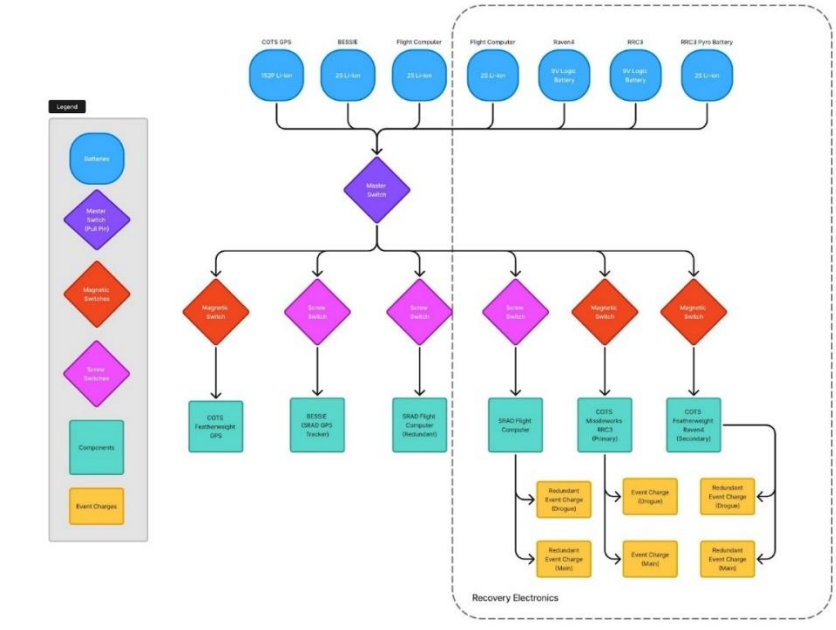
Each SRAD module is connected via dedicated XT30 connectors to the Master Switch. Downstream of the switch, each line is routed through a Screw Switch for individual activation of the following:

1. Sirius Flight Computer (Primary)
2. Sirius Flight Computer (Redundant)
3. Bessie GPS Tracker

COTS modules also interface with the Master Switch through their own XT30 inputs. Each line is then routed through a Magnetic Switch, allowing touchless activation for:

1. Featherweight GPS Module
2. Missileworks RRC3 (Primary Recovery)
3. Featherweight Raven4 (Secondary Recovery)

The power flow throughout the avionics bay is shown below:



Avionics Bay:

The modular avionics and power bay is engineered to house the rocket's primary flight electronics, battery modules, and supporting hardware within a compact and structurally integrated unit. Acting as the central intelligence hub of the rocket, the system ensures stable power delivery and reliable mission-critical data acquisition throughout the duration of the flight.

The bay consists of an upper battery pack housing and a lower avionics section, separated by a removable interface. This separation enhances accessibility during integration and testing while maintaining structural integrity.:

1. Upper Power Section: Battery bay with COTS-electronics
2. Lower Avionics Section: Multi-PCB stack for core flight operations

These two sections are connected via a removable planar bulkhead, enabling rapid assembly and testing during integration and launch preparations.

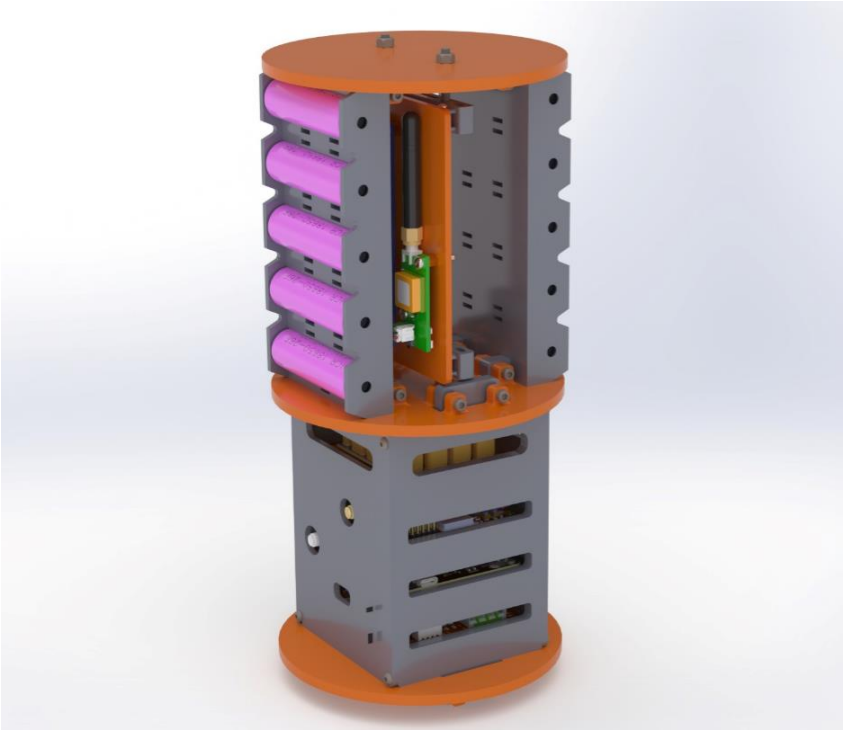


Image 81: Avionics Bay

1. Upper Section: Battery Bay + Modular COTS Carrier

The battery bay contains five horizontally mounted ICR 18650 lithium-ion cells per face, secured in custom-cut mounting slots that offer precise fit and vibration isolation. A modular internal rail system supports a sliding carrier plate, which houses the Raven 4 altimeter, Featherweight GPS module, and RRC3 flight computer. This plate is designed to be removed independently of the battery holders, allowing for straightforward hardware replacement or debugging. Additionally, all connectors are organized along a vertical rail, providing easy plug-and-play access to each component.

3. Lower Section: Avionics Stack

The avionics section features a stack of four discrete PCBs mounted on custom precision standoffs, each serving a critical function in the flight system. It includes two Sirius flight computers—one as the primary controller and the

other as a redundant data logger—a Bessie GPS module for real-time positioning and timestamping, and a Master Switch PCB that manages power routing and integrates safety interlocks. To streamline maintenance, the stack includes side slots and cutouts for probing and wiring access without requiring full disassembly. The mechanical layout ensures secure multi-axis mounting and clear electrical separation between layers.

4. Mechanical and Structural Integration

All internal assemblies are mounted along a central stainless steel lead screw. This configuration ensures axial rigidity and efficient transmission of mechanical loads through the rocket body during launch and recovery. The central mounting system contributes to the robustness of the bay under high dynamic stress conditions.

5. Key Advantages

The design allows for rapid module replacement, with the sliding COTS carrier and stackable PCB configuration enabling quick hardware swaps during field testing. Modular debugging is also facilitated, as each subsystem can be isolated and accessed independently without impacting the rest of the system. Additionally, the structure is rated for high-G conditions exceeding 30g acceleration, with low centre-of-mass mounting and a mechanically isolated PCB stack enhancing both performance and survivability during flight.

G. CS Subsystem

1. Flight Software

The Flight Software System is developed to autonomously log and transmit telemetry during flight. Running on the SIRIUS flight computer, the software integrates real-time sensor data collection and storage, onboard decision-making, and LoRa-based telemetry for reliable communication with the ground station.

Key Objectives:

- Track altitude, orientation, position, and velocity in real-time.
- Identify key flight phases (launch, ascent, apogee, descent, touchdown).
- Log and transmit mission-critical data using redundant storage methods like SPI Flash, SD Card and LoRa.
- SIRIUS (flight computer), used as a tertiary recovery system to fire the deployment charges in the Recovery Module and Tender Descender.

Workflow and architecture:

SOFTWARE WORKFLOW:

State 1: Launch_Pad

In the Launch_Pad state, the system begins by initializing all sensors and starts sending telemetry to the ground control station, it checks whether the altitude exceeds 5 meters. If so, a launchTrust counter is incremented. Only when this counter confirms the altitude condition multiple times (launchTrust ≥ 3) does the system confidently transition to the Ascent state, ensuring that the detection is reliable and not due to noise or temporary glitches.

State 2: Ascent

During the Ascent state, the system continually monitors the current altitude against the maximum recorded altitude. If a new higher altitude is reached, it updates the maximum and resets the ascentTrust counter. If the altitude starts dropping or remains the same for consecutive checks, the ascentTrust counter increases. Once this counter exceeds a threshold (ascentTrust ≥ 3), indicating that the rocket has likely reached peak height, the system transitions to the Apogee state.

State 3: Apogee

In the Apogee state, the system checks if the current altitude is less than or equal to the previous altitude. If this condition is confirmed for multiple consecutive cycles (descentTrust ≥ 3), it is assumed that the rocket has begun descending. At this point, the descent phase is considered valid, and the system progresses to the Descent state.

State 4: Descent

The Descent state confirms that the rocket has landed motion by checking if the current altitude remains the same as the previous reading. When this happens, the LandTrust counter is incremented. If the counter reaches or exceeds 3 consistent confirmations (LandTrust ≥ 3), it is assumed that the rocket has landed, triggering the transition to the final state - Touchdown.

State 5: Touchdown

Once in the Touchdown state, the system activates a buzzer to audibly indicate landing. The system stops telemetry transmissions and ends the flight routine, marking the completion of the mission.

STATE DIAGRAM**:**

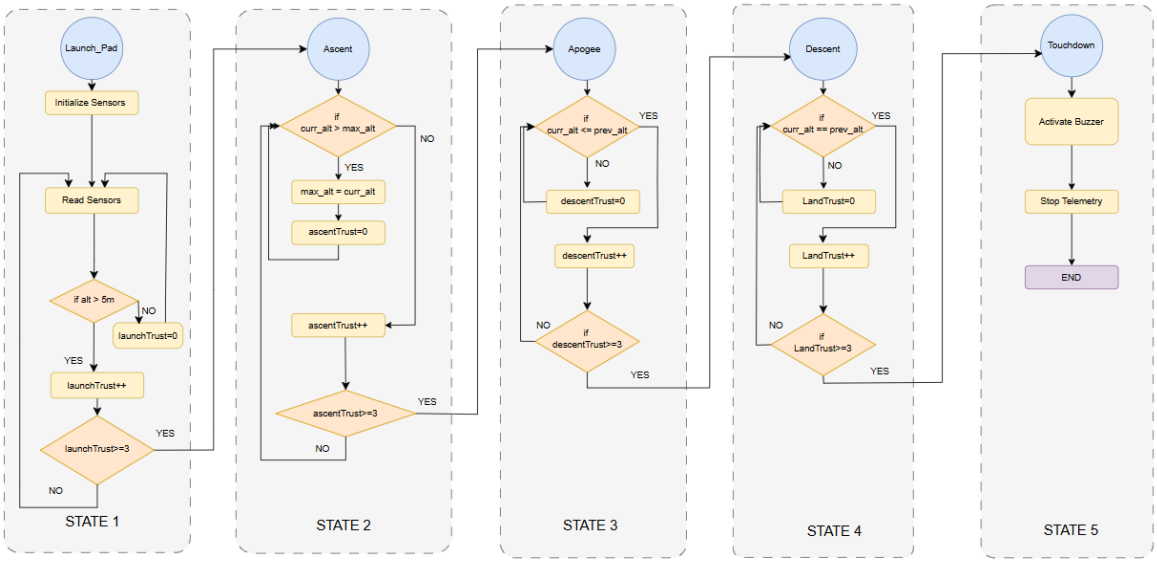


Image 82: State Diagram

Technologies and Concepts:

TECH STACK:

The flight software is written in C++ using the Arduino framework. During development, we rely on the Arduino IDE for code editing, compilation and uploading. The flight computer, Sirius, will manage and control all operations. To interface with the sensors, we use various libraries like: Wire and SPI handle I²C and SPI communications, LoRa to provide long range radio telemetry, TinyGPS++ for NEO-M8N, Adafruit BMP280, Adafruit HMC5883_U, DS3231-RTC for real-time clock functions, SimpleKalmanFilter to implement noise-reduction algorithms, Adafruit_SPIFlash for external flash memory.

DATA STORAGE FORMAT:

TEAM_ID, TIME, PACKET_COUNT, ALTITUDE, PRESSURE, TEMPERATURE, VOLTAGE, VELOCITY, GPS_SPEED, GPS_LATITUDE, GPS_LONGITUDE, GPS_ALTITUDE, GPS_SATS, GPS_TIME, MAGNETOMETER_X, MAGNETOMENTER_Y, MAGNETOMETER_Z, GYROSCOPE_X, GYROSCOPE_Y, GYROSCOPE_Z, TRIGGER, ANGLE_ROLL, ANGLE_PITCH, ANGLE_YAW, ACCELEROMETER_R, ACCELEROMETER_P, ACCELEROMETER_Y, CURRENT_STATE

DATA TRANSMISSION FORMAT:

Segment 1 (fundamental flight data): *TEAM_ID, TIME, PACKET_COUNT, ALTITUDE, PRESSURE, TEMPERATURE, VOLTAGE*

Segment 2 (GPS and Velocity data): *VELOCITY, GPS_SPEED, GPS_LATITUDE, GPS_LONGITUDE, GPS_ALTITUDE, GPS_SATS, GPS_TIME*

Segment 3 (IMU sensor data): *MAGNETOMETER_X, MAGNETOMENTER_Y, MAGNETOMETER_Z, GYROSCOPE_X, GYROSCOPE_Y, GYROSCOPE_Z, TRIGGER*

Segment 4 (Orientation and Acceleration data): *MAGNETOMETER_X, MAGNETOMENTER_Y, MAGNETOMETER_Z, GYROSCOPE_X, GYROSCOPE_Y, GYROSCOPE_Z, TRIGGER*

ALGORITHMS:

The Kalman Filtering algorithm plays a pivotal role in ensuring the accuracy and stability of sensor readings by smoothing out noise from raw inputs such as gyroscope and acceleration. It is used to fuse these readings to get the orientation values.

A carefully designed state machine governs the identification of distinct flight phases including Launch Pad, Ascent, Apogee, Descent, and Touchdown, by applying logic-based thresholds and time-validated confirmation checks. Each transition between states is determined not only by altitude data but also through trust-based mechanisms that prevent false triggers due to transient anomalies or noise. This approach ensures that critical events like apogee detection and parachute deployment occur precisely when needed. Sensor fusion is employed to merge data from the accelerometer, gyroscope, and magnetometer using a Kalman or Complementary filter.
To ensure efficient and reliable data transmission, telemetry encoding compresses and formats critical flight parameters into compact packets optimized for LoRa communication. The total data packet contains 28 fields of size 210 to 220 bytes, carefully structured to balance comprehensive flight state reporting with minimal bandwidth usage. To enhance transmission integrity and reduce the likelihood of packet loss, this 28-field packet is divided into four smaller sub-packets of 7 fields each. These sub-packets are transmitted sequentially, improving reception reliability, especially under fluctuating signal conditions or high-noise environments.

Redundant Storage methods safeguard mission-critical data through dual logging mechanisms. All telemetry is written to an onboard SD card in a structured CSV format for post-flight analysis. Simultaneously, the same data is stored in an external SPI Flash, from where the last state is recovered in case of system reset.

2. FLAIR – SRAD GROUND STATION SOFTWARE

FLAIR is Team Sammard's custom-built Ground Control Station (GCS), developed to monitor rocket flights in real time and support critical decision-making during test campaigns. Designed using Python and PyQt5, FLAIR interfaces with the SIRIUS (flight computer) via SHROTA 2.0 to receive live telemetry throughout the mission.
It goes beyond basic visualization by offering real-time plotting of key flight parameters through a clean, intuitive interface. These visuals—primarily line graphs—allow the team to quickly interpret trends, monitor system health, and react to anomalies during flight. FLAIR also handles reliable data logging, storing telemetry both onboard and on the ground to ensure no information is lost.
This archived data becomes invaluable during post-flight analysis, helping the team refine systems, improve rocket performance, and prepare more effectively for future launches. By combining real-time insights with long-term data storage, FLAIR plays a central role in driving mission success.

Workflow and Architecture:

The FLAIR system is designed to efficiently manage real-time telemetry reception and processing during flight missions. It begins with continuous telemetry reception via LoRa modules, which transmit live data from the rocket to the ground station. This data is received through a serial interface over USB, which is exposed by the operating system as a virtual COM port, using SHROTA 2.0, which connects directly to a laptop running the FLAIR software. PySerial is employed to handle the low-level serial communication, ensuring a reliable stream of telemetry data into the system.
Once received, telemetry data undergoes parsing and validation. FLAIR analyses each data string, verifies it using predefined formats and checksum validation, and filters out any corrupted packets caused by wireless interference. The system is also capable of detecting packet loss, helping maintain continuity and accuracy in the data stream. To ensure seamless and lag-free operation, FLAIR uses a multithreaded architecture where one thread handles telemetry reception and others take care of data parsing, processing, and updating the user interface. This design ensures high responsiveness and real-time feedback without compromising system performance.
To give mission operators both clarity and control, FLAIR offers two synchronized views of the incoming telemetry: a tabular and a graphical interface. The tabular view provides structured, real-time access to key flight parameters

such as temperature, pressure, velocity, and altitude, facilitating detailed monitoring and analysis. Simultaneously, the graphical view, built using Matplotlib, delivers live-updating plots of critical variables, offering a quick and intuitive visual representation of the system's behaviour. These two views work in tandem to provide both depth and clarity, enabling fast decision-making during mission-critical events.
In parallel with real-time monitoring, FLAIR ensures comprehensive data logging by saving all telemetry to a CSV file throughout the mission. This log serves as a complete record for post-flight analysis, helping teams evaluate performance, diagnose issues, and verify system behaviour. Additionally, the system supports replay functionality, which allows the telemetry data to be visualized as if it were being received live. This feature is essential for debugging, training new team members, and simulating different mission scenarios. Altogether, these tools make FLAIR not only a powerful live-monitoring platform but also an effective resource for continuous improvement and learning.

Various Functionalities Of FLAIR:

1. **Home Page:**

The GCS homepage serves as the central dashboard for monitoring real-time flight data, providing clear, actionable insights throughout the rocket's mission. It features four essential live-updating graphs that track altitude vs time, GPS speed vs time, pressure vs time, and voltage vs time allowing operators to monitor the rocket's performance and orientation at a glance. Alongside these plots, the dashboard displays a comprehensive set of key telemetry parameters, including accelerometer and gyrometer readings, Euler angles, RSSI, GPS coordinates, altitude, course, and satellite count. Together, these visual and tabular data elements offer a complete overview of the flight's dynamics. Additionally, the real-time GPS location tracking is crucial for post-flight recovery, enabling the team to pinpoint the rocket's landing site with precision. This layout ensures intuitive navigation, quick diagnostics, and informed decision-making during all mission phases.

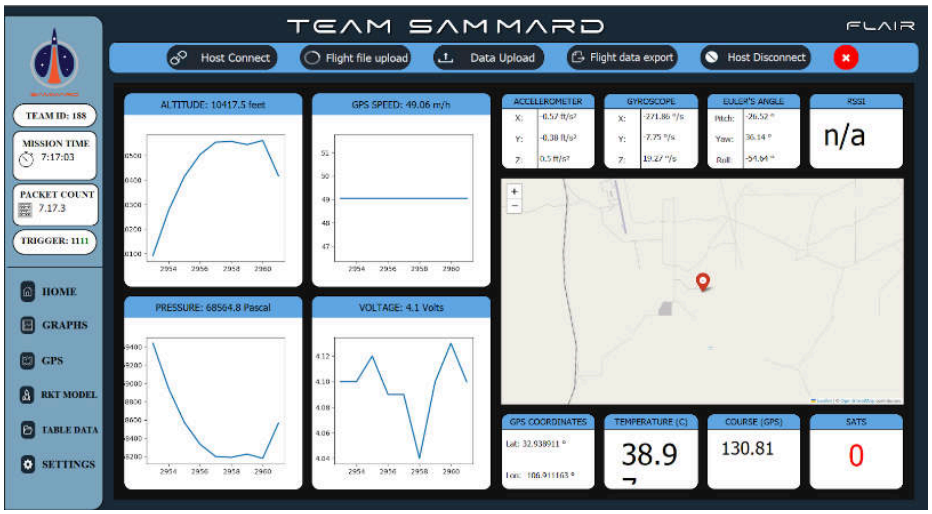


Image 83: FLAIR Home Page

2. Graphs Page:

The graphs page is a crucial part of FLAIR as it is the hub of meticulously curated data which is featured in four different tabs. While switching tabs, users can analyse two graphs, every tab including essential metrics such as Altitude, Pressure, Voltage, Velocity, Velocity vs. GPS Speed and Altitude vs. GPS Altitude. These graphs, displayed prominently provide users efficient visualization and a detailed overview of flight dynamics throughout the journey. The flight data panel displays few of the crucial fight parameters such as Altitude, Speed and Temperature, Acceleration alongside the X, Y and Z axes, as well as the gyrometer values along the X, Y and Z axes.

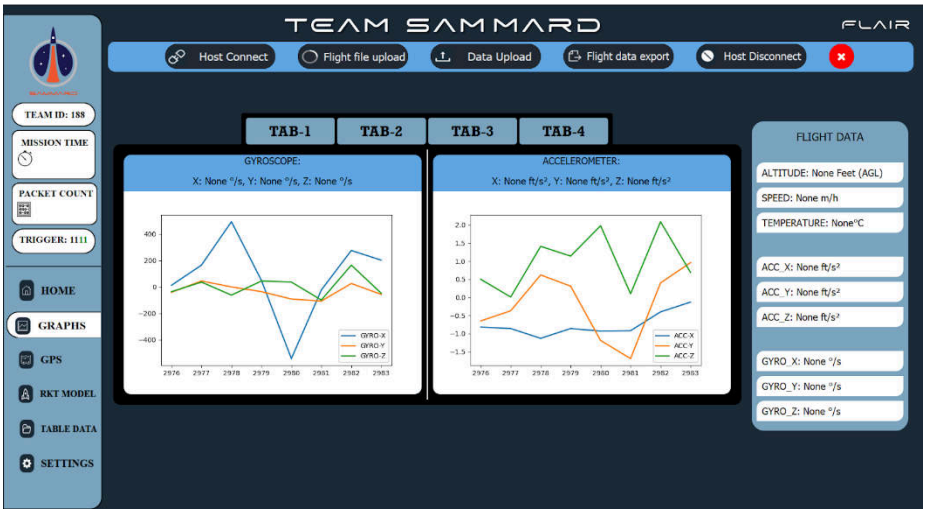


Image 84: FLAIR Graphs Page

3. GPS Page:

The GPS page is designed to support efficient and accurate rocket recovery. With a larger GPS display, tracking the rocket's position becomes quick and effortless, allowing for faster and more precise retrieval. The dedicated GPS data panel shows all essential information including time, date, number of satellites, latitude, longitude, altitude, and speed, giving a full picture of the rocket's current state. These details help ensure smooth monitoring and recovery, both during flight and after landing.

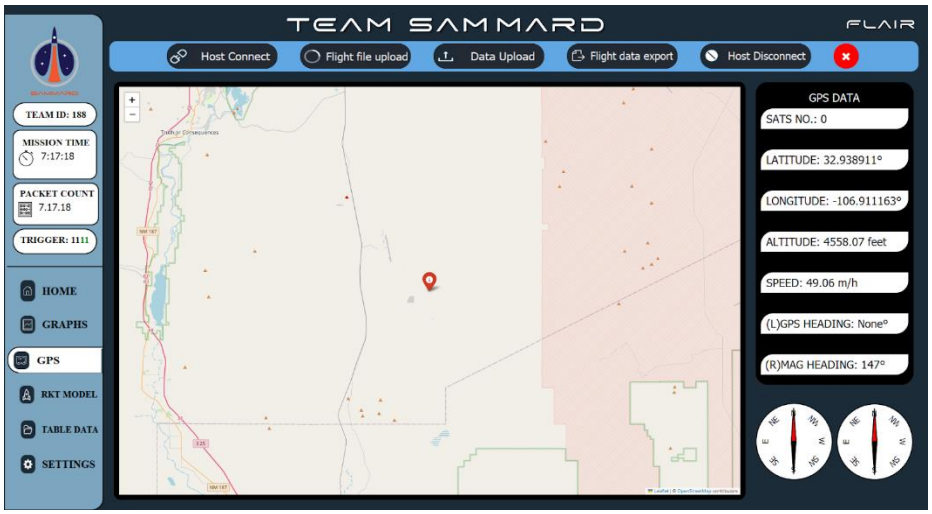


Image 85: FLAIR GPS Page

4. Table Data Page:

The Table Data Page serves as a detailed interface for viewing all flight parameters captured by the flight software system. It features a tabular layout that displays every key data point received, making it easy to track and review information both during the flight in real-time and after the mission. Users can easily scroll through the data for quick checks or thorough analysis, ensuring that nothing is missed during monitoring or post-flight review.

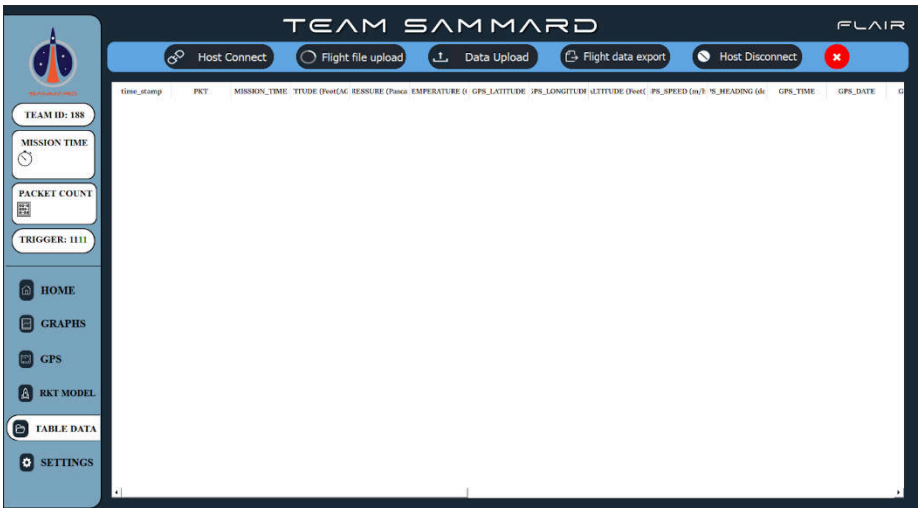


Image 86: FLAIR Table Page

5. Setting Page:

The Settings page acts as the central control panel for unit conversions and communication port management. It allows users to convert selected flight parameters in the saved CSV file, producing a new file with updated units based on user preferences. After selecting the desired units and clicking the APPLY ALL CHANGES button, the system quickly processes the changes and saves the modified CSV. Additionally, the Comm Link section displays the currently active communication port, providing easy access to connection status and ensuring reliable serial communication.

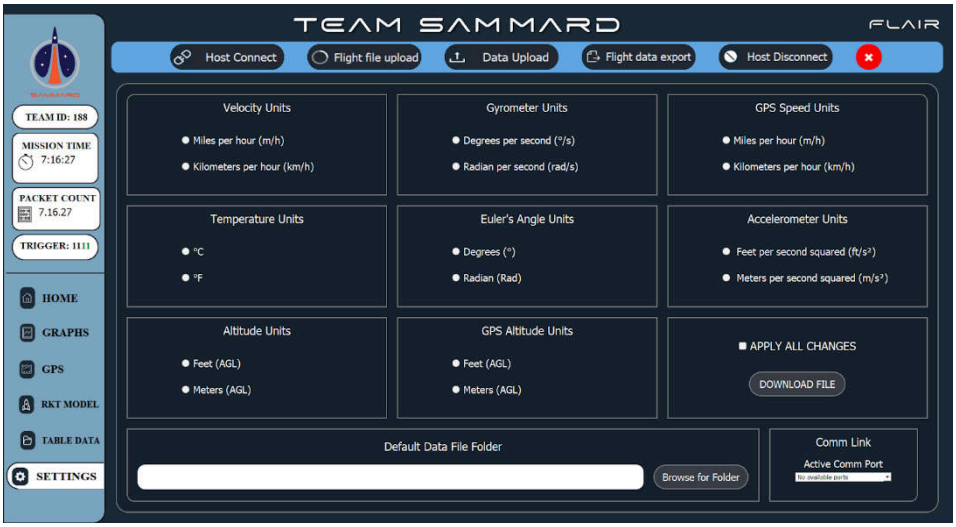


Image 87: FLAIR Settings Page

Technologies, Concepts, and Hardware

TECHNOLOGIES AND CONCEPTS USED:

The FLAIR system is primarily developed using Python, chosen for its simplicity, readability, and strong ecosystem of libraries that support rapid development and prototyping. Python's versatility makes it ideal for handling both the backend logic (like serial communication and data processing) and the frontend interface (such as plotting and GUI design), thereby enabling a unified development workflow.

To maintain a responsive and smooth user interface while continuously receiving live telemetry data, FLAIR leverages multithreading. This architecture separates data acquisition from UI updates and processing logic. One thread is dedicated to handling real-time data reception from the LoRa module, while other threads manage parsing, updating the GUI, and storing data. This separation ensures that the graphical user interface remains fluid and responsive, even under heavy data throughput conditions.

FLAIR enforces strict sequential validation, processing packets only when received in their original transmission order. If components of earlier packets arrive after subsequent ones—such as Part B of Packet 1 appearing during Packet 2's processing—the system automatically archives these fragments to CSV for post-mission analysis but excludes them from real-time visualization to prevent order-of-operations errors. This ensures live displays exclusively reflect chronologically verified data streams, while the CSV maintains a complete, unaltered record of all received transmissions, including disordered or partial packets. By prioritizing sequential integrity for visualization while preserving raw data fidelity, FLAIR eliminates real-time display inaccuracies during signal interruptions while retaining full telemetry transparency for reconstruction and diagnostics.

For data visualization and GUI development, FLAIR uses a combination of Matplotlib and PyQt5. Matplotlib enables the creation of dynamic, real-time plots for visualizing critical flight parameters. The GUI is built using PyQt's framework and is structured around QMainWindow, QTabWidget, QPushButton, QLabel, and other widgets to give operators full control over the interface. This structured design ensures clarity, ease of use, and flexibility in managing telemetry display.

Telemetry reception from the LoRa module is handled via serial communication, implemented using PySerial. The LoRa module is connected to the ground station computer via a USB port, and PySerial facilitates reading the incoming telemetry packets efficiently. The system also incorporates CSV logging, where all flight data—including timestamps and structured telemetry fields—are stored locally. This log enables post-flight analysis, offline plotting, replay functionality, and serves as an archive of mission data for diagnostics or training purposes.

TECH STACK:

The FLAIR system is developed using Python 3.10+, a modern and robust version of the language that brings in performance improvements, cleaner syntax, and enhanced features like pattern matching—making the codebase more maintainable and efficient. Python is well-suited for telemetry applications due to its ease of use, rich ecosystem, and rapid development capabilities. A variety of libraries are employed to support different aspects of the system: pyserial is used for reliable serial communication with the LoRa module, enabling efficient reading of incoming data through USB-connected interfaces; threading allows the application to run multiple tasks concurrently—such as reading data, processing it, and updating the GUI in real time—without freezing or lagging. matplotlib is used to generate dynamic, real-time plots for critical telemetry metrics, offering intuitive visual insights during flight. pandas is utilized for structured data handling, especially when logging and replaying CSV files, due to its powerful data frame management and easy-to-use syntax for manipulating large datasets.

The user interface (UI) is a custom-built GUI designed using PyQt5, structured with a tabbed interface that allows operators to switch between different views such as live telemetry, data logs, graphs, and system status pages. This design provides a clean and organized experience, enabling operators to quickly find and focus on the data they need during fast-paced mission operations. The system uses a structured telemetry format—specifically, comma-separated values (CSV)—for both real-time data parsing and persistent logging. This format is lightweight, easy to generate and read, and widely compatible with tools like Excel, Google Sheets, and Python libraries like pandas. Using CSV ensures

that each telemetry parameter is clearly segmented and timestamped, facilitating reliable analysis, plotting, and debugging both during and after the mission.

HARDWARE REQUIREMENTS:

The hardware requirements for running the FLAIR telemetry system are minimal yet essential to ensure smooth operation and reliable data handling. At the core of the communication setup is the LoRa module, which is responsible for receiving telemetry data transmitted from the rocket. This module is connected to the ground station via USB or UART, allowing it to interface directly with the laptop or PC running the FLAIR software. The LoRa module serves as the primary link between the rocket's onboard sensors and the ground control system, making it a critical component for real-time monitoring.

DATA FLOW DIAGRAM:

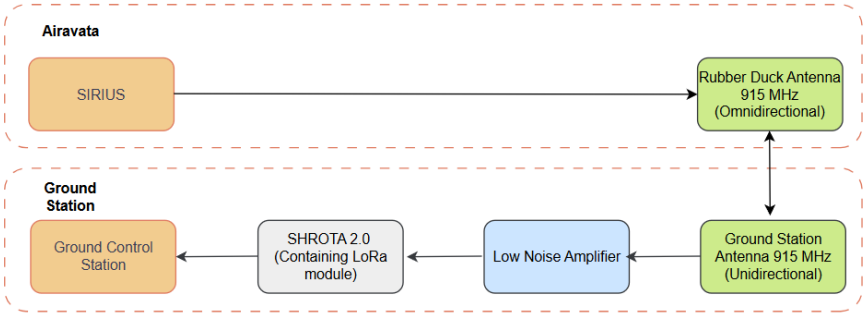


Image 88: Dataflow Diagram

The frequencies for the above-mentioned antennas range between 902-928 Mhz.

To ensure efficient and reliable data transmission, telemetry encoding compresses and formats critical flight parameters into compact packets optimized for LoRa communication. The total data packet contains 28 fields of size 210-220 bytes, carefully structured to balance comprehensive flight state reporting with minimal bandwidth usage. To enhance transmission integrity and reduce the likelihood of packet loss, this 28-field packet is divided into four smaller sub-packets of 7 fields each. These sub-packets are transmitted sequentially, improving reception reliability, especially under fluctuating signal conditions or high-noise environments.

Data Storage Format:
TEAM_ID, TIME, PACKET_COUNT, ALTITUDE, PRESSURE, TEMPERATURE, VOLTAGE, VELOCITY, GPS_SPEED, GPS_LATITUDE, GPS_LONGITUDE, GPS_ALTITUDE, GPS_SATS, GPS_TIME, MAGNETOMETER_X, MAGNETOMENTER_Y, MAGNETOMETER_Z, GYROSCOPE_X, GYROSCOPE_Y, GYROSCOPE_Z, TRIGGER, ANGLE_ROLL, ANGLE_PITCH, ANGLE_YAW, ACCELEROMETER_R, ACCELEROMETER_P, ACCELEROMETER_Y, CURRENT_STATE

Data Transmission Format:
Segment 1 (45 bytes) *: TEAM_ID, TIME, PACKET_COUNT, ALTITUDE, PRESSURE, TEMPERATURE, VOLTAGE*
Segment 2 (49 bytes): *VELOCITY, GPS_SPEED, GPS_LATITUDE, GPS_LONGITUDE, GPS_ALTITUDE, GPS_SATS, GPS_TIME*
Segment 3 (43 bytes): *MAGNETOMETER_X, MAGNETOMENTER_Y, MAGNETOMETER_Z, GYROSCOPE_X, GYROSCOPE_Y, GYROSCOPE_Z, TRIGGER*
Segment 4 (49bytes): *ANGLE_ROLL, ANGLE_PITCH, ANGLE_YAW, ACCELEROMETER_R, ACCELEROMETER_P, ACCELEROMETER_Y, CURRENT_STATE*

A laptop or PC is required to operate the FLAIR software, manage serial communication, visualize real-time telemetry, and log flight data. For this purpose, the ThinkPad P15v Gen3 has been selected as the ground control system (GCS) platform. This machine is well-suited for mission-critical operations, offering excellent performance, durability, and battery life. It can operate for over 8 hours on a full charge, even at maximum brightness. To further ensure visibility and prevent overheating due to environmental exposure, the laptop will be shielded with an umbrella during outdoor operations to protect it from direct sunlight. The FLAIR software itself is lightweight and optimized for efficiency, capable of running smoothly on this system. Running on Windows or Linux, the setup requires Python 3.10 or higher along with essential libraries to support multithreading, real-time plotting, and data logging. These combined hardware and environmental readiness considerations make the system robust, reliable, and portable, perfectly suited for both field testing and active mission control.

IV. Mission Concept of Operations (CONOPS)

On the day of launch, the aerostructures, avionics, and payload sections are assembled concurrently and are integrated into the rocket. The upper section is completed by sliding the deployer into the nose cone and forward couplers, routing shock cords, and inserting the cube-sat and guided recovery unit. The avionics bay, pre-assembled with electronics, is finalized by installing charges and Raptor units, aligning the bay using the camera mount, and sealing the camera hole. The Recovery hardware is connected and verified to be in working order. Following which it is integrated in the upper body-tube. The parachutes are inserted appropriately and the pre-assembled Avionics Bay and Payload Section is integrated to their respective places.
The Motor, assembled prior under the guidance of the Flyer of Record is inserted and locked in place using a UNC Bolt to the Thrust Plate. At the launch pad, the rocket is mounted on the rail and set upright. The avionics and payload systems are armed via master switches, and system readiness is confirmed on our Ground Station GUI. Upon receiving clearance, the ignition wire is connected, and the rocket is declared pad nominal for launch.

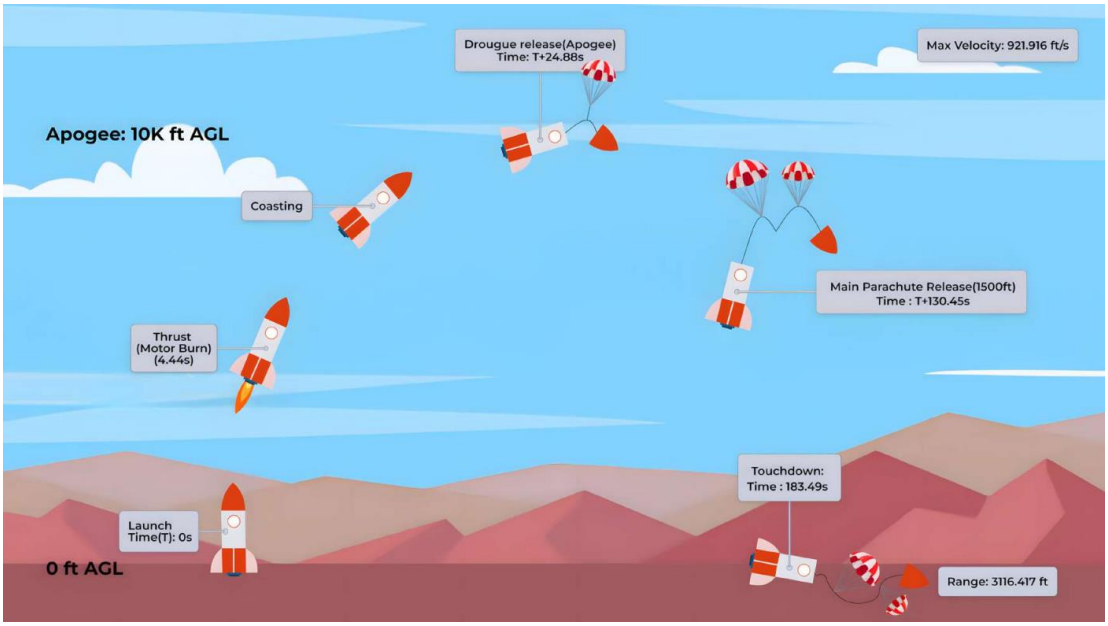


Image 89: CONOPS GRAPHIC

The flight of the rocket is subdivided into 7 mission events, which makes up of the Launch and Recovery Operations. There are multiple subsystems which control and guide the rocket during all its phases.

SYSTEM CATEGORY	SYSTEM NAME	ABBREVIATION	COMPONENTS	FUNCTION
PROPULSION	FLIGHT PROPULSION SYSTEM	FPS	SOLID MOTOR	PROVIDES THRUST FOR FLIGHT
	PROPULSION IGNITION SYSTEM	PIS	ELECTRIC MATCH	INITIATES MOTOR BURN
RECOVERY	VEHICLE RECOVERY SYSTEM	VRS	ALTIMETER, PARACHUTES, RECOVERY	ENSURES SAFE DESCENT AND TRACKING

SOFTWARE	FLIGHT SOFTWARE SYSTEM	FSS	BMP280 DS3231 RTC MS5611 BMI088 LSM6DSL LIS3MDLTR HMC5883L-TR LoRa NEO-M8N UBLOX - for GPS	COLLECTS ALTITUDE DATA
AVIONICS	FLIGHT COMPUTERS	FC	GPS	COLLECTS DATA AND TRANSMITS TO GROUND LEVEL
MECHANICAL SAFE	MECHANICAL SAFE	MS	PULL PIN SWITCH	ENABLES FUNTIONING OF AVIONICS BAY

Table 12: System Definition for CONOPS

IGNITION INITIATION: Launch at T = 0s from 0 ft AGL

ABBREVIATION	ARM STATUS	OPERATION	DESCRIPTION
FPS	ARMED	Pressurized	Solid Rocket Motor is Ignited
VRS	ARMED	Powered	The Recovery System is Active and Ready to Deploy Parachutes
FSS	ARMED	Receiving	Flight Software constantly receiving data
PIS	ARMED	Powered	Initiates Motor Burn
FC	ARMED	Transmitting	Collects Data and Transmits to Ground Station
MS	ARMED	Activated	The Pull Switch is pulled to activate the avionics bay.

Table 13: Ignition Table

THRUST PHASE: Motor burn for 4.44 seconds

ABBREVIATION	ARM STATUS	OPERATION	DESCRIPTION
FPS	COMPLETED	Consuming	The Solid Rocket Propellent is being Consumed during the thrust phase
VRS	ARMED	Powered	The Recovery System is Active and Ready to Deploy Parachutes
FSS	ARMED	Receiving	Flight Software constantly receiving data
PIS	COMPLETED	Utilised	The Ignitor has been utilized prior to ignite motor

FC	ARMED	Transmitting	Collects Data and Transmits to Ground Station
MS	COMPLETED	Activated	Avionics Bay is Active

Table 14: Thrust Phase Table

Coasting Phase: Rocket coasts upward after burnout

ABBREVIATION	ARM STATUS	OPERATION	DESCRIPTION
FPS	COMPLETED	Consumed	The Solid Rocket Propellent is completely consumed during this phase
VRS	ARMED	Powered	The Recovery System is Active and Ready to Deploy Parachutes
FSS	ARMED	Receiving	Flight Software constantly receiving data
PIS	COMPLETED	Utilised	The Ignitor has been utilized prior to ignite motor
FC	ARMED	Transmitting	Collects Data and Transmits to Ground Station
MS	COMPLETED	Activated	Avionics Bay is Active

Table 15: Coasting Phase Table

Apogee (10,000 ft AGL): Reaches maximum altitude

ABBREVIATION	ARM STATUS	OPERATION	DESCRIPTION
FPS	COMPLETED	Consumed	The Solid Rocket Propellent is completely consumed
VRS	COMPLETED PART-1	Deploying Drogue	The Recovery System is Active and has deployed the drogue parachute
FSS	ARMED	Receiving	Flight Software constantly receiving data
PIS	COMPLETED	Utilised	The Ignitor has been utilized prior to ignite motor
FC	ARMED	Transmitting	Collects Data and Transmits to Ground Station
MS	COMPLETED	Activated	Avionics Bay is Active

Table 16: Apogee Table

Main Parachute Deployment (1,500 ft AGL): Main parachute released at T+130.45s

ABBREVIATION	ARM STATUS	OPERATION	DESCRIPTION
FPS	COMPLETED	Consumed	The Solid Rocket Propellent is completely consumed
VRS	COMPLETED	Deploying Main	The Recovery System is Active and has deployed the main parachute
FSS	ARMED	Receiving	Flight Software constantly receiving data
PIS	COMPLETED	Utilised	The Ignitor has been utilized prior to ignite motor
FC	ARMED	Transmitting	Collects Data and Transmits to Ground Station
MS	COMPLETED	Activated	Avionics Bay is Active

Table 17: Main Parachute Table

Descent & Landing: Touchdown at T+183.49s

ABBREVIATION	ARM STATUS	OPERATION	DESCRIPTION
FPS	COMPLETED	Consumed	The Solid Rocket Propellent is completely consumed
VRS	COMPLETED	Utilised	The Recovery System is dormant post parachute deployment
FSS	ARMED	Receiving	Flight Software constantly receiving data
PIS	COMPLETED	Utilised	The Ignitor has been utilized prior to ignite motor
FC	ARMED	Transmitting	Collects Data and Transmits to Ground Station
MS	COMPLETED	Activated	Avionics Bay is Active

Table 18: Decent and Landing Table

Post-Landing Position: Final location range: 3116.417 ft from launch point

ABBREVIATION	ARM STATUS	OPERATION	DESCRIPTION
FPS	COMPLETED	Consumed	The Solid Rocket Propellent is completely consumed
VRS	COMPLETED	Utilised	The Recovery System is dormant post parachute deployment
FSS	COMPLETED	Receiving	Flight Software constantly receiving data
PIS	COMPLETED	Utilised	The Ignitor has been utilized prior to ignite motor
FC	COMPLETED	Transmitting	Collects Data and Transmits to Ground Station
MS	COMPLETED	Activated	Avionics Bay is Active

Table 13: Post Landing Table

V. Conclusion

After the launch of Agneya last year in the Spaceport America Cup 2024, and other previous rocket launches in prior competitions, the team had multiple areas to focus on. Even though the changes in paradigm and way of thinking came with its own set of challenges, Team Sammard's absolute resolve on resilience and learning always had a priority in the minds of the members of the team whilst designing and manufacturing Airavata.

Lessons Learned

1. **Logistics**
   From our previous year's participation with Agneya, one of the key challenges we faced was managing international logistics within tight timelines, which resulted in customs delays and limited flexibility upon arrival. These constraints affected our overall preparation and readiness during the competition.
   In response to these challenges, we have significantly improved our logistics planning for Airavata. This year, our team initiated the logistical process well in advance, allowing us ample time for documentation, packaging, and coordination with freight services. A major step forward in this direction has been our decision to apply for an ATA Carnet for the first time. This international customs document will allow us to temporarily export and re-import our rocket hardware without incurring additional duties or delays. By using the ATA Carnet, we expect to streamline customs procedures both in India and in the United States.
   Additionally, the financial benefits of using the ATA Carnet—by eliminating the need for import taxes and security deposits—enable us to allocate resources more effectively across other critical aspects of the project. To further support logistical efficiency, we implemented a structured checklist system for material procurement, clearly separating high-priority purchases from those that could be deferred. Overall, these improvements ensure a more reliable, efficient, and cost-effective logistics pipeline for Airavata compared to our experience with Agneya.

2. **Payload**
   The payload being developed for launch onboard Airavata marks Team Sammard's third payload project. In our previous rocket Agneya, the team had many innovative payload ideas. However, most of them could not be implemented due to limited resources, high costs in India, and tight development timelines.
   This year, we followed a more structured and research-driven approach. We studied various scientific experiments done by other student teams and research institutions. As part of our R&D, we explored concepts like an Active Stabilized Damping System and an Aerosol Particle Sizing System. While these ideas were not fully implemented this year, the preliminary testing we conducted helped us learn and opened new possibilities for future missions.
   We aimed to design a futuristic and socially relevant payload, with the potential to collaborate with external organizations for advanced research. The integration process was also improved, with a modular payload structure designed for easier assembly and validation.

3. **Mechanical**
   The mechanical design of Airavata reflects several improvements based on the lessons learned from Agneya. The fin thickness was increased from 3 mm to 4.5 mm for better strength, and the material was upgraded from fiberglass to carbon fiber for improved rigidity. A full-body wet layup process was introduced and completed entirely in-house for better quality control. We retained the sectional design from last year as it simplified the assembly. One of the issues we faced in Agneya was the placement of rail buttons, which has now been correctly addressed. The arming system was upgraded from magnetic switches to a master switch, improving safety and accessibility in the avionics bay.
   A significant improvement was the change from a metallic coupler to a fiberglass coupler, manufactured experimentally in-house. The fin can was redesigned to allow easier alignment and installation. The body tube thickness was optimized from 5.5 mm to 3 mm using structural simulations, reducing weight without compromising strength. Finally, the recovery system was made more compact and lightweight by applying thermal analysis to minimize bulk while maintaining performance.

4. Avionics and Computer Science

Significant advancements have been made in the electronics domain for Airavata, driven by the limitations observed in our previous mission. The earlier avionics bay design, though compact, lacked modularity and posed accessibility challenges—particularly with battery housing and direct soldered connections. In the redesigned version, modularity and serviceability were prioritized. PCBs are now arranged to allow easier troubleshooting, while the battery bay uses solderless connectors for simpler replacements. The introduction of crimp connectors has streamlined wire routing and improved overall accessibility. Telemetry performance has also been addressed through refined RF tuning. Impedance matching has been incorporated into both the Bessie 2.0 and Sirius systems to reduce signal loss. Additionally, Low-Noise Amplifiers (LNAs) have been implemented on the ground station receiver to improve signal strength and minimize data corruption, leading to more reliable data acquisition.

In recovery electronics, the Sirius flight computer introduces notable upgrades. Comparators have been added for precise state recognition in the pyro channels, while MOSFETs are now used for pyro triggering. Continuity state indicators have been included to provide visual confirmation of channel readiness. Unlike previous years, this cycle emphasized early testing and sensor integration. To address the sensor data inconsistencies experienced during Agneya, sensor fusion was implemented in Sirius, improving the accuracy and reliability of flight data.

5. Miscellaneous

We maintained detailed documentation throughout the development of Airavata, enabling us to track progress, manage pending tasks, and maintain alignment across subsystems. This has enhanced overall accountability and workflow clarity.

To support efficient collaboration, we instituted regular interdepartmental meetings for synchronized timelines and transparent communication. Furthermore, recognizing the need for extensive testing of multiple subsystems, we secured early permissions for component testing, including access to labs during extended hours. Finally, Airavata placed a stronger emphasis on in-house R&D, leading to developments like an antenna tracking system, SRAD antenna, and an indigenously built rocket motor. These efforts not only improved our technical capabilities but also enriched our team's SRAD experience. A major step forward has also been the streamlining of project timelines by respecting the unique resource and schedule requirements of each domain, enabling focused and efficient progress.

VI. Acknowledgement

We would like to express our heartfelt gratitude to Dr. Senthil Kumar M, Faculty Coordinator of Team Sammard, for his constant support, guidance, and mentorship throughout our journey to the International Rocket Engineering Competition (IREC) 2025. His belief in our potential and constant encouragement played a pivotal role in shaping our project and pushing the team toward excellence.

We also extend our sincere gratitude to Vellore Institute of Technology (VIT), our academy, for providing us with the infrastructure, resources, and support that made this ambitious endeavor possible. We would especially like to thank Dr. Naiju C D, Director of Students' Welfare, and Dr. Rajasekhara Reddy Mutra, Assistant Director of Students' Welfare, for their continued support in facilitating our work by providing us with dedicated space and ensuring we received the necessary permissions to carry forward our project. The university's unwavering commitment to innovation and student-driven research has greatly supported and inspired teams like ours to dream big, push boundaries, and bring engineering ideas to life.

We would also like to extend our heartfelt gratitude to Dr. Anil Kumar Gourishetty and Mr. Vivek Anand from IIT Roorkee for their invaluable guidance and support in the development and manufacturing of our muon detection scintillator, one of the key payloads of our rocket. We are also thankful to Nikhil Uppin from Proact IMS and to Endeavour Engineering for their generous support in providing us with cost-effective materials, which significantly contributed to the overall development of our rocket.

We would also like to extend our sincere gratitude to Convergent Science India LLP, our Platinum Sponsor, for providing us with access to their powerful simulation software. This tool played a pivotal role in allowing us to carry out advanced Computational Fluid Dynamics (CFD) analyses, which were essential for understanding the complex aerodynamic and thermodynamic behaviors of our rocket under various flight conditions. These insights significantly enhanced the accuracy and reliability of our design decisions, particularly in the development of our propulsion system, which is still underway. Their contribution has not only supported our technical progress but has also enriched our learning experience as aspiring engineers.

We would also like to thank Mr. Edward Samuel, Senior Manager Purchase at VIT, and Mr. Avinash from Flomic Global Logistics for their constant and prompt support in facilitating the transfer and shipment of our rocket from India to the USA for the competition. Their assistance played a vital role in ensuring the timely and safe delivery of our project.

We are especially thankful to Mr. Robert Dickinson for serving as our Flyer of Record. His critical insights and steadfast support throughout the review and certification process were instrumental in ensuring our compliance with launch and safety protocols.

Finally, we would like to thank the Experimental Sounding Rocket Association (ESRA) for organizing IREC and fostering a global platform for aspiring rocketry teams. The opportunity to participate has been both challenging and immensely rewarding, allowing us to engage deeply with the global aerospace community.

Appendix A.
Systems Weights, Measures and Performance Data

A) Rocket Information

Parameter	Value	Units
Vehicle Length	2820	mm
Airframe Diameter	140	Mm
Vehicle Weight	23.7	Kg
Payload Weight	2	Kg
Liftoff Weight	25.7	Kg
Centre of Pressure	2010	mm
Centre of Gravity	1721	mm
Number of Fins	4	
Fin Height	109	mm
Root Chord	195	mm
Tip Chord	70	mm
Fin Thickness	4	mm

B) Propulsion Information

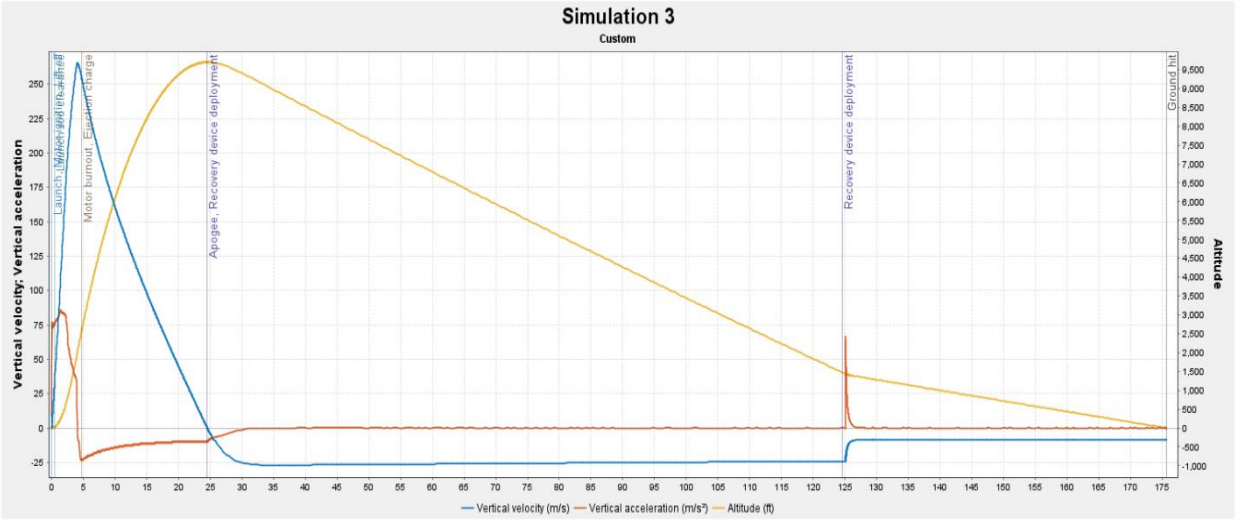
Parameter	Value	Units
Type	Solid	
COTS / SRAD	COTS	
Class	M Class	
Manufacturer	Aerotech	
Designation	M1845-NT PS	
Motor Length	597	mm
Casing Diameter	98	mm
Empty Motor Case Mass	2,910	grams
Propellant Mass	3,772	grams
Motor Wet Mass	6,682	grams
Impulse of Motor	8,307	Newton seconds
Specific Impulse	225	Newton seconds
Average Thrust	1,875	Newton
Maximum Thrust	3,081	Newton
Burn Time	4.4	Seconds

C) Flight Data Information

Parameter	Value
Launch Rail Length	5600m
Lift off Thrust-to-weight ratio	12.1
Launch Rail Departure Velocity	30
Static Margin	1.95
Maximum Acceleration	9.05 m/s^2
Maximum Velocity	271 m/s
Fin Flutter Velocity	1.59 mach
Target Apogee	10000 ft

Predicted Apogee	9640 ft

D) Flight Profile Graph



E) Recovery Information

Parameter	Value
Primary Altimeter	Missileworks RRC3
Redundant Altimeter	Featherweight Raven
Drogue primary deployment charge	12g
Drogue redundant deployment charge	10g
Main primary deployment charge	7g
Drogue deployment altitude	3048m (10000ft)
Drogue descent rate	25 m/s
Main deployment altitude	452m (1500ft)
Main descent rate	8.5 m/s

Appendix B.
Project Test Reports

Test I:

Recovery Testing 1

Test Lead: Vishvak Narayanan

Test Date: March 26, 2025

Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Vishvak Narayanan Shivam Halan Dhruv Murarka Joseph Thomas Sumit Ranjan Guru Mayukhi Sengupta Sanjay Ganesh	Recovery mechanism SRAD e-matches Batteries	SJT Ground, VIT Vellore	Partial Success

Test Purposes:

The test was conducted to thoroughly evaluate and ensure the proper functioning, reliability, and responsiveness of the recovery mechanism during deployment. This was done under controlled conditions to simulate actual operational scenarios and to verify that all components performed as expected throughout the recovery sequence.

Desired Result:

The recovery mechanism should open completely and reliably because of the black powder explosion, ensuring that all deployment components are fully released and that the recovery system activates without obstruction.

Test Procedure:

1. The SRAD e-matches are securely installed within the recovery mechanism.
2. A packet of black powder is then placed inside the recovery system.
3. The base, through which the wiring passes, is sealed using M-seal to prevent any leakage.
4. The e-match is manually triggered by supplying voltage from the battery.

Test Observation:

The recovery mechanism failed to deploy fully due to incomplete combustion of the black powder charge, tolerance problems and gas leakage from the containment area. This resulted in insufficient pressure generation, causing only partial actuation of the mechanism and preventing the recovery system from functioning as intended.

Project Lead Comments:
Failure of this test was primarily due to improper machining of our Recovery Mechanism. New Recovery mechanism must be machined.

Additional Photographs:





Test II:

Recovery Testing 2

Test Lead: Vishvak Narayanan

Test Date: March 31, 2025			
Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Vishvak Narayanan Shivam Halan Dhruv Murarka Sumit Ranjan Guru Mayukhi Sengupta Sanjay Ganesh	Recovery mechanism SRAD e-matches Batteries	SJT Ground, VIT Vellore	Partial Success

Test Purposes:

The test was conducted to thoroughly evaluate and ensure the proper functioning, reliability, and responsiveness of the recovery mechanism during deployment. This was done under controlled conditions to simulate actual operational scenarios and to verify that all components performed as expected throughout the recovery sequence.

Desired Result:

The recovery mechanism should open completely and reliably because of the black powder explosion, ensuring that all deployment components are fully released and that the recovery system activates without obstruction.

Test Procedure:

1. The SRAD e-matches are securely installed within the recovery mechanism.
2. A packet of black powder is then placed inside the recovery system.
3. The base, through which the wiring passes, is sealed using M-seal to prevent any leakage.
4. The e-match is manually triggered by supplying voltage from the battery.

Test Observation:

The recovery mechanism failed to deploy fully due to gas leakage or incomplete pressure generation which caused only partial actuation of the mechanism and preventing the recovery system from functioning as intended.

Project Lead Comments:
Recovery Mechanism must be optimized more to avoid gas and pressure leakage. Design needs to be worked on and verified better.

Additional Photographs:



Test III:

Recovery Testing 3

Test Lead: Shivam Halan

Test Date: April 6, 2025			
Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Shivam Halan Dhruv Murarka Sanjay Ganesh Aditya Santosh	Recovery mechanism SRAD e-matches Batteries	SJT Ground, VIT Vellore	Success

Test Purposes:

The test was conducted to thoroughly evaluate and ensure the proper functioning, reliability and responsiveness of the recovery mechanism during deployment. This was done under controlled conditions to simulate actual operational scenarios and to verify that all components performed as expected throughout the recovery sequence.

Desired Result:

The recovery mechanism should open completely and reliably because of the black powder explosion, ensuring that all deployment components are fully released and that the recovery system activates without obstruction.

Test Procedure:

5. The SRAD e-matches are securely installed within the recovery mechanism.
6. A packet of black powder is then placed inside the recovery system.
7. The base, through which the wiring passes, is sealed using M-seal to prevent any leakage.
8. Some weight is put over the recovery mechanism.
9. The e-match is manually triggered by supplying voltage from the battery.

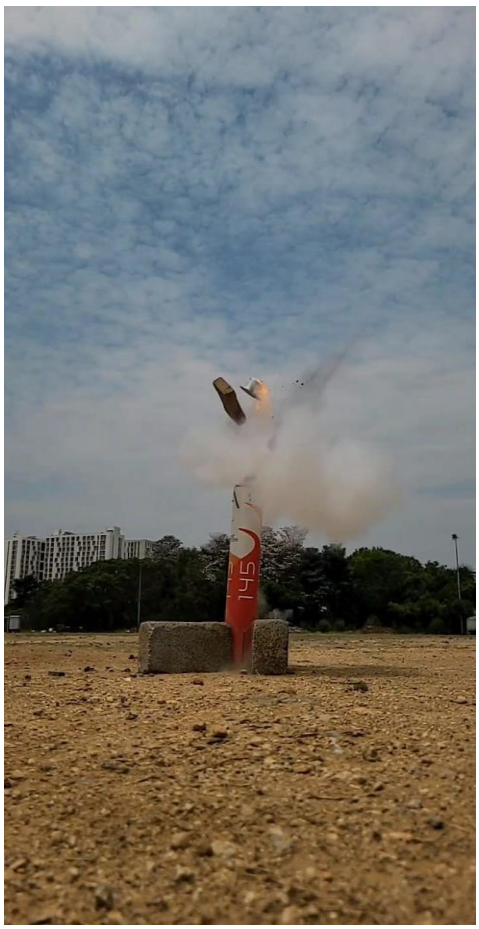
Test Observation:

The recovery mechanism deployed fully and operated as designed, indicating successful ignition of the black powder charge, proper pressure buildup, and effective mechanical release. All components of the system functioned in accordance with the intended recovery sequence.

Project Lead Comments:
Proper functioning of system has been verified.

Additional Photographs:





Test IV:
Main Parachute Testing

Test Lead: Simone Framroze

Test Date: April 4, 2025

Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Om Unnithan Simone Framroze Sumit Guru Joseph Thomas	Main Parachute Shroud Lines Body tubes	VIT Vellore	Successful

Test Purposes:

The shroud lines and the parachute canopy were carefully packed together and stored inside a body tube, simulating the configuration of a compact ejection system. This setup was designed with the intention that the parachute would deploy effectively upon the application of a sudden force, thereby mimicking the dynamics of ejection mechanism used in our rockets.

Desired Result:

The parachute should inflate properly upon deployment, ensuring that the canopy fully opens and fills with air to create the necessary drag force.

Test Procedure:

1.The parachute and its shroud lines are neatly folded and packed inside the body tube as a single unit.
2.A person initiates the deployment by running, applying a sharp pulling force to the shroud lines.
3.This sudden tension, combined with the motion generated by the runner, causes the parachute canopy to deploy and inflate with air.

Test Observation:

Upon deployment, the parachute successfully achieved full inflation, with the canopy expanding completely and effectively capturing the surrounding air. This optimal inflation was crucial in generating the necessary aerodynamic drag to significantly reduce the descent velocity. As a result, the system experienced improved stability during its descent phase, ultimately leading to a smooth, controlled, and safe landing. The proper functioning of the parachute also demonstrated the reliability of the deployment mechanism and the effectiveness of the overall recovery system.

Project Lead Comments:
Parachutes have been deployed properly, giving assurance on the design of the system and its ability to deploy without hinderance.

Test V:
Main Parachute Opening Test

Test Lead: Simone Framroze

Test Date: February 9, 2025			
Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Om Unnithan Simone Framroze Sumit Guru Joseph Thomas	Main Parachute Shroud Lines	VIT Vellore	Successful

Test Purposes:

The shroud lines and the parachute canopy were carefully packed together in a compact configuration, designed to deploy upon the application of a sudden force. This setup was intended to replicate the mechanism of an ejection system

Desired Result:

The parachute should inflate properly upon deployment, ensuring that the canopy fully opens and fills with air to create the necessary drag force.

Test Procedure:

1.The parachute and its shroud lines are neatly folded and stored together as a single unit.
2.A person initiates the deployment by running, applying a sharp pulling force to the shroud lines.
3.This sudden tension, combined with the motion generated by the runner, causes the parachute canopy to deploy and inflate with air.

Test Observation:

The parachute successfully inflated upon deployment, with the canopy fully expanding and capturing air as intended. This proper inflation ensured the generation of sufficient drag to slow down the descent, stabilize the system, and allow for a smooth and controlled landing.

Project Lead Comments:
Success, nothing of significance to note.

Additional Photographs:



Test VI

Tender Descender Testing

Test Lead: Vishvak Narayanan

Test Date: April 6, 2025			
Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Vishvak Narayanan Shivam Halan Dhruv Murarka Sanjay Ganesh Aditya Santosh	Tender descender SRAD e-matches Batteries	SJT Ground, VIT Vellore	Successful

Test Purposes:

The Tender Descender plays a critical role in the dual deployment recovery system of high-powered rockets. Upon receiving the signal, the ignition of electric matches triggers a black powder charge, producing a controlled explosion. This explosive force is precisely calculated to fully activate and open the Tender Descender mechanism. Once deployed, it allows for the sequential release of the drogue and main parachutes, ensuring a stable and safe descent of the rocket. Its reliable performance is essential to the success of the recovery operation, minimizing the risk of damage upon landing.

Desired Result:

Both the sections of the tender descender should separate completely and detach from each other for the deployment of the main parachute.

Test Procedure:

1.The SRAD e-match is inserted into the lower cup and sealed using M-seal.
2.The top and bottom cups are secured together using shear pins.
3.Tension is applied to the assembly from both sides
4.The e-match is ignited by manually applying voltage from a battery.

Test Observation:

The tender descender separation mechanism functioned successfully, as both halves of the assembly separated as intended upon activation. This confirmed the effectiveness of the separation system in initiating the deployment sequence and ensured that the payload components could continue their descent without obstruction. The clean and reliable separation also validated the design and timing of the mechanism, which is critical for the overall success of the recovery process.

Project Lead Comments:
The tender descenders reliable design has been verified to work ideally with the estimated amount of black powder. Giving us confidence on its operation in real time.

Additional Photographs:



Test VII

LoRa testing with Yagi UDA

Test Lead: Shawn SJ

Test Date: December 19, 2024			
Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Sanjith SNN Shawn SJ	Shrota (SRAD LoRa) Tx Shrota (SRAD LoRa) Rx Yagi UDA Laptop Mobile	VIT Vellore Campus	Failure

Test Purposes:

To determine the maximum communication range achievable using the RFM-95W LoRa module paired with a Yagi antenna.

Desired Result:

To successfully receive LoRa packets at the farthest possible distance, thereby identifying the maximum effective range of the system.

Test Procedure:

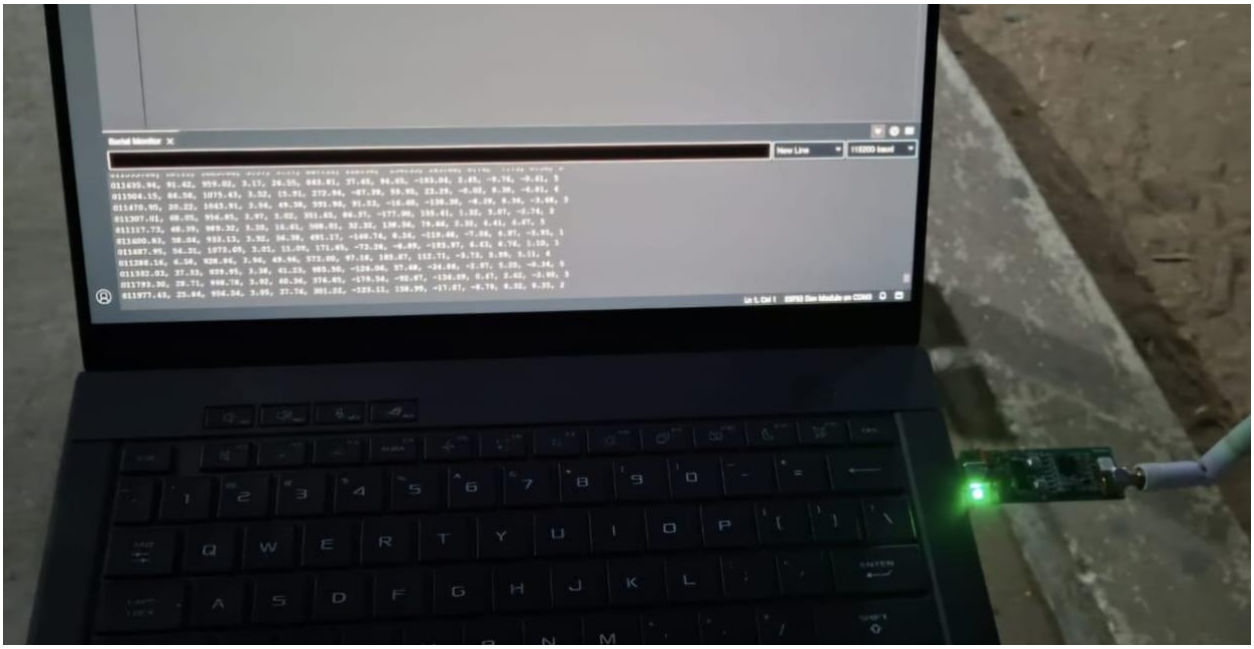
1. Configure two Shrota LoRa modules—one as a transmitter and the other as a receiver.
2. Connect the transmitter to a smartphone and the receiver to a laptop for monitoring.
3. Attach Yagi antennas to both modules to improve signal directionality and strength.
4. Move the transmitter incrementally away from the receiver and monitor for packet reception.
5. Record the distance at which packet reception ceases.

Test Observation:

The system maintained consistent communication up to approximately **750–800 meters**. Beyond this range, no packets were received, establishing the effective operational limit with the current configuration.

Project Lead Comments:
The limited range gave us a realization on the quality of hardware existing to receive data from rocket. New Yagi Antennas must be looked into or other ways to increase gain of existing antenna must be looked into to increase range.

Additional Photographs:



TEST VIII

RRC3 Testing

Test Lead: Karun Joseph

Test Date: December 19, 2024			
Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Arudhran K. Karun Joseph	RRC3 9V battery 8V pyrotechnic battery mDACS	VIT Vellore Campus	Successful

Test Purposes:

To verify the functionality of the RRC3 module in igniting SRAD e-matches using pyrotechnic batteries, triggered through simulated commands from the mDACS software.

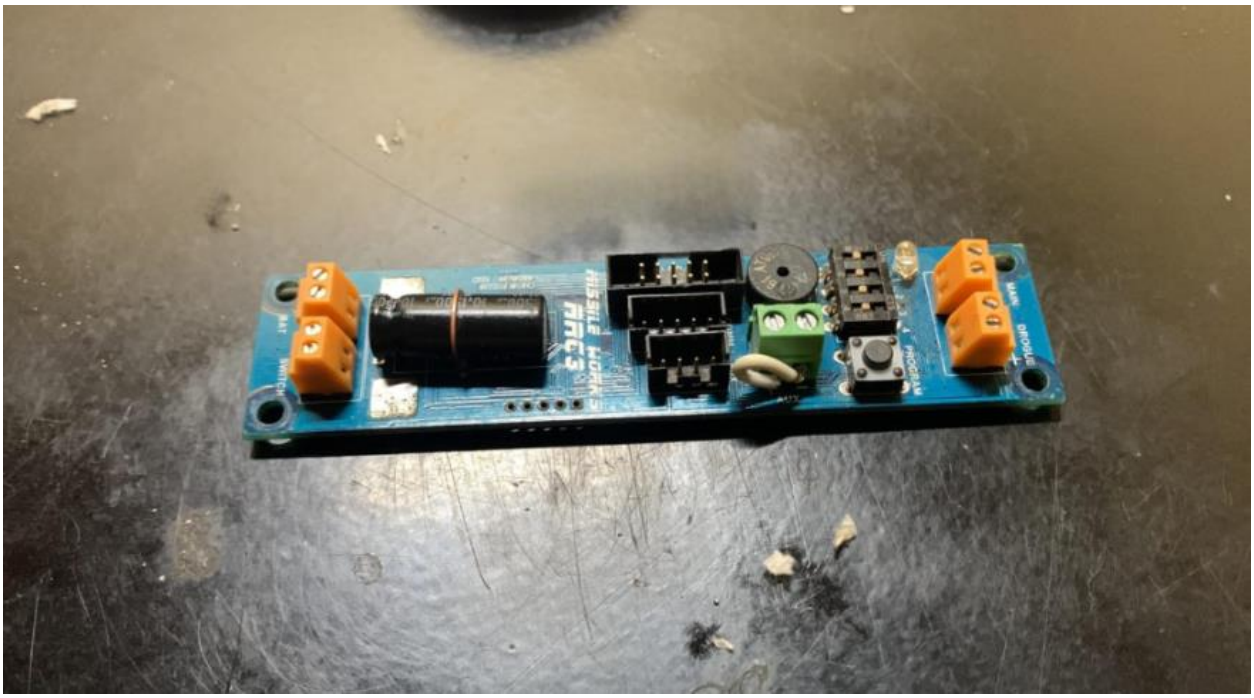
Desired Result:

The RRC3 should successfully receive firing signals from the mDACS simulation and activate the pyrotechnic output to ignite the e-matches.

Test Procedure:

1. Set up the RRC3 with a 9V battery and connect it to the SRAD e-matches.
2. Integrate the system with the mDACS simulation environment.
3. Initiate the firing sequence through mDACS software.
4. Monitor the RRC3's response and e-match activation.

Test Observation:
The RRC3 responded correctly to the simulated firing command from the mDACS software. The pyrotechnic output successfully ignited the SRAD e-matches, confirming the system's readiness for deployment.
Project Lead Comments:
Proper functioning of system observed. Nothing of significance to note.



Test IX

Raven4 Testing

Test Lead: Arudhran K

Test Date: December 19, 2024			
Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Arudhran K. Karun Joseph	Raven4 9V battery Featherweight wFIP 2.0 e-matches	VIT Vellore Campus	Successful

Test Purposes:

To validate the operation of the Raven4 module in triggering SRAD e-matches using a 9V battery, controlled via simulated firing signals from the Featherweight wFIP 2.0 software.

Desired Result:

The Raven4 should correctly interpret the simulated command from the wFIP 2.0 software and deliver sufficient output to ignite the connected e-matches.

Test Procedure:

1. Connect the Raven4 to a 9V battery and interface it with SRAD e-matches.
2. Configure and run the Featherweight wFIP 2.0 simulation environment.
3. Issue the firing command via the software.
4. Observe the module's behaviour and check for ignition of the e-matches.

Test Observation:

The Raven4 successfully received the firing command from the Featherweight wFIP 2.0 simulation and ignited the SRAD e-matches as expected, confirming full functionality

Project Lead Comments:
Proper functioning of system observed. Nothing of significance to note.

Test X

Featherweight GPS Tracker Testing

Test Lead: Arudhran K.

Test Date: December 19, 2024			
Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Arudhran K.	Featherweight GPS Tracker Featherweight Ground Station 3.7V battery Featherweight User Interface Phone App	VIT Vellore Campus	Successful

Test Purposes:

To verify the functionality of the Featherweight GPS Tracker in transmitting real-time GPS location data to the Featherweight Ground Station and displaying it on the Featherweight User Interface Phone App.

Desired Result:

The GPS Tracker should successfully communicate its positional data to the ground station, and the data should be accessible on the mobile app interface within a reasonable operational range.

Test Procedure:

1. Power the Featherweight GPS Tracker using a 3.7V battery.
2. Set up the Featherweight Ground Station and pair it with the GPS Tracker.
3. Connect the ground station to the Featherweight User Interface Phone App.
4. Move the GPS Tracker to increasing distances and monitor GPS data transmission to the app.

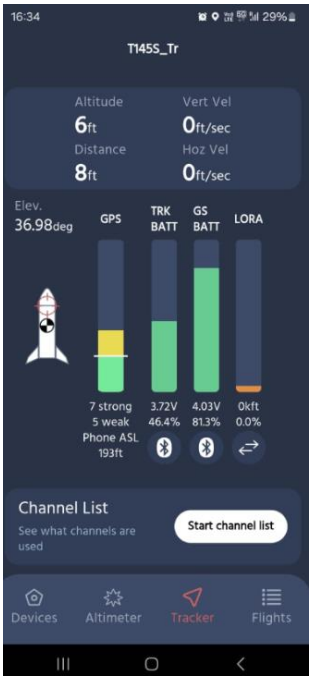
Test Observation:

The GPS data was successfully transmitted from the Featherweight GPS Tracker to the Ground Station and displayed on the mobile app. Data reception was confirmed up to a distance of 900 meters, validating effective communication and tracking range.

Project Lead Comments:
Range is still a glaring issue but GPS functionality has been verified. Range improvement techniques must be continued to looked into.

Additional Photographs:





TEST XI

Battery Drain Test (COTS)

Test Lead: Robin Phillip

Test Date: December 19, 2024			
Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Robin Philip Khyati Kejriwal Arudhran K.	RRC3 Raven4 Featherweight GPS Tracker 9V battery * 2 3.7V battery	VIT Vellore Campus	Successful

Test Purposes:

To evaluate the battery consumption of COTS modules (RRC3, Raven4, and Featherweight GPS Tracker) over a 2-hour continuous operation period and determine their power efficiency for mission readiness.

Desired Result:

Each module should maintain sufficient operational voltage after 2 hours, with minimal voltage drop to ensure reliability during actual deployment.

Test Procedure:

1. Power RRC3 and Raven4 modules with separate 9V batteries.
2. Power the Featherweight GPS Tracker with a 3.7V battery.
3. Leave all three systems running under idle/standard test conditions for 2 continuous hours.
4. Measure and record the voltage levels of each battery before and after the test.

Test Observation:

RRC3: Initial 9V → Final 7V (2V drop)
Raven4: Initial 9V → Final 4V (5V drop)
Featherweight GPS Tracker: Initial 3.7V → Final 2.7V (1V drop)
All modules remained operational throughout the test. However, Raven4 showed a significantly higher voltage drain, indicating the need for more frequent battery checks or a higher-capacity power source during extended missions.

Project Lead Comments:
Power source needs to be improved and possibly a resource management system can be explored upon.

TEST XII

LoRa multiple frequency test

Test Lead: Khyati Kejriwal

Test Date: December 19, 2024			
Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Khyati Kejriwal Karun Joesph	Shrota (Tx) Shrota (Rx) Laptop	VIT Vellore Campus	Successful

Test Purposes:

To validate the operation of two independent LoRa transmitter-receiver (Tx-Rx) pairs functioning on adjacent frequencies (915 MHz and 916 MHz), and assess potential interference or signal contamination when antennas are placed in close proximity—simulating their actual configuration in the avionics bay.

Desired Result:

Each LoRa Tx-Rx pair should communicate only with its designated counterpart without any cross-talk or garbage data being received, ensuring stable multi-frequency operation in confined spaces.

Test Procedure:

1. Set up two Shrota LoRa transmitters on separate frequencies: 915 MHz and 916 MHz.
2. Pair each transmitter with a corresponding receiver on the same frequency.
3. Position the two transmitter-receiver pairs in close physical proximity to each other, simulating avionics bay placement.
4. Use a laptop to monitor serial outputs for any interference or invalid/garbage values on either receiver.

Test Observation:

Both LoRa pairs communicated successfully on their respective frequencies. No garbage data, cross-talk, or interference was observed despite the close placement of antennas. The setup confirmed reliable simultaneous operation of multiple LoRa links in the same environment.

Project Lead Comments:
Nominal functionality of system has been verified.

Test XII

Range Test

Test Lead: Aditi Sharma

Test Date: February 12 2025			
Participant List:	**Material List:**	**Test Location:**	**Test Result:**
Aditi Sharma Saksham Singh Robin Phillip Yash Poddar	Lora (rfm95w) Esp32 915 MHz dipole rubber duck antenna	**From:** Naipaul avenue, VIT Rd, near TT building, Katpadi, Vellore, Tamil Nadu 632014] **To**: SJT, Katpadi, Vellore, Tamil Nadu 632014	Failure
Test Purpose:			
The purpose of this test is to evaluate the maximum achievable range of the LoRa communication system operating at 915 MHz and determine if it meets the required 4000m range. The test assesses signal quality by measuring SNR and RSSI at different distances while analysing interference and connection stability. It also evaluates the performance of the Rubber Duck (transmitter) and Rubber Duck (receiver) antennas.			
Desired Result:			
The desired result is to achieve a stable and reliable LoRa communication range of 4000m with minimal signal degradation. The connection should maintain high SNR and low RSSI, ensuring strong signal quality without significant interference or dropouts. The system should demonstrate consistent performance across the entire range, enabling effective long-distance communication between the transmitter (Rubber Duck antenna) and receiver (Rubber Duck antenna) at 915 MHz			
Test Procedure:			

Hardware Setup:

- Upload LoRa transmit code to one transmitter module and connect it with a Rubber Duck antenna.
- Upload LoRa receiver code to another Shrota module and connect it with a Rubber Duck antenna.
- Ensure both modules are operating at 915 MHz

Test Execution:

- Establish a connection between the transmitter and receiver.
- Gradually increase the separation distance between the two modules while monitoring signal strength.
- Record values of SNR (Signal-to-Noise Ratio) and RSSI (Received Signal Strength Indicator) at different distances.

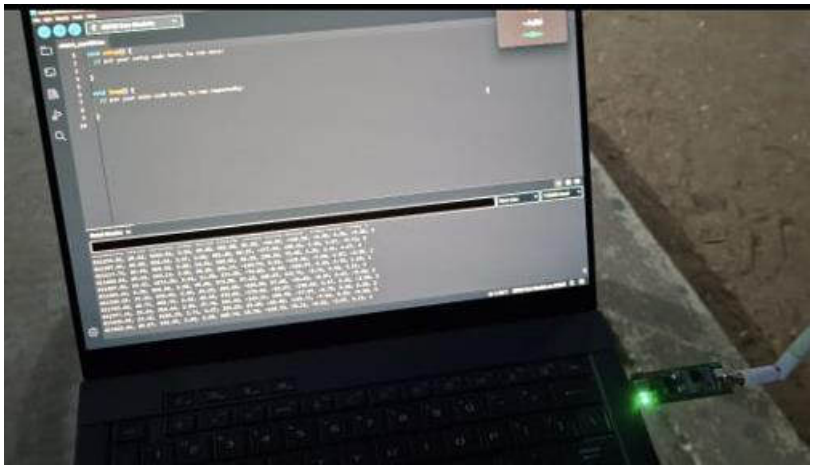
Test Observations:

1. The maximum range achieved was 700m, significantly lower than the required 4000m.
2. Beyond 800m, heavy noise interference was observed for a few meters, followed by a complete loss of connection.
3. Throughout the test, low SNR and high RSSI values were recorded, indicating a poor signal quality.

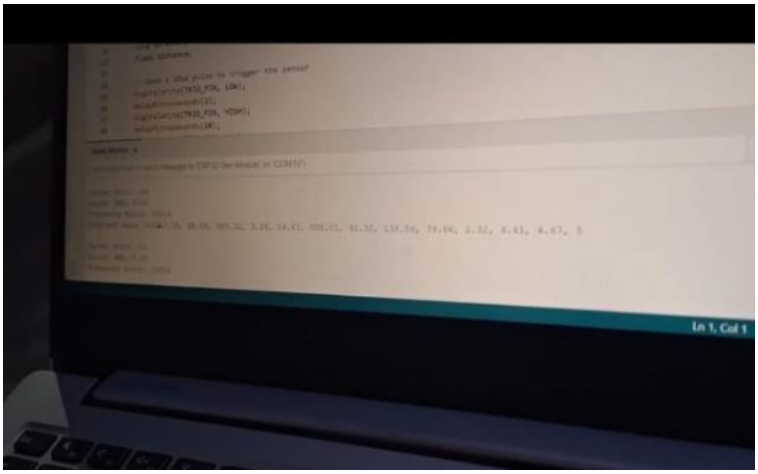
Project Lead Comments:

Range issue needs to be fixed still, all current improvement methods have been proven to be a failure.

Additional Photographs:



Transmitter End
(Sends data wirelessly using a
Rubber Duck antenna at 915 MHz)



Receiver End
(Receives data using a rubber duck
antenna for better signal
gain and directionality.)

Appendix C.
Hazard Analysis

This appendix outlines the identification and preliminary assessment of hazards associated with the design, manufacturing, transportation, and launch phases of Airavata. A structured risk analysis approach is employed, beginning with the formulation of a preliminary Hazard Scale (HS) to quantify risks in terms of individual safety, operational integrity, and potential damage to property. This scale provides a standardized framework to evaluate the severity and likelihood of each hazard. Utilizing this framework, a comprehensive Hazard Matrix is developed, detailing the nature of each hazard, its potential causes, the assigned risk level based on the HS, and the corresponding mitigation strategies. This process ensures early recognition and management of risks, enabling proactive safety planning and reinforcing the robustness of mission-critical systems throughout the lifecycle of Airavata.

Table [No] Hazard Scale

Risk	Severity
Minimal	Possess minimal hazard to personnel and property.
Low	Possess low hazard and a possibility for minor injury.
Intermediate	Possess intermediate hazard to personnel and property.
High	Possess high hazard to personnel and a slight possibility to sustain a critical injury or death.
Severe	Possess severe hazard to personnel and a significant possibility to sustain a critical injury or death.

Table [No] Hazard Matrix

Hazard	Possible Cause	Hazard Level	Mitigation Approach
General Safety Guidelines	Any inexperienced team member utilizing any of the following without proper training or practice: i) Equipment ii) Tools iii) Materials iv) Electronics	Minimal to Severe	Be aware of your own actions and the actions of others. i) Only handle any of the aforementioned articles with proper training and practice. i) Point out risks and mitigate them. ii) Review procedures before beginning potentially dangerous actions. Safety Equipment: i) Only wear hardened, sturdy shoes in the laboratory

Hazard	Possible Cause	Hazard Level	Mitigation Approach
			ii) Always wear safety glasses wherever possible iii) Always use respirators, i.e. face masks, respirators, etc., wherever possible
Ignitor related hazards	Improper handling and storage of the ignitor	Minimal	Igniter has to be stored in a cool, dry place and within its proper packaging until the final loading happens into the motor. Only experienced personnel must be authorised to handle it.
Exposure to epoxy	Improper or non utilization of PPE during Lay-Up or storage.	Low (Irritation of skin, eyes, and respiratory tract by contact and/or inhalation of hazardous vapors, sensitization of the skin)	Use nitrile or latex gloves when handling epoxy, and only use it in a well ventilated area. If epoxy gets in your eyes, flush them immediately with water. Wash any skin that has encountered epoxy to avoid sensitization and irritation. If excessive fumes are inhaled, call the nearby hospital and take the patient to an ambulance.
Spray Paint related hazards	Improper utilization of PPE while using Spray Paint	Low to Intermediate (Irritation to eyes and respiratory system from contact and/or inhalation of hazardous vapors)	Use spray paint only in a well ventilated area. If spray paint gets in your eyes, flush immediately with water. If excessive vapors are inhaled, call a nearby hospital and transport the patient to an ambulance.
Fiberglass related hazards	Improper or non utilization of PPE whilst handling or storage.	Low to High (Irritation of the skin, eyes, nose, throat, respiratory tract, and digestive system from contact and/or inhalation of particles. Miniature cuts to	Use gloves and wear long sleeves when handling fiberglass. Cover mouth and nose with an examination mask or respirator and wear safety glasses when cutting. If fiberglass particles get in your eyes, flush with water immediately. Wash any skin that has come in contact with fiberglass particles to prevent irritation. If excessive particles

Hazard	Possible Cause	Hazard Level	Mitigation Approach
		body parts after Lay-Up.)	are inhaled, call a nearby hospital and transport the patient to an ambulance.
Carbon fibre related hazards	Improper or non utilization of PPE whilst handling or storage.	Low to High (Irritation of the skin, eyes, nose, throat, respiratory tract, and digestive system from contact and/or inhalation of particles. Miniature cuts to body parts after Lay-Up.)	Use gloves and wear long sleeves when handling carbon fiber. Cover mouth and nose with an examination mask or respirator and wear safety glasses when cutting.Wash any skin that has come in contact with carbon fiber particles to prevent irritation. If excessive particles are inhaled, call a nearby hospital and transport the patient to an ambulance.
Motor Adhesive	Improper utilization of PPE while handling and motor integration.	Intermediate (Skin/eye irritation)	Use gloves and wear long sleeves when handling the adhesive. OSHA and manufacturer guidelines must be adhered to. Cover mouth and nose with an examination mask or respirator..Wash any skin that has come in contact with the adhesive to prevent irritation. If excessive irritation occurs, call a nearby hospital and transport the patient to an ambulance.
Dremel	Improper utilization of the Dremel itself and non utilization of PPE during its use.	Intermediate to Severe (Burns and lacerations)	Avoid burns and lacerations by wearing long sleeves and gloves when operating a Dremel. Always wear eye protection. Keep a first aid kit ready to treat burns and lacerations. Flush any burns immediately with cool water. Depending on the level of the emergency, call a nearby hospital and transport the patient to an ambulance.
Drills	Improper utilization of the Drill itself and	Intermediate to Severe	Avoid lacerations by wearing long sleeves and gloves when

Hazard	Possible Cause	Hazard Level	Mitigation Approach
	non utilization of PPE during its use.	(Lacerations)	operating a Drill. Keep precaution by staying away from the drill bit and drill only on a steady surface. Always wear eye protection. Keep a first aid kit ready to treat burns and lacerations. Flush any burns immediately with cool water. Depending on the level of the emergency, call a nearby hospital and transport the patient to an ambulance.
Recovery Failure	Non-Deployment of recovery due to failure of altimeter or poor packing of the parachute.	Intermediate to Severe (Blunt Force Trauma Injuries or Cuts/Lacerations during recovery)	Keep aware of your surroundings and have basic presence of mind. The altimeter must be calibrated properly and ensured to be in working order, and the parachute will be packed properly. In case of any injuries call a nearby hospital and transport the patient to an ambulance.
Black Powder related hazards	Improper care demonstrated while handling black powder and non utilization of PPE during handling, storage and recovery testing.	Severe (Burns or blunt force trauma injuries)	Avoid damaging lungs by wearing appropriate face masks and respirators. Fire Extinguisher should be always available. Use of Proper gloves and lab coats to avoid contact with the skin. Depending on the level of the emergency, call a nearby hospital and transport the patient to an ambulance.
Propellent	Unwanted and unauthorised handling by team members without having pre-requisite experience and without utilizing proper safety precautions.	Severe (Burns, blunt force trauma from shock wave or engine case impact, lacerations from shrapnel)	Extreme care must be taken when handling the engine. Only trained personnel should handle and assemble the engine. The detonator will remain sealed and away from the engine until the vehicle is properly mounted on the rail and all personnel, except essential, have left the area. Ensure that the ignition circuit

Hazard	Possible Cause	Hazard Level	Mitigation Approach
			is open until all personnel have moved to the designated safety area. The engine will not be assembled until it is ready to be loaded into the body of the vehicle. Until then, it will remain in a cool, dry place in a sealed and locked container away from any flame or heat source. A first aid kit with bandages and burn treatment options must always be handy when handling the engine. 5 gallons of water and a fire extinguisher must be kept in reserve to extinguish fires and treat burns. All personnel should be trained in the treatment of burns and cuts. Should an emergency occur, immediately alert the nearby hospital while first aid is administered to the victim

Appendix D.
Risk Assessment

This appendix provides a detailed assessment of potential risks associated with the pre-launch, launch, recovery, and landing stages of *Airavata*. Each risk is systematically analyzed based on its likelihood of occurrence and the extent of its impact across key parameters—namely, health and safety, system performance, timeline adherence, regulatory compliance, and financial burden. To facilitate a consistent evaluation methodology, a structured Probability Scale and a comprehensive Implication Scale have been defined. These tools support the development of a robust Risk Assessment Matrix, which enumerates each risk, its root causes, estimated probability before and after mitigation, and its multi-parameter implications. This risk assessment approach enables the identification of operational weaknesses and supports the implementation of effective mitigation strategies, thereby promoting mission success and ensuring a high level of safety and reliability across all phases of *Airavata*'s deployment.

Table [No] Probability Scale

Probability	Implication
Very Low	Aforementioned issue has a very low probability to occur (unlikely to occur)
Low	Aforementioned issue has a low probability to occur, (slight chance to occur).
Medium	Aforementioned issue has a medium probability to occur, (might occur).
High	Aforementioned issue has a high probability to occur, (very likely will occur).

Table [No] Implication Scale

	Concerned Parameter	Level 1	Level 2	Level 3	Level 4	Level 5
A	Health and Safety	No or Minor injuries fixable by first aid.	Moderate injuries or illness.	Significant injuries and/or incapacitation and/or impairment	Injuries which can subtend to permanent injuries.	Loss of life
B	System Operation	Issue to components unimportant to flight	Loss of components unimportant to flight	Issue/Damage to components important to flight	Loss of components important to flight	Loss of complete system operation
C	Schedule	Minor operational delays	Delays < 3hrs	3hrs < Delays < 10hrs	Delays of over a day	Significant Delay pertaining to launch
D	Cost	< 10$	10$ - 30$	30$ - 100$	100$ - 500$	>1000$

Table [No] Risk Assessment Matrix

Phase	Risk	Probable Cause	Probability Before Mitigation	Mitigation Method	Probability After Mitigation	Implication			
						A	B	C	D
Pre-launch	Communication failure from the altimeters and Flight Computer	Antenna misalignment or improper configuration	Medium	Ensure proper line of sight between antenna and rocket and actively point towards the rocket. Place it in a higher location for clear signal	Low	L1	L3	L2	L1
	Improper function of Flight Software and Ground Station GUI	Mismatch in telemetry format in FSW and GCS. Errors in the code and improper optimization.	Low	Repeated testing debugging to ensure there is no misalignment in telemetry format.	Very Low	L1	L1	L1	L1
	Loss of GPS signal before launch	GPS sensor not connected to the flight computer properly.. Satellites not acquired properly,	Medium	Ensuring proper connections and testing the gps sensor prior.	Very Low	L1	L1	L1	L2
	Inferno in the Avionics Bay	Short circuits and High temperature	High	Ensuring proper insulation and connections	Low	L3	L5	L5	L4
	Non ignition of the motor starter mechanism	Shorted Ematch	Low	Ensuring proper continuity across Ematch	Very Low	L1	L4	L1	L2
	Detachment of Rail Buttons prior to launch	Improper handling of rocket.	Low	Handling only by specific qualified personnel.	Very Low	L1	L5	L1	L1
	Battery Overheating	Short circuits and High temperatur e	Very High	Ensuring proper insulation and connections	Medium	L1	L4	L1	L2
Launch	Premature Ignition	Unaccounted stray current passing through Ematch	Low	Ensuring no short circuits persists, proper insulation and connections	Very Low	L1	L3	L1	L1

	Failure to Liftoff	Motor failure, improper attachment of rail buttons.	Low	Handling only by specific qualified personnel.	Low	L1	L5	L4	L1
	Motor Failure during launch	Unfavorable weather conditions, improper connections, manufacturing defects.	Low (COTS motor)	Proper loading, installation in rocket by certified mentor.	Very Low	L1	L5	L4	L1
	Deviation from nominal flight path	Fin defects, improper anchoring of launch rod to the ground, failure of rail buttons during launch.	Medium	Checking fins for defects before launch, the rail buttons must be tightly secured and placed in to a sturdy launch rod	Low	L1	L5	L1	L1
	Body Tube Failure	Manufacturing defects	Low	Manufactured by reliable manufacturer, proper inspection and testing before launch.	Very Low	L1	L5	L1	L3
	Fins Failure	Manufacturing defects, improper integration	Medium	Checking fins for defects, proper integration of fin can, epoxying gaps between fin and body tube.	Low	L1	L5	L1	L3
Recovery	Premature Recovery Deployment	Premature ignition of black powder charge, shear pins shear off prematurely.	Medium	Extensive ground testing, proper integration of recovery mechanism, use of appropriate shear pins.	Low	L1	L5	L1	L3
	Altimeter Failure	Improper connections, incorrectly configured	Very high	Ensure proper and tight connections. Configure and test prior flight.	Medium	L5	L5	L5	L5
	Non-Deployment of Recovery	Failure of black powder charge to ignite, nose cone fails to eject, parachutes or shock cord	Medium	Extensive ground testing to determine right amount of black powder, proper packing of	Low	L1	L5	L1	L3

		gets caught in the airframe.		parachutes, use of correct shear pins.					
	Recovery Body Tube- Nose Cone Separation Failure	Use of wrong shear pins, failure of black powder recovery mechanism.	Low	Correct shear pins and required amount of black powder chosen based on calculations, extensive ground testing.	Very Low	L1	L5	L1	L3
	Failure of Drogue Parachute	Incorrect material choice, improper stitching, improper packing, premature or delayed deployment.	Medium	Proper material choice based on research, ensuring quality stitching, proper packing and extensive ground testing.	Low	L1	L5	L1	L3
	Failure of Main Parachute	Incorrect material choice, improper stitching, improper packing, premature or delayed deployment	Medium	Proper material choice based on research, ensuring quality stitching, proper packing and extensive ground testing.	Low	L1	L5	L1	L3
	Failure of Shroud Lines or Shock Cords	Shock cord and shroud lines may break, improper integration, incorrect material selection.	Low	Ensure proper integration by qualified team members, perform calculations to ensure the shock cord can handle expected load.	Very Low	L1	L5	L1	L3
Landing	Damage to Rocket post Impact	Hard landing, parachute failure, excessive descent speed, impact on rough terrain	Medium	Dual-redundant parachute system to ensure controlled descent. Shock-absorbing materials in key areas (e.g., nose cone, avionics bay, fins). Selection of	Low	L1	L4	L3	L4

				soft landing zones (grassy fields, sand, or designated recovery areas).					
	Damage to property post landing	Uncontrolled descent, strong winds, parachute failure, Improper placement of the launch tower	Medium	Controlled launch and designated landing zones to avoid populated areas. Real-time GPS tracking during descent.Public and regulatory notifications about launch windows and potential risks. Ensure that the launch tower is in proper position prior to launch	Low	L1	L4	L3	L4
	Injury to people while landing	Uncontrolled descent, strong winds, parachute failure, Improper placement of the launch tower	Low	Redundant parachutes: Dual deployment systems to prevent failure. Automated flight termination: System to disable rocket in case of severe trajectory deviation. Safety announcements & restricted zones: Keep spectators and personnel at a safe distance. Ensure that the launch tower is in proper position prior to launch	Very Low	L5	L4	L4	L4
	Landing of Rocket out of the recovery zone/ in a restricted area	Strong winds, trajectory deviation, parachute failure, GPS failure, miscalculated	Medium	Launch from locations with a large, clear recovery area to minimize risks. Live GPS tracking	Low	L2	L5	L3	L4

		descent, Improper placement of the launch tower		with real-time alerts. Dual-redundant parachute system for controlled descent. Regulatory clearance and coordination with authorities before launch. Ensure that the launch tower is in proper position prior to launch					
	No GPS Signal Obtained for recovery	Range, faulty gps sensor, improper antenna orientation.	Very High	Ensuring proper line of sight orientation at all times, prior testing of gps sensor in similar conditoos	Medium	L2	L1	L3	L3
	Injury to Recovery team while locating rocket	Rough terrain, environmental hazards (extreme weather), fatigue, improper safety gear	Medium	Provide proper safety gear (gloves, boots, first aid kits, GPS trackers). Pre-assess the landing area before launch to identify hazards. Establish a communication system (radio/GPS tracking) between recovery members.	Low	L3	L4	L4	L4

Appendix E.
Assembly, Preflight, Launch, Recovery and Off-Nominal Checklists

TEAM SAMMARD GROUND OPERATIONS

PROJECT AIRAVATA CHECKLIST FOR ASSEMBLY, PRE-FLIGHT, RECOVERY AND OFF-NOMINAL FLIGHT

Version 1.1

Updated 30/03/2025

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IREC 2025

Midland, Texas, USA

TEAM 188 - Vellore Institute of Technology
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1. Material List

The following list is a checklist for all the required components, items, parts, tools, and materials required for IREC 2025. The team members are mandated to follow the checklist before leaving for the USA from India and in all the days of the competition.

Item Name	Quantity	Category		Item Name	Quantity	Category	
Soldering Iron +	1	Avionics		18650 Batteries	10	Avionics	
Antennae	5	Avionics		9V Batteries	2	Avionics	
Flight Computer	2	Avionics		Eyebolt M10	4	Part	
Bessie	1	Avionics		Swivel	4	Part	
Shrota	1	Avionics		Quicklink	6	Part	
Power Distribution board	1	Avionics		M4X10mm screw	12	Part	
RRC3	1	Avionics		M4X20mm screw	42	Part	
Raven 4	1	Avionics		M4X16mm screw	12	Part	
SRAD Recovery	1	Avionics		M3X16mm screw	84	Part	
Jumpers	-	Avionics		M3X12mm screw	8	Part	
Jettson Nano	1	Avionics		Shear-pins M4X15mm	50(pack)	Part	
Magnetic Switches	7	Avionics		L bracket	16	Part	
Camera module	1	Avionics		Epoxy and hardener	1	Material	
Screws (Size?)		Avionics		Drogue parachute	1	Component	
Master Key	1	Avionics		Main parachute	1	Component	
Shrota	2	Avionics		Shock Cord	1	Component	
Wire Cutter	1	Avionics		Drill	1	Tool	
Avionics Bay Printed	1	Avionics		Allen Keys Kit	2	Tool	
Extra Connectors	-	Avionics		Angle cutter	1	Tool	
Micro USB	2	Avionics		Vernier calliper	2	Tool	

cables							
Posters		Miscellaneous		Measuring tape	1	Tool	
Brochures		Miscellaneous		Tender Descender	5	Component	
Carbon fibre		Material		Markers	10	Miscellaneous	
Fiberglass		Material		Black Powder	1 lb	Component	
1515 large rail buttons	2	Component		Fin Can	1	Component	
Pliers	2	Tool		Battery pack	1		
Mallet	1	Tool		Binoculars	2	Miscellaneous	
Clamp	2	Tool		Rubber Bands		Miscellaneous	
Masking tape	5	Miscellaneous		Zip Ties		Miscellaneous	
Drill bits		Tool		Extension boxes	2	Miscellaneous	
Sandpaper	5	Material		Orbital Sander	1	Tool	
Adapter	5	Miscellaneous		Camera	1	Miscellaneous	
Fevi-quick	10	Miscellaneous		Gloves		Miscellaneous	
Hot glue gun	1	Tool		Paper plates/bowls		Miscellaneous	
Weighing Scale	2	Tool		Electrical tape	3	Tool?	
Screw Driver	3	Tool		Masks		Miscellaneous	
Brush	5	Miscellaneous		M seal	3 boxes	Material	
X stands	2 sets	Miscellaneous		Araldite	1 box	Material	
Boric powder	1 box	Miscellaneous		Wire cutter	3	Tool	
Spray Paint		Miscellaneous					
Ats components							
Antenna	1	Component		Antenna Stand (3d printed)	1	Component	
Cu wires		Part		Antenna holder (3d printed)	1	Component	
Antenna Base	1	Component		Bevel gears	2	Component	

plate (3d printed)				(3d printed)			
Spur gears (3d printed)	4	Component		Holding Rod (3d printed)	1	Component	
Gear shaft (3d printed)	1	Component		Nema 17 stepper motor	2	Component	
Hollow gear shaft/holder (3d printed)	1	Component		Stepper motor driver	2	Component	
Wires		Part		Ball bearing	1	Component	
Microcontrollers	1	Component		Thrust bearing	1	Component	
ESP 32	1	Component					

2. Assembly Checklist

1. Camera Assembly
1.1. Mark the location of the holes on the body tube, using the camera mount and Ensure the mount is fitting snug.
1.2. Drill the holes, and tap them, mount the camera print using screws.
1.3. Pass the wiring through the mount and tape the connection wire inside the tube to the walls.
1.4. Ensure the length of the wire in enough on the inside, to connect to the avionics
1.5. Connect the cameras to the wires and the mount using glue, and epoxy
1.6. Test the cameras by connecting to the avionics.
2. Avionics Assembly
2.1. SIRIUS
2.1.1 Visually inspect for any physical abnormalities or damages on flight computer
2.1.2 Connect the RUBBER DUCK antenna to the SMA connector
2.1.3 Check the Potential of the 2S batteries and connect it to the flight computer
2.1.4 Wait 2–5 seconds for two buzzer beeps confirming normal operation.

2.1.5 Repeat the same procedure for the secondary flight computer
2.2. BESSIE V2.0
2.2.1 Visually inspect for any physical abnormalities or damages on BESSIE
2.2.2 Connect the RUBBER DUCK antenna to the SMA connector
2.2.3 Check the Potential of the 2S batteries and connect it to the BESSIE
2.2.4 Wait 2–5 seconds for two LED flashes confirming normal operation.
2.3. MASTER PIN SWITCH
2.3.1 Visually inspect for any physical abnormalities or damages on MASTER PIN SWITCH
2.3.2 Connect the correct battery terminals in the correct input side designators (LEFT SIDE)
2.3.3 Connect the correct output terminals with the right component designators(RIGHT SIDE)
2.3.4 Check the LED are OFF or ON if the switch is PRESSED or RELEASED respectively
2.4. RRC3
2.4.1 Visually inspect for any physical abnormalities or damages on RRC3
2.4.2 Check the Potential of the 9V alkaline battery and connect it to the RRC3
2.4.3 After 3-10 secs wait for a long beep for 5 secs indicating there is no post-fault error
2.5. RAVEN4
2.5.1 Visually inspect for any physical abnormalities or damages on RAVEN4
2.5.2 Check the Potential of the 9V alkaline battery and connect it to the RAVEN4
2.5.3 Check for 9 fast beeps followed by slow beeps every 2 seconds
2.6.FEATHERWEIGHT GPS
2.6.1 Visually inspect for any physical abnormalities or damages on FEATHERWEIGHT GPS
2.6.2 Connect the RUBBER DUCK antenna to the SMA connector
2.6.3 Check the Potential of the 1S battery and connect it to the FEATHERWEIGHT GPS
2.6.4 After 3-5 secs connect the FEATHERWEIGHT GPS to the FeatherWeightUI App
3. Payload Assembly
3.1. Payload electronic components are checked for any physical abnormalities or damages.
3.2. Payload structure is assembled ground up with its structural parts and components. (Example: PCBs, Cooler Fan, 3D-Printed Parts).

3.3. The strain gauge is attached to the lower body-tube and is connected to the health monitoring pcb in the bottom of the payload through screw terminal.
3.4 The scintillator (BC408) has to be attached to the 3d printed holder and the muon-counting pcb has to be assembled on top of it.
3.5. The additive-manufactured scintillator has to be attached to the holder along with the muon-counting pcb has to be assembled on top of it.
3.6. The 3d printed holder for VisNav has to be assembled in the payload and the Nvidia Jetson should be assembled after assembling the batteries in the battery holder.
4. Wiring and Connectors
4.1. Battery to MASTER PIN SWITCH
4.1.1 Connect the batteries terminal in the required series and parallel configuration
4.1.2 Connect the required batteries to the input side of the MASTER PIN SWITCH (LEFT)
4.1.3 Make sure the batteries are connected in the correct designated connectors
4.2. MASTER PIN SWITCH to Components
4.2.1 Connect all the components to the output side of the MASTER PIN SWITCH (RIGHT)
4.2.3 Make sure the batteries are connected in the correct designated connectors
5. Rocket Integration
5.1. Fincan Assembly
5.1.1. Fix the flat bars with the fin tab, in between them, with nyloc nut and screws.
5.1.2. Fix the flat bar to the quarter arcs using L brackets & nyloc screws at the precut holes.
5.1.3. Complete the assembly of all four fins and arcs.
5.2. Thrust Plate Integration
5.2.1. Take the lower body tube and mark out the location of the thrust plate
5.2.2. Drill 6 equally spaced holes on the body tube at marked locations with the help of the jig.
5.2.3. Place the thrust plate inside and align the holes with the ones on the body tube.
5.2.4. Screw the thrust plate with the help of M4 Screws.
5.3. Bulkhead Integration
5.3.1. Take both the body tubes and mark out the locations of the bulkheads.
5.3.2. Drill 4 equally spaced holes at the location with the help of the jig.
5.3.3. Place the bulkhead inside and align the holes with the ones on the body tube.

5.3.4. Screw the bulkheads using M4 screws.
5.3.5. Repeat the same at the other locations and ensure all the bolts
5.4. Nose Cone Preparation
5.4.1. Take a bulkhead and drill an M10-sized hole at the centre.
5.4.2. Screw an M10 eyebolt and fix it using a thread locker.
5.4.3. Place the bulkhead as deep within the nose cone with the eyebolt accessible.
5.4.4. Use metal epoxy to fix the bulkhead to the nose cone and let it dry.
5.4.5. Tug on the eyebolt to check it is fixed properly.
5.4.6. Drill 2 equally spaced holes on the nose cone shoulder for the shear pins.
5.4.7. Take the recovery lip, fix it inside the shoulder with metal epoxy, and dry it.
5.4.8. Check if the lip is fixed and rigid.
5.5. Parachutes Preparation
5.5.1. Lay the parachutes out flat forming a semicircle
5.5.2. Fold both parachute's ends to the centre and repeat until a satisfied size is reached.
5.5.3. Fold the shroud lines inside with the end remaining out.
5.5.4. Fold the parachutes along their vertical axis fold the length of the parachute in a zig
5.5.5. Fold the length of the parachute in a zig-zag manner until the packing length is reached.
5.5.6. Tie the end opposite to the shroud lines with the help of a rubber band.
5.5.7. Connect all the shroud lines to a quick link for easy attachment.
5.5.8. Apply talcum powder as a greasing agent to the parachute.
5.6. Recovery Mechanisms Preparation
5.6.1. Main Recovery Mechanism
5.6.1.1. Coat inner surfaces of the piston mechanism, and charge cups with plaster of Paris.
5.6.1.2. Fill the charge cup with black powder and insert e-match through the bottom
5.6.1.3. Stuff the remaining space with cotton, and repeat the same with redundant charge.
5.6.1.4. Seal the hole in the charge cup and pass the wires through the main mechanism
5.6.1.5. Again seal the bottom hole, and seal the mechanism by placing the top half.
5.6.1.6. Verify, if the mechanism is loose or not airtight.

5.6.2. Tender Descender Mechanism
5.6.2.1. Coat the inner surfaces of charge cups with plaster of Paris.
5.6.2.2. Fill the charge cup with black powder and insert e-match through the bottom
5.6.2.3. Stuff the remaining space with cotton and pass the wires through the bottom
5.6.2.4. Seal the bottom hole, and seal the mechanism by placing the top half.
5.6.2.5. Pass a short wire through the top and bottom mechanism.
5.6.2.6. Tie off each end to two separate shock cords and knot each end to the halves.
5.6.2.7. Place the shear pins in and verify if all parts are proper.
5.7. Recovery Mechanism Integration
5.7.1. Attach the flat shock cord to the eyebolt of the nose cone bulkhead with a bowman knot.
5.7.2. Pass the cord through the main piston recovery mechanism hole.
5.7.3. Place the main recovery inside the shoulder of the nose cone.
5.7.4. Provide some slack and use a bowline knot under the main recovery.
5.7.5. Attach the shroud lines of the drogue parachute to the flat shock cord using a quick link and a swivel.
5.7.6. Make two connections from the drogue to the tender descender and the main parachute.
5.7.7. Attach the shock cord from the drogue to one side of the tender descender and extend the other line.
5.7.8. Tie another shock cord at the other end of the tender descender and attach to the eyebolt in body tube.
5.7.9. Ensure the other connection is longer than the TD line and attach the main parachute to it using a swivel and quick link.
5.7.10. Extend the shock cord and connect it to the eyebolt in the body tube.
5.7.11. Bolt the eyebolt to the bulkhead using a nut and use a thread locker to fix it.
5.7.12. Tape the electrical wires along the body tube walls, with masking tape and grease the walls with talcum.
5.7.13. Pack the shock cords with the connections and place the main parachute into the body tube.
5.7.14. Place the tender descender above the main parachute and cover it with the remaining shroud line
5.7.15. Place the drogue and push the contents deep into the body tube.
5.7.16. Place the nose cone on the body tube and using shear pins attach it to the body tube.
5.8. Rail Buttons, Fincan & Motor Integration
5.8.1. Mark the location of the Rail Buttons on the lower body tube.

5.8.2. Drill the holes at marked locations and attach the rail buttons
5.8.3. Take the assembled fin-can and insert it into the lower body tube through the slits.
5.8.4. Take the motor, pass it through the fin can, and screw it to the thrust plate with a bolt.
5.8.5. Attach the lower ring for the fin can.
5.8.6. Use epoxy, to apply fillets at the fins.
5.8.7. Verify if all parts are fixed.
5.9. Avionics Bay Integration
5.9.1. Attach The COTS components RRC3, RAVEN4, FEATHERWEIGHT GPS in the intended location using brass standoff
5.9.2. Attach The SRAD components 2 x SIRIUS, BESSIE, MASTER PIN SWITCH in the intended location using brass standoff
5.9.3. Insert the batteries in the battery bay and make wire connections using bolts and ring terminal
5.9.4.Connect the batteries in the necessary predetermined configuration for powering components
5.9.5.Connect the battery terminal to the input side of MASTER PIN SWITCH (LEFT SIDE) and make sure to connect it to the correct designated pin
5.9.6.Connect the output of MASTER PIN SWITCH (RIGHT SIDE) to the components and make sure to connect it to the correct designated pin
5.9.7 Ensure that there are no loose connections during the assembly process.
5.9.8 Verify that all systems are powering on and operating nominally.
5.9.9 Shake the fully assembled avionics bay and confirm the nominal operation of components
5.10. Payload Integration
5.10.1. Take the payload and bolt it to the bulkhead above the thrust plate using screws.
5.10.2.Take another bulkhead and attach it to the placed payload and then to the body tube using screws.
5.10.3.Mark and drill a hole on the body tube to expose the payload and the switch to turn it on is reachable
5.10.4. Check if the payload is operational by turning it ON.
5.10.5. Once Verified Turn off the Payload.
5.11. Body Tube Integration
5.11.1. Take the coupler and drill 4 holes equidistant from each other and 2 inches from both ends.
5.11.2. Place the coupler in the body tubes and mark the places for the holes.
5.11.3. Drill out the locations and tap these ends, and screw the coupler to the lower body and ensure it sits flush.

5.11.4. Place the avionics bay inside the coupler and attach another bulkhead to the top of the avionics.
5.11.5. Mark the holes on the coupler and body tube for the bulkhead to be attached and drill them
5.11.6. Align the holes for the bulkhead and screw them to the body tubes.
5.11.7. Drill 2 Holes through the coupler, such that the avionics is able to be turned ON.
5.11.8. Verify if the Avionics is able to be turned ON and check telemetry, once verified Turn to OFF position.
5.12. Final Integrated Check
5.12.1. Verify the flexibility of fins.
5.12.2. Verify all external screws are flush.
5.12.3. Verify the Telemetry is being received.
5.12.4. Verify the Shear pins are properly attached.
5.12.5. Verify the Payload, Camera and Avionics is turned ON.

3. Pre-Flight Checklist

1. Ensure the completion of the following checklists.
1.1. Assembly
2. Inspect rail buttons.
2.1. Verify that rail buttons are securely attached.
2.2. Verify that the rail buttons are positioned correctly.
2.3. Inspect for any indications of wear or damage.
3. Transport the rocket to the launch pad.
4. Prepare the launch rail.
4.1. Loosen all screws on the launch rail and position it horizontally.
4.2. Examine the launch rail for any damage or obstructions.
5. Apply dry graphite spray to lubricate the rail.
6. Position the rocket on the launch rail.
6.1. Carefully slide the rail button into the launch rail until the rocket reaches the end stop.
6.2. Be cautious to prevent twisting or lifting of the rocket once the rail guides are in position.

7. Elevate and adjust the launch rail.
7.1. Lift the launch rail with care.
7.2. Modify the launch angle as necessary, according to conditions
8. Proceed to the launch checklist.
9. Ground Control Station: System Setup
9.1 Power up the laptop
9.2 Run the GCS Software, check for any errors
10. Telemetry Link
10.1 Confirm downlink telemetry packets are being received
10.2 Confirm packet rate is stable.
11. Data Visualization & UI
11.1 Ensure the map view displays the current rocket location
11.2 Confirm 3D Rocket model is updating with telemetry
11.3 Confirm whether all graphs are updating dynamically with the telemetry data.
12. Flight Software
12.1 Confirm system time is synced with GCS
12.2 start telemetry broadcast
12.3 confirm all sensors are reporting correctly
13. Run a dry test for state transitions.

4. Launch

1. Verify that all pre-flight checklists have been completed.
2. Activate the Avionics.
2.1. Monitor the sound occurring every 5 seconds.
3. Activate the Camera.
4. Activate the Payload.
5. Verify the establishment of a connection between Avionics and Telemetry.

5.1. If the connection has not been established, examine the channel and baud rate values on the ground station interface, and rectify them accordingly.
6. Verify that Avionics is connected to a minimum of 4 satellites.
7. Ensure the altimeter is operational.
8. Final Checks: GCS
8.1 Confirm telemetry link is active
8.2 Reconfirm GPS Fix is valid
8.3 Final Status on UI: all sensors are sending values.

5. Recovery Checklist

1. Approach the rocket cautiously and ensure it is safe to handle.
2. Examine the rocket splash zone for damage sustained or missing parts during flight or landing.
3. Cut any exposed wires, to disconnect the recovery mechanisms.
4. Document the splash zone as is and observe any damages, existing and new.
5. Gather all rocket components, including any detached parts or debris.
6. Provide competition officials with pertinent flight data, including achieved altitude, flight duration, and recovery system performance.
7. Dispose of used motors and igniters safely, and clean blackened surfaces to check for damage.
8. Safely store the rocket and other equipment for future use.
9. Open the recovery application on team members' mobile phones
10. Input or receive the last known rocket coordinates
11. The arrow points towards the coordinates of the rocket
12. Once recovered, take screenshots of recovered components and submit the final recovery log for the final report

6. Off-Nominal Checklist

1. Launch Scrub
1.1 Scrub the launch in case of failure to pass Final Flight Safety Review.
1.2 Scrub the launch in case if COTS Altimeter (main and redundant) are non-responsive.
1.3 Scrub the launch in case of improper/non-ignition of rocket motor.
1.4 Scrub the launch in case of any structural damage to the body of the rocket impacting its aerodynamics.
2. Safety Violation
2.1 In case of any safety violations pointed out in the Flight Review, address all issues promptly following required safety protocols and re-attempt FSR.
2.2 In case of any critical system malfunction which cannot be handled within necessary time frame, disengage all active mechanisms and prepare for launch scrub.
3. Meteorological Issues
3.1 Consult with meteorological experts and local weather app to predict impact of weather on the rocket trajectory.
3.2 Make necessary predictions and precautions for abnormal meteorological conditions.
3.3 In case of conditions being too drastic for launch, proceed to scrubbing launch.

Appendix G.
Rocket Navigation Application

1. 1. Introduction

The Rocket Navigation App is a custom mobile application built with Flutter that aims to track and locate a rocket after it has landed, particularly in remote environments such as deserts. The primary purpose of the app is to assist the recovery team of Team SAMMARD by calculating the real-time distance and direction from the user's current location to the rocket's final landing coordinates. The app achieves this by integrating GPS and Bluetooth functionalities, enabling it to communicate with a location-transmitting module onboard the rocket. Additionally, the app also has a components page, the functionality of which is to store images of the recovered parts of the rocket, the state they are in and a description if necessary. Apart from this there is a Terminal page which keeps in check the fact that the data received from Bluetooth is not garbage value.

2. 2. Workflow and System Architecture

Overall Workflow:

a. Rocket Launch and Tracking:

During or after the rocket's descent, a GPS module called Bessie 2.0 is integrated into the rocket which begins collecting its geographical coordinates. These coordinates include precise latitude and longitude values, which are essential for identifying the exact landing location of the rocket. To ensure the recovery team can track and locate the rocket efficiently, the onboard GPS continuously updates its location data.

The Bessie 2.0 data format is in the following form:
Packet no, RTC time, GPS time, GPS X, GPS Y, GPS Z, Acceleration X, Acceleration Y , Acceleration Z, Gyro X, Gyro Y, Gyro Z, Magnetic X, Magnetic Y, Magnetic Z.

Once the rocket has landed or is close to the ground, the collected GPS location is transmitted to the recovery team's device. This transmission is carried out via a Bluetooth connection, typically facilitated by a microcontroller called ESP32 S3 , which is capable of handling both Bluetooth communication and GPS data processing.

b. Data Reception:

To retrieve the location data from the rocket, the mobile application used by the recovery team establishes a connection with the Bluetooth module on the rocket (or the rocket's tracking device). Once the connection is established, the app receives the rocket's GPS coordinates Shrota 2.0. Shrota 2.0 collects data from Bessie 2.0. Bessie 2.0 is a SRAD GPS tracker. It uses the Lora module operating at 915MHz to collect data from the rocket. These coordinates, in the form of latitude and longitude, are crucial for the next steps in locating the rocket physically.

c. Current Location Acquisition:

In addition to receiving the rocket's location, the mobile app also gathers the current geographical position of the user (i.e., the person attempting to recover the rocket). This is achieved by accessing the phone's internal GPS sensor. By obtaining the user's real-time coordinates, the app can calculate relative positioning, which is necessary for navigation.

d. Distance Computation:

To determine how far the user is from the rocket and in which direction they should head, the app performs a calculation using data from multiple onboard sensors—specifically the gyro meter, accelerometer, and magnetometer. This combination of data sources is often referred to as sensor fusion. By blending readings from these sensors, the app can accurately estimate both the distance and bearing (direction) between the user and the rocket's location. This helps guide the user toward the rocket, even in challenging environments such as deserts or open fields where landmarks may be absent.

e. User Interface Output:

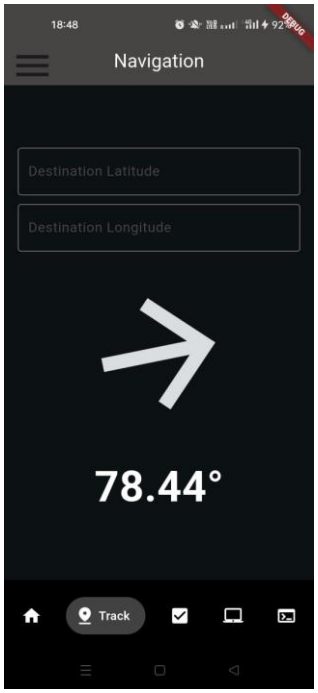
All the processed information is presented to the user through a clean and intuitive interface in the app. The interface displays three key pieces of information:

1. The rocket's last known coordinates (as received via Bluetooth).
2. The user's current GPS location.
3. The calculated distance and direction to the rocket.

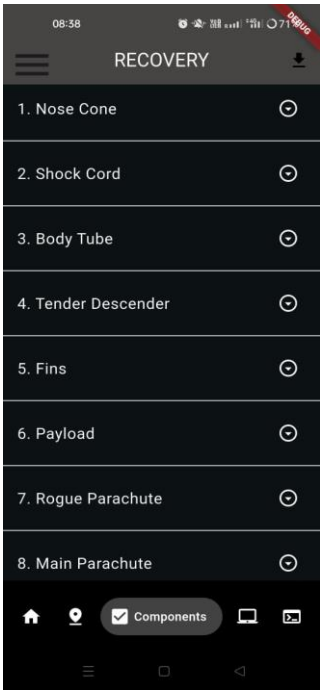
f. Recovered Components checklist:

An additional feature of this app is that along with real-time tracking, there is also a page where the recovery team gets to update the recovered components along with its state of condition. Further it also enables the recovery team to provide a description of the recovered part.

This enables the recovery team to effectively navigate toward the rocket's landing point, improving the chances of successful and timely recovery. The demonstration of the pages is attached below.



a. Navigation Page

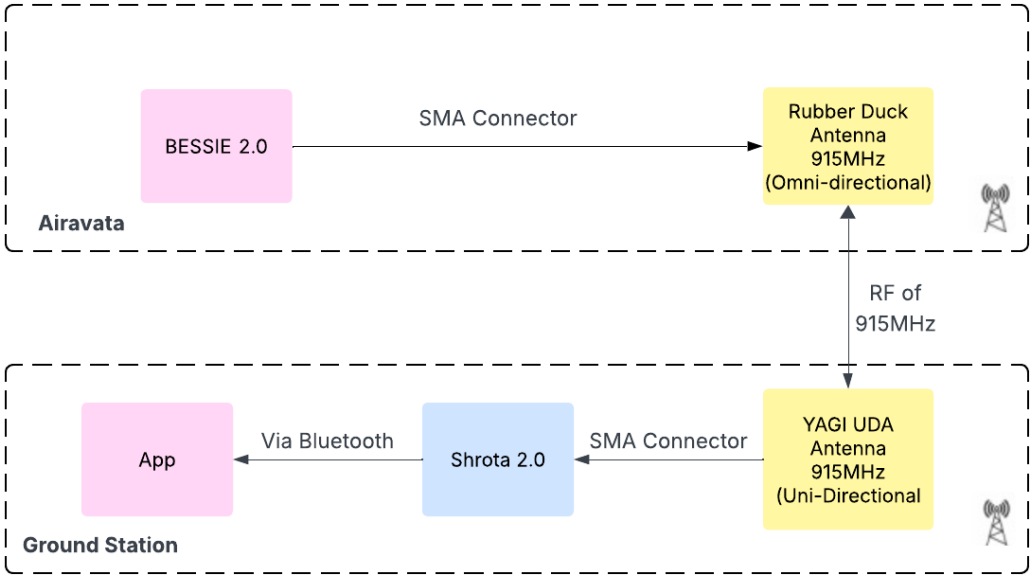


b. Recovery Page

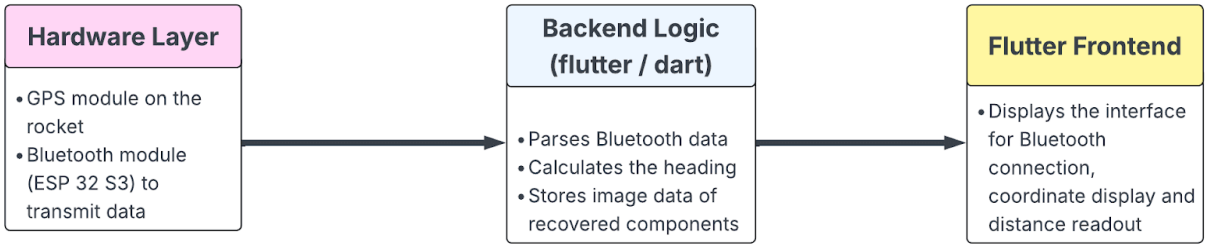


c. Terminal Page

System Architecture:



Architecture Layers:



3. 3. Technologies, Concepts, and Requirements

The development of this system relies on a carefully selected set of technologies and frameworks, each chosen to fulfil specific functional and performance requirements. The architecture is designed to ensure seamless interaction between hardware and software components while maintaining efficiency, scalability, and reliability. Below is an in-depth breakdown of the key technologies and their roles within the system.

Technology Stack

The foundation of the application is built upon Flutter, a versatile and powerful open-source framework developed by Google. Flutter enables cross-platform development, allowing the application to run smoothly on both iOS and Android devices without the need for separate codebases. The backend logic is also integrated within the Flutter environment, leveraging Dart as the primary programming language. Dart's strong typing, asynchronous capabilities, and robust performance make it an excellent choice for handling complex operations while maintaining clean and maintainable code.

Bluetooth Communication

To facilitate wireless communication between devices, the system incorporates specialized Flutter plugins designed for Bluetooth connectivity. Depending on specific use cases and compatibility requirements, different packages can be employed, including:

- flutter_blu

e: A comprehensive plugin that provides extensive Bluetooth Low Energy (BLE) functionality, supporting device discovery, connection management, and data transmission.

- flutter_bluetooth_serial: Ideal for classic Bluetooth (RFCOMM) communication, enabling serial data exchange between paired devices.
- Other Bluetooth plugins: Depending on project needs, alternative packages may be explored to optimize performance, power efficiency, or platform-specific features.

These plugins ensure reliable device pairing, stable data transfer, and real-time interaction, which are crucial for applications requiring peripheral communication.

Location Services

Accurate location tracking is achieved using the geolocator package, a widely adopted Flutter plugin that interfaces with the device's native GPS and network-based positioning systems. This package provides precise latitude and longitude coordinates. supports continuous location updates, and allows customization of accuracy levels based on power-saving or high-precision requirements. Whether the application needs real-time tracking, geofencing, or location-based triggers, the geolocator package ensures robust and efficient location data retrieval.

Distance Calculation

To measure proximity between devices or objects, the system utilizes the sensor_plus plugin, which enhances sensor fusion capabilities. This plugin aggregates data from multiple device sensors—such as the accelerometer, gyroscope, and magnetometer—to improve the accuracy of distance estimation. By combining Bluetooth signal strength (RSSI) with sensor data, the system can calculate relative distances more reliably, even in dynamic environments where signal interference or movement may affect measurements. This approach is particularly useful for applications like indoor navigation, asset tracking, or proximity-based alerts.

Key Concepts & Formulas:

- Bluetooth Serial Communication: Data is sent from the rocket via UART to a Bluetooth module and received as strings by the app.
- GPS Coordinates: Both the rocket and the user provide latitude and longitude for tracking purposes.

Hardware Requirements:

The successful operation of this system relies on a combination of specialized hardware components, each serving a critical role in ensuring seamless functionality. The hardware architecture is divided into two primary segments: the onboard rocket hardware (Shrota 2.0 and the user device Each segment is equipped with specific modules and capabilities to facilitate real-time data gathering, transmission, and processing.

Rocket Hardware (Shrota 2.0)

The Shrota 2.0 module is designed as an embedded system that integrates several key components to collect and transmit essential flight data. At its core is a microcontroller called ESP32 S3, chosen for its reliability, low power consumption, and ability to interface with multiple sensors. The microcontroller acts as the central processing unit, responsible for aggregating data from connected modules and transmitting it to the user device.

A GPS module is integrated into the system to provide precise real-time location tracking. This module captures critical telemetry data, including latitude, longitude, altitude, and velocity, which are essential for monitoring the rocket's trajectory and ensuring accurate recovery. The GPS module must offer high sensitivity and fast signal acquisition to maintain accuracy even during high-speed flight.

Additionally, a Bluetooth module is incorporated to establish a wireless communication link between the rocket and the user's smartphone. This module enables the transmission of GPS and sensor data in real time, allowing users to monitor flight parameters remotely. The choice of Bluetooth ensures a balance between range, power efficiency, and data transfer speed, making it ideal for short-range telemetry applications.

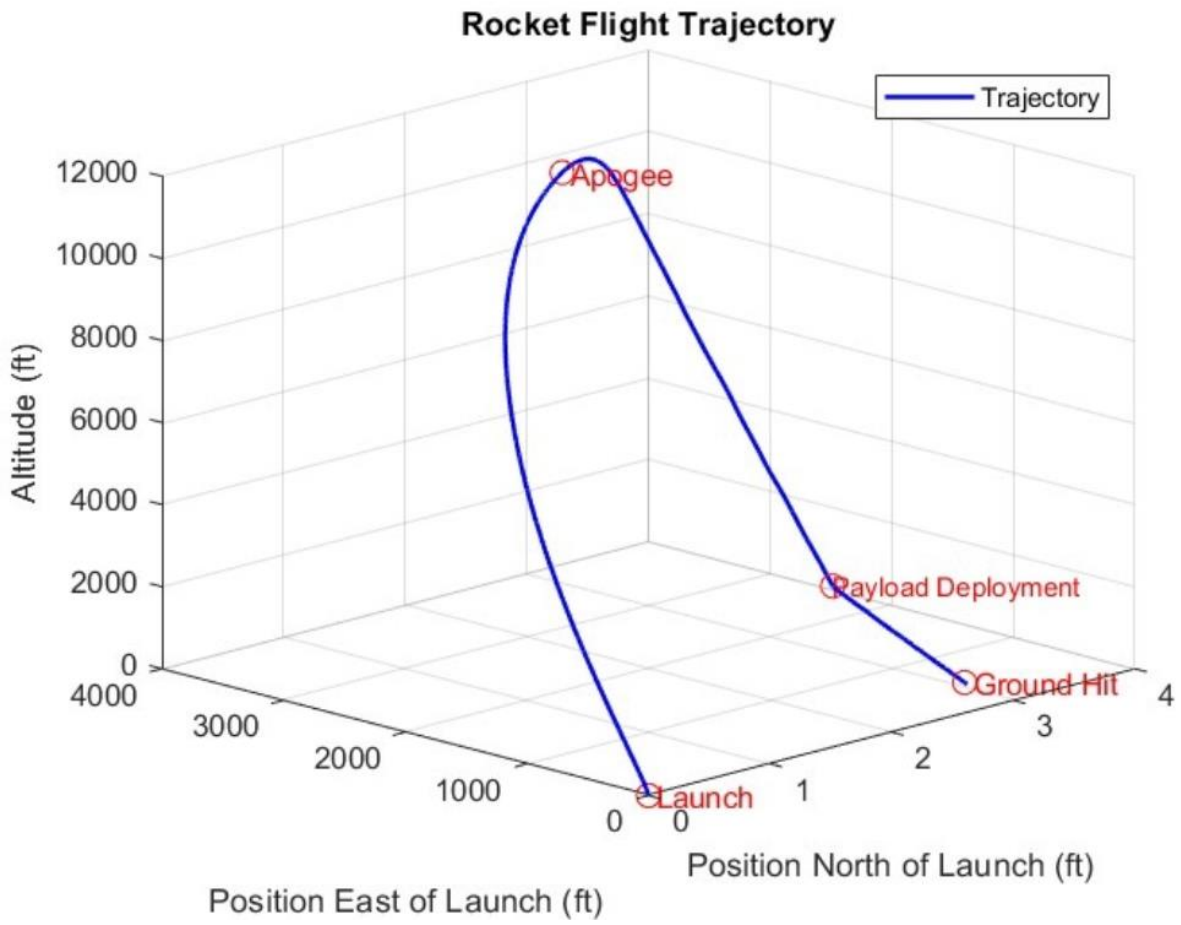
User Device

The system is designed to interact with an Android or iOS smartphone, serving as the primary interface for data visualization and control. The user device must meet several hardware and software specifications to ensure compatibility and optimal performance. First, the smartphone must have built-in GPS support to supplement the rocket's telemetry data. This allows for redundancy in location tracking, ensuring that positional data remains available even if the onboard GPS experiences temporary signal loss. The device's GPS should support high update rates for real-time tracking accuracy. Second, the smartphone must feature Bluetooth capability (preferably Bluetooth Low Energy for power efficiency) to maintain a stable connection with the onboard hardware. This ensures seamless reception of flight data, including location updates and sensor readings. Lastly, the device must run a Flutter-compatible operating system version (e.g., Android 5.0+ or iOS 8.0+) to support the cross-platform application framework. This ensures that the app functions smoothly across different devices while maintaining consistent performance and user experience.

Conclusion:

The hardware requirements for this system are carefully selected to ensure reliability, real-time data accuracy, and seamless communication between the rocket and the user's device. By combining robust embedded components (microcontroller, GPS, and Bluetooth) with a modern smartphone, the system achieves a balance of performance, portability, and usability, making it well-suited for aerospace tracking and recovery applications.

Appendix H.
Rocket Flight Trajectory



MATLAB Code:

```
% Load the data from the CSV file
data = readtable('Airavataforitr2.csv', 'CommentStyle', '#', 'ReadVariableNames', false);
Time = data.Var1;      % Time (s)
Altitude = data.Var2;  % Altitude (ft)
East = data.Var3;      % Position East of Launch (ft)
North = data.Var4;     % Position North of Launch (ft)

% Find indices of key events based on time
[~, idxLaunch] = min(abs(Time - 0));        % Launch at t = 0 sec
[~, idxApogee] = min(abs(Time - 24.941));   % Apogee at t = 25.079 sec
[~, idxRecovery] = min(abs(Time- 131.168)); % Updated Recovery Device Deployment at t = 131.484 sec
[~, idxGroundHit] = min(abs(Time - 182.11)); % Ground Hit at t = 182.125 sec

% Create a 3D plot
figure;
plot3(North, East, Altitude, 'b-', 'LineWidth', 1.5); % Plot the trajectory
hold on;

% Highlight key events
```

```
plot3(North(idxLaunch), East(idxLaunch), Altitude(idxLaunch), 'ro', 'MarkerSize', 8, 'DisplayName', 'Launch');
text(North(idxLaunch), East(idxLaunch), Altitude(idxLaunch), ' Launch', 'Color', 'red', 'FontSize', 10);

plot3(North(idxApogee), East(idxApogee), Altitude(idxApogee), 'ro', 'MarkerSize', 8, 'DisplayName', 'Apogee');
text(North(idxApogee), East(idxApogee), Altitude(idxApogee), ' Apogee', 'Color', 'red', 'FontSize', 10);

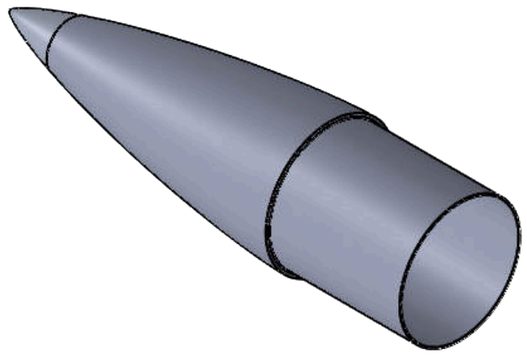
% Updated Recovery Device Deployment event
plot3(3.135, 1619.3, 1432.536, 'ro', 'MarkerSize', 8, 'DisplayName', 'Payload Deployment');
text(3.135, 1619.3, 1432.536, 'Payload Deployment', 'Color', 'red', 'FontSize', 9, 'Interpreter', 'none');

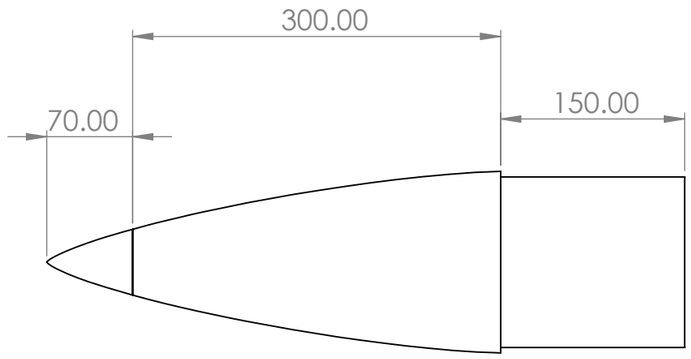
plot3(North(idxGroundHit), East(idxGroundHit), Altitude(idxGroundHit), 'ro', 'MarkerSize', 8, 'DisplayName',
'Ground Hit');
text(North(idxGroundHit), East(idxGroundHit), Altitude(idxGroundHit), ' Ground Hit', 'Color', 'red', 'FontSize', 10);

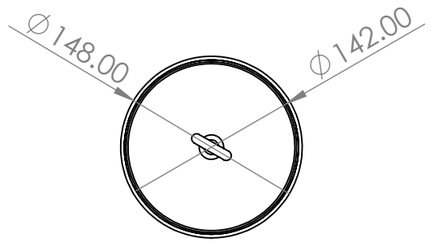
% Customize the plot
xlabel('Position North of Launch (ft)');
ylabel('Position East of Launch (ft)');
zlabel('Altitude (ft)');
title('Rocket Flight Trajectory');
grid on;
legend('Trajectory', 'Location', 'northeast');
hold off;

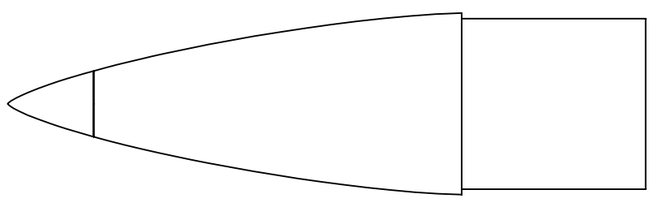
% Adjust view for better visualization
view(-45, 20);
```

Appendix I.
Engineering Diagrams

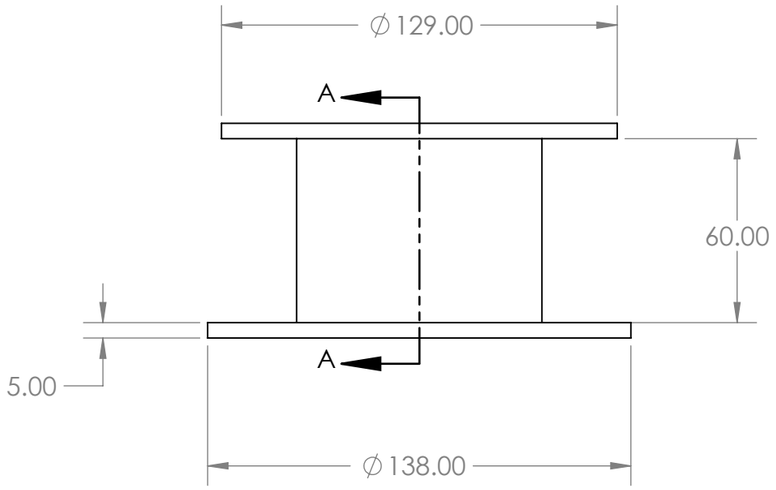


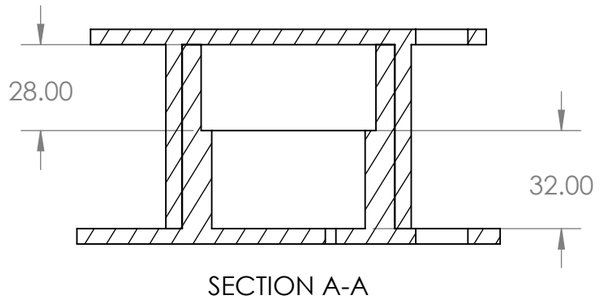




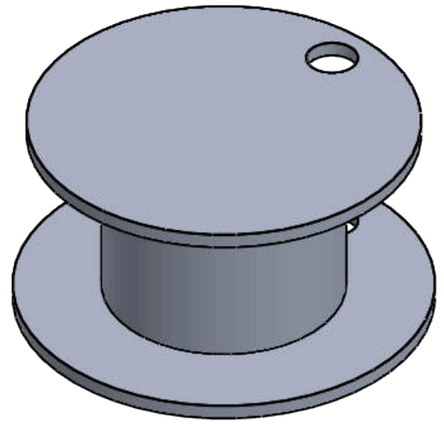


UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:			DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION
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CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL:		DWG NO.	A3
						NOSECONE ASSEMBLY	
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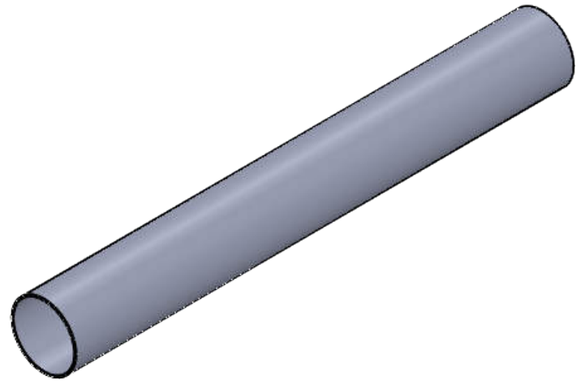


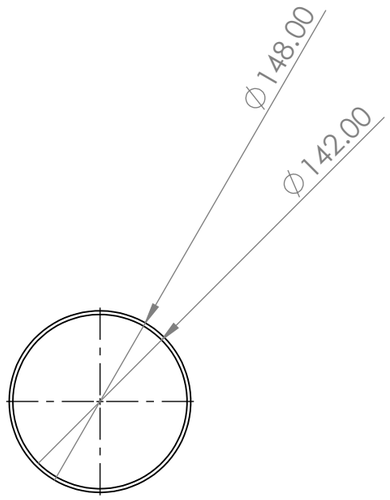


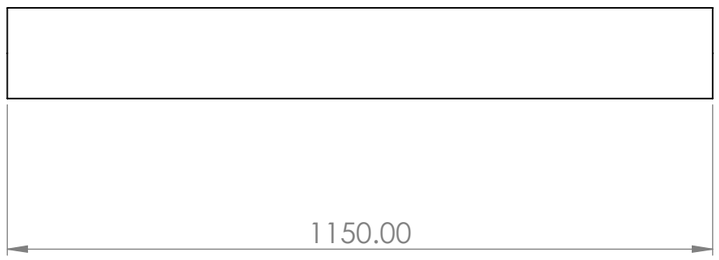
SECTION A-A



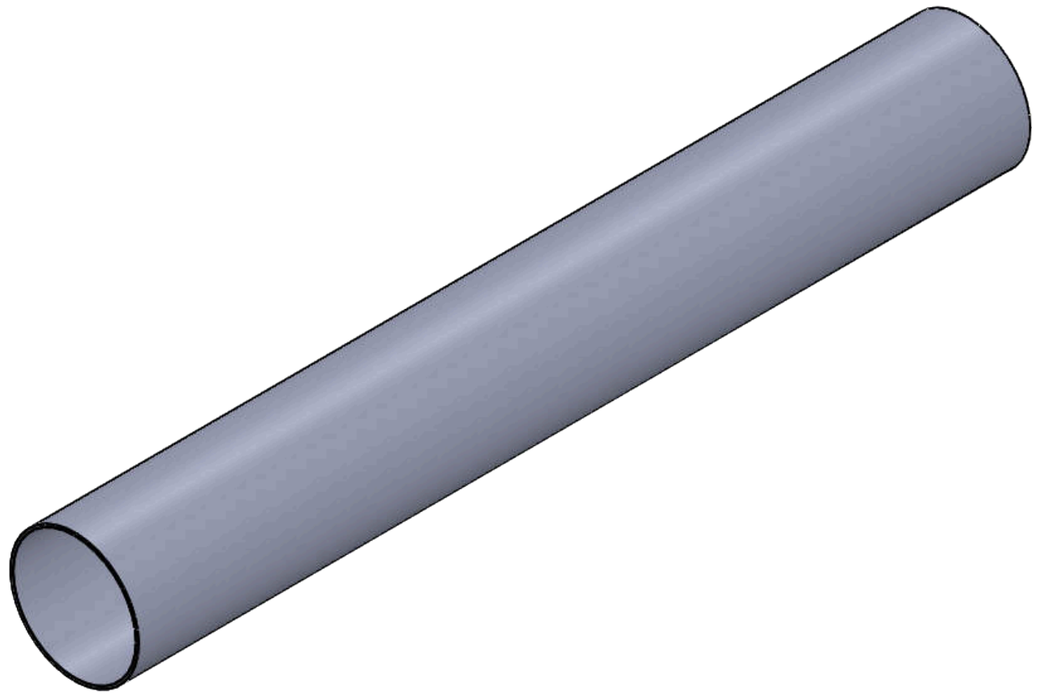
UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:			DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION
	NAME	SIGNATURE	DATE			TITLE:	
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CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL:		DWG NO. Recovery Mechanism	A3
				WEIGHT:		SCALE:1:2	SHEET 1 OF 1

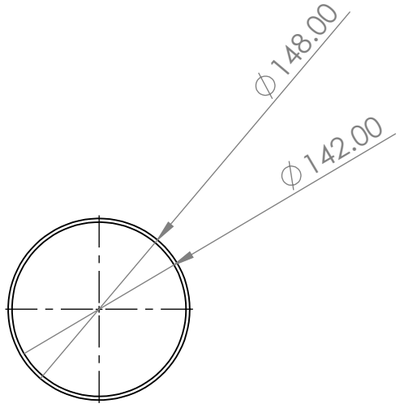


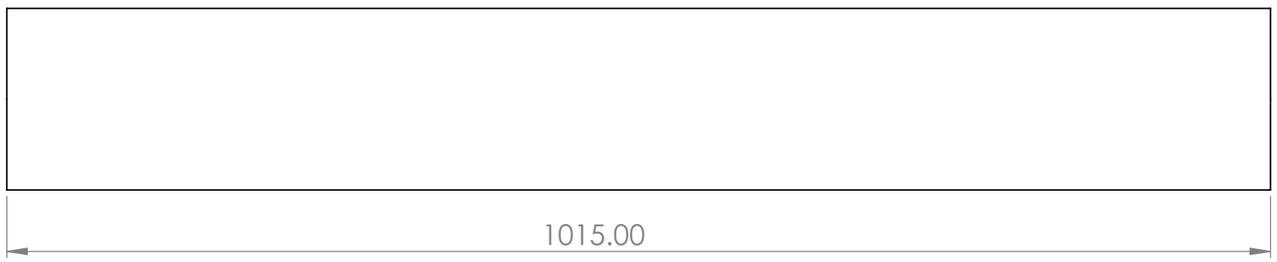




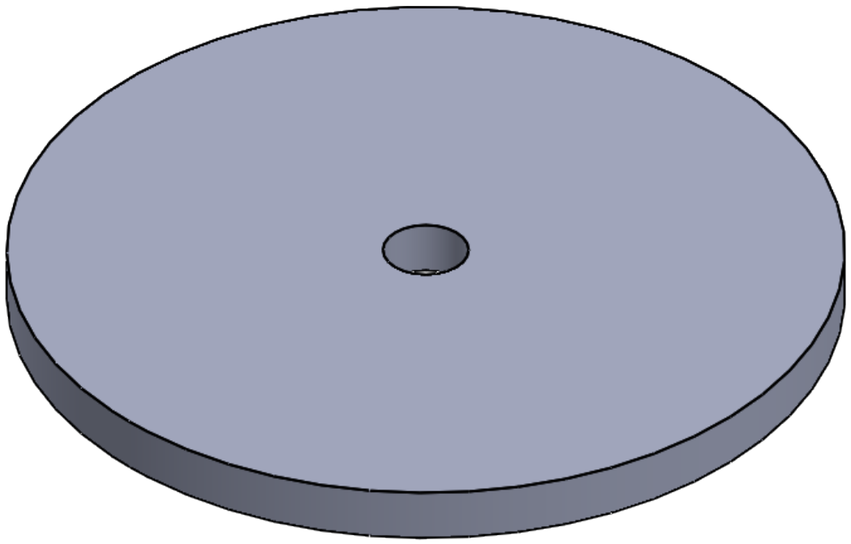
UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:			DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION
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DRAWN							
CHK'D							
APPV'D							
MFG							
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				WEIGHT:		SCALE:1:5	SHEET 1 OF 1

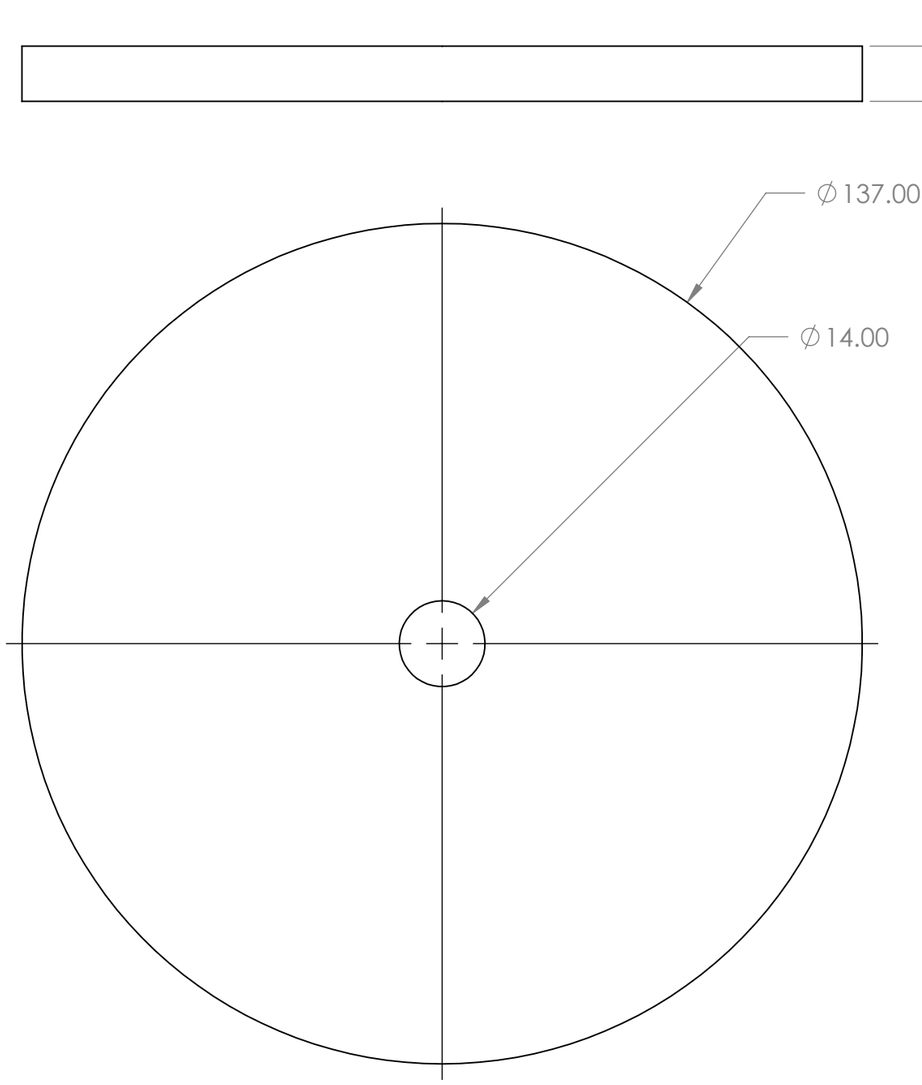




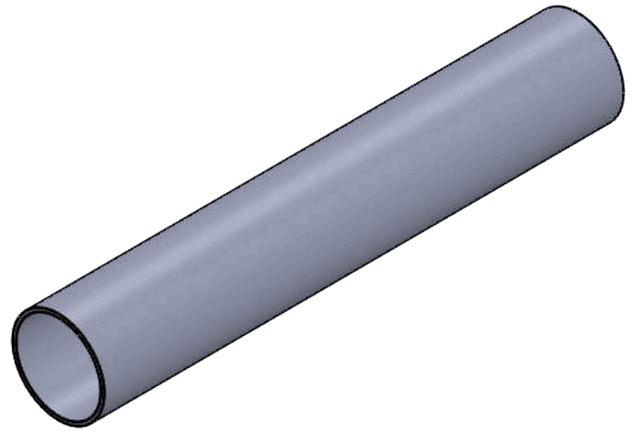


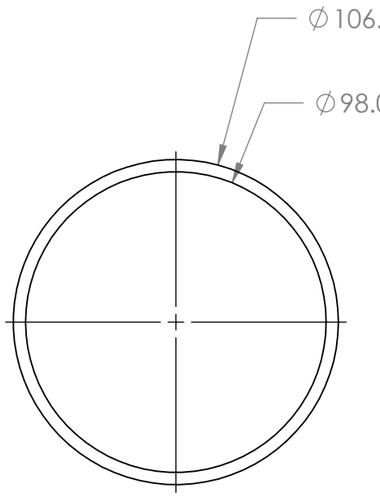
UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION
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DRAWN							
CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL:		DWG NO. Lower Body Tube	A3
				WEIGHT:		SCALE:1:5	SHEET 1 OF 1





UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:			DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION
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CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL:		DWG NO. Thrust Plate	A3
				WEIGHT:		SCALE:1:2	SHEET 1 OF 1





UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:			DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION
	NAME	SIGNATURE	DATE			TITLE:	
DRAWN							
CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL:		DWG NO. Motor Case	A3
				WEIGHT:		SCALE:1:5	SHEET 1 OF 1